

Mod 5: Heredity

IQ1: How does reproduction ensure the continuity of a species?

5.1.1 explain the mechanisms of reproduction that ensure the **continuity of a species**, by analysing sexual and asexual methods of reproduction in a variety of organisms, including but not limited to:

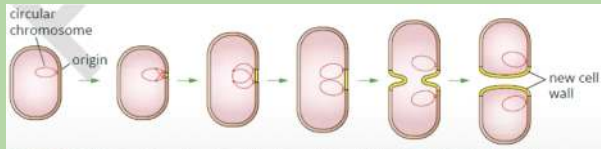
1. animals: advantages of external and internal fertilisation
2. plants: asexual and sexual reproduction
3. fungi: budding, spores
4. bacteria: binary fission (ACSBL075)
5. protists: binary fission, budding

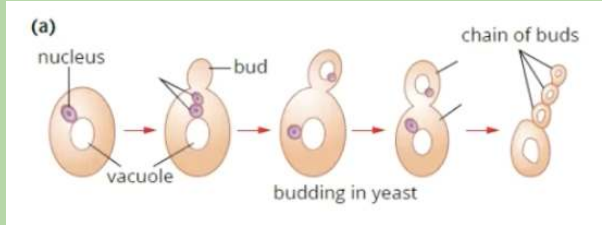
5.1.1 Reproduction

	Sexual vs Asexual	
Definition	Involves the fusion of the sperm and egg to form offspring that are genetic mixture of both parents	Involves a single parent, which produces a new individual from part of itself
Advantages	- Means a population is better able to adapt and survive, especially in a changing environment - Allows for genetic variation	- Population can increase in size at a faster rate - Involves only one parent
Disadvantages	- In changing environmental conditions, the population lacks the genetic diversity to evolve. - Slow reproductive rate	- Costs the parents time, energy and resources - No genetic variation in offspring

	Internal vs External	
Definition	Occurs when the male transfers his gametes directly into the female's body through a tube in his penis, using the process of copulation	A male's sperm fertilises a female's egg outside of the female's body
Advantages	- More offspring survive due to protection and paternal care - Successful fertilisation more likely	- Little energy needed to find a mate - Many offspring produced - Offspring not in competition with parents
Disadvantages	- Requires more energy to find a	- Many gametes do not survive or are

	mate - Fewer offspring produced - Large investment from female parent, may leave her vulnerable	not fertilised - Zygotes and embryos are protected - No parental care mean many offspring do not survive to adulthood
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Fission	Splitting of one cell into two (equal size)
Binary fission	The process by which prokaryotic organisms divide and reproduce (very rapidly at warm temperatures and with plenty of nutrients)
Budding	Outgrowths from a parent cell, each smaller than the parent
Fragmentation of body parts	Where a piece of the body, or fragment, of the parent breaks off and develops into an independent offspring
Spore formation	Where spores germinate and create a new plant under favourable conditions
Parthenogenesis	Where the offspring develops from the egg without the prior fertilisation from the male gamete
Bacteria	- Binary fission: a single parent cell splits into <u>two approximately equal daughter cells</u> which occurs more quickly in <u>prokaryotes</u> due to the lack of <u>organelles</u> and smaller amounts of DNA  FIGURE 2.1.4 Division of a prokaryotic cell by binary fission. Replication of the single circular chromosome begins at a point called the origin. When the parent cell has grown large enough, a new cell wall and membrane are laid down to divide the cell into two genetically identical smaller daughter cells.
Protists	<u>Binary fission:</u> - The body of an individual is pinched into two parts/halves; the parent cell disappears and is replaced by a pair of daughter nuclei in two new cells (need to be mature to be recognisable as members of the parental species)

	- A membrane-bound nucleus that needs to be replicated means that it is a different type of binary fission <u>Budding:</u> - Occurs when a new identical organism grows from the body of the parent - Usually occurs on the outside of the cell from which it detaches to live independently or sometimes remains in contact to form a colony
Fungi	<u>Budding:</u> - First the parent yeast cell produces a small outgrowth that grows larger and forms a bud - The nucleus of the parent cell splits off a smaller daughter nucleus, which migrates into the daughter cell. The bud detaches from the parent by pinching inwards at the base, although it may remain in contact with the parent cell - The bud starts off much smaller than the parent but is genetically identical  <u>Spores:</u> - Can be produced asexually and are called mitospores or sexually and are called meiospores - A cluster of spores can be produced inside a sporangium which can be dispersed into the environment
Mitospores	Haploid reproductive cells capable of developing into an adult without fission with a second cell. They are genetically identical and produced in large numbers.
Plants	<u>Vegetative reproduction:</u> - The growth of specialised plant tissues that can grow into a new plant if it

	becomes separated from the parent plant - Can produce a rapid increase in the number of plants growing in an area so that they can outcompete, or displace, neighbouring species. However, under constant competition from sister and parent plant for resources - Some plants also undergo asexual reproduction when broken off pieces of branch regenerate into identical new plants (fragmentation)
Animals	<u>Regeneration:</u> - When a detached part of an individual grows into another individual <u>Fission:</u> - Can occur in multicellular animals to produce 2 new offspring <u>Budding:</u> - Producing genetically identical but smaller buds as outgrowths from the parent body <u>Fragmentation:</u> - The body of an organism breaks into two or more parts, each of which generates the missing pieces to form a new, complete individual <u>Parthenogenesis:</u> - The development of an egg in the absence of fertilisation

Reproduction: the process where an organism passes on their genetic material to another generation by having offspring.

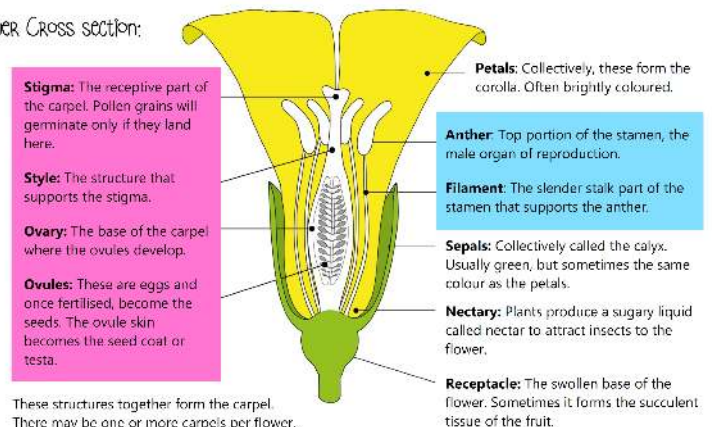
Plant reproduction

Flowers are the asexual reproductive organs of angiosperms.

Self pollination: pollen from the stamen, lands on the stigma on the same plant

Cross pollination: pollen from the stamen lands on the stigma of another plant of the same type

Flower Cross section:

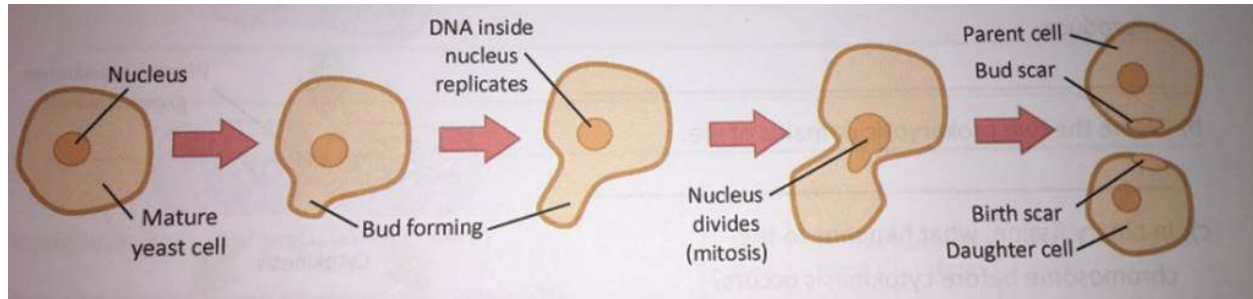


Pollination: the transfer of pollen to a stigma ovule, flower, or plant to allow fertilisation

Fertilisation: the process of fertilising a plant, involving the fusion of male and female gametes

Fungi reproduction

- Most species of fungi are multicellular
- Unicellular species of fungi are referred to as yeasts
- Yeasts reproduce asexually mainly through budding



IQ1: How does reproduction ensure the continuity of a species?

5.1.2 analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals

5.1.2 Fertilisation, Implantation and Birth

Fertilisation: the fission of **two haploid gametes** to form a single haploid zygote cell. The zygote contains genetic material from both the egg and sperm

- **Equal** distribution of genetic material from both parents
- Sperm cells move in one direction towards the egg

Order of reproduction:

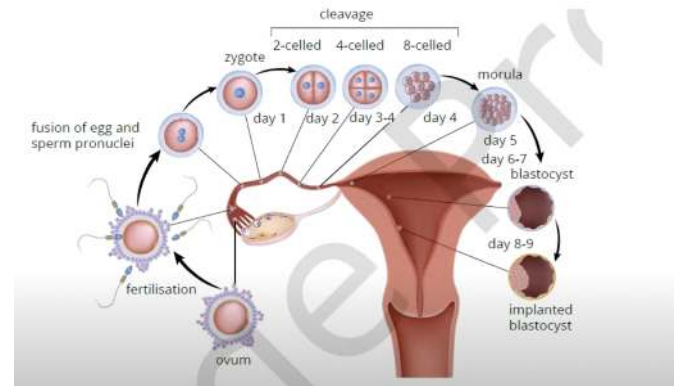
1. Ovulation
2. Fertilisation
3. Implantation
4. Placental formation

Steps of fertilisation:

1. The sperm uses **enzymes** from the **acrosome** to dissolve and penetrate the protective layer (zona pellucida) surrounding the egg to reach the cell membrane
2. Molecules on the sperm surface bind to receptors (specialised proteins) on the egg's cell membrane to ensure that a sperm of the same species fertilises the egg, then the nucleus of the sperm enters the cytoplasm in the egg cell
3. Changes at the surface of the egg occur to prevent the entry of multiple sperm nuclei into the egg
4. Fusion of the haploid egg and sperm nuclei results in a diploid zygote cell (the fertilised egg)

Steps of implantation:

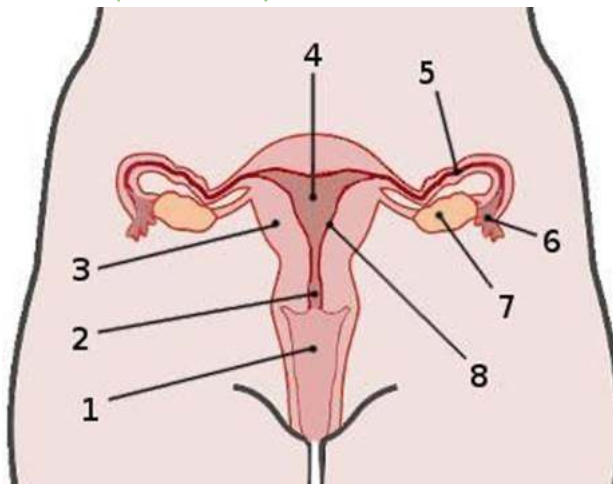
1. The zygote continues to travel down the oviduct until it reaches the uterus
2. The first stage of development is cleavage- a period of rapid cell proliferation during which the single-celled zygote is divided into many hundreds of smaller cells by mitosis
3. This mitosis transitions the cell to a morula. This is a ball of unspecialised embryonic stem cells
4. The continued mitosis causes the morula to become a blastocyst as its cells begin to differentiate. The inner cell mass will give rise to the embryo and the outer layer of cells will help the placenta develop
5. After implantation, gastrulation occurs to form a gastrula which eventually becomes a fetus



3 layers of the embryo:

- Ectoderm
 - ◆ Forms epidermis, hair, brain and spinal cord cells
- Mesoderm
 - ◆ Forms muscle, cartilage, kidney and gonad cells
- Endoderm
 - ◆ Forms the lungs, bladder and lining of the digestive system

Female reproductive system



Part	Description
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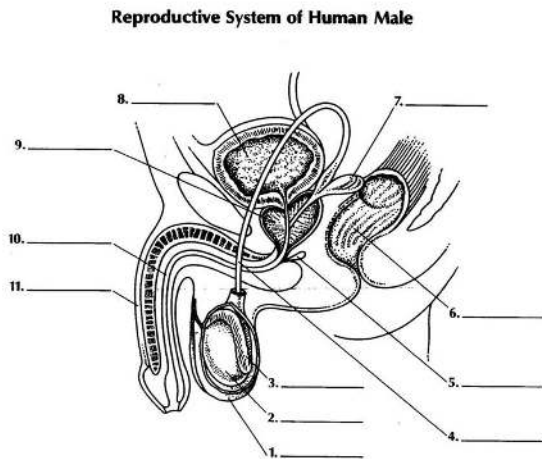
Label 1	Structure: vagina
	Function: the start of sexual reproduction (where the sperm and egg meet)
Label 2	Structure: cervix
	Function: allows fluids from the vagina to flow to the uterus
Label 3	Structure: myometrium
	Function: expands during pregnancy to hold the growing baby
Label 4	Structure: uterus
	Function: where a fertilised egg is placed during pregnancy and where it develops
Label 5	Structure: fallopian tube
	Function: a pathway for the egg to travel from the ovaries to the uterus
Label 6	Structure: fimbriae
	Function: catches the egg that one of the ovaries releases during ovulation
Label 7	Structure: ovaries
	Function: makes eggs and hormones
Label 8	Structure: endometrium
	Function: where the blood from the menstrual cycle comes out

Role of hormones in the female reproductive system

- **Hormones:** chemical messengers that are produced by the body, which travel in the blood to other cells and is responsible for communication between organs and tissues
 - Transported by the circulatory and lymphatic system or diffused through extracellular fluid
 - Important in pregnancy for the transfer of nutrients and support the developing baby
 - Produced by particular groups of cells within an organism (gland), that regulates the growth of activity of the cells capable of responding (target cells)
 - Regulate physiological and behavioural processes
 - Synthesised by specialised cells (usually one of the endocrine glands)
 - Transported by the circulatory and lymphatic system or diffused through extracellular fluid to where they're needed
- **Oestrogen**
 - Stimulates **ovulation** through producing egg cells for fertilisation
 - Aids blood flow in mammals and helps stimulate and aid the development of different organs (lungs and liver)

- Stimulates progesterone production
- **Progesterone**
 - Stimulates the thickening of the uterus lining
 - Aids placenta function and relaxes the uterus, helping the placenta work properly
 - Helps the mother immune system tolerate the growing offspring which is recognised as foreign by the mother's body

Male reproductive system



Part	Description
Label 1	Structure: scrotum
	Function: holds and helps to protect the testicles
Label 2	Structure: testicle
	Function: makes the sperm and is also involved in producing testosterone (hormone)
Label 3	Structure: epididymis
	Function: attached to each of the testicles, the sperm cells finish maturing and are stored here
Label 4	Structure: vas deferens
	Function: carries sperm from the epididymis to the prostate gland and urethra
Label 5	Structure: bulbourethral gland (Cowper's gland)
	Function: produces a pre-ejaculate that cleanses and lubricates the urethra prior to the arrival of the semen

Label 6	Structure: rectum
	Function: temporary storage site for faeces
Label 7	Structure: seminal vesicle
	Function: produces fluids that turn into semen
Label 8	Structure: urinary bladder
	Function: temporarily stores urine conveyed by the ureters from the kidneys until the body is ready to expel it through the urethra
Label 9	Structure: prostate
	Function: produces the fluid that nourishes and transports sperm
Label 10	Structure: urethra
	Function: carries the urine and sperm
Label 11	Structure: penis
	Function: urination and sexual intercourse

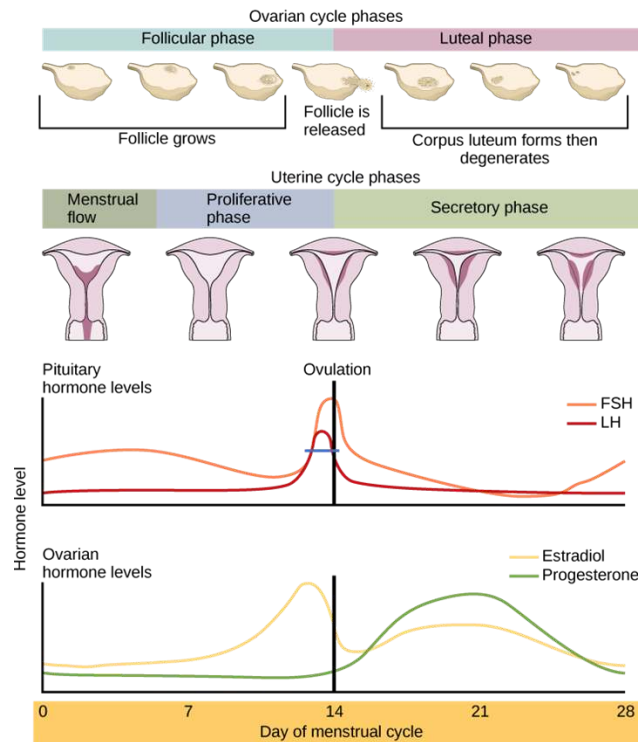
Process of fertilisation and implantation (hormones)

- **Production of sperm**
 - **Spermatogenesis:** the production of sperm cells in males (controlled by hormones)
 - **Gonadotropin-releasing hormone (GnRH):** responsible for the release of follicle-stimulating hormone (FSH) and luteinising hormone (LH) from the anterior pituitary
 - **Luteinising hormone (LH):** stimulate the production of testosterone
 - **Follicle stimulating hormone (FSH):** stimulates the production of protein in the testes to maintain levels of testosterone
 - Once the testosterone is sufficient and maintained, sperm cells are produced and stored until they are matured
 - Inhibin is then secreted to reduce the levels of FSH in the body, reducing the level of testosterone and inhibits sperm production
- **Production of Ovum (egg): ovarian cycle**
 - **Ovarian cycle:** the monthly cycle of the ovum development in the ovary with follicular phase, ovulation and the luteal phase (all regulated by hormones)
 - During puberty the ovarian and menstrual cycles begin which is when the ova begins to mature
 - The ova in the ovaries become surrounded by a single layer of cells that envelop them and begin to divide → formation of the primary dormant follicles
 - Hormones are secreted after puberty begins until menopause occurs (~40 years later)

- The hormones trigger the development and maturation of ova each month
- At the start of each cycle a few follicles start to develop but usually only one enlarges more than the others to reach maturity to be released during ovulation
 - Average cycle lasts 28 days
- Key terms:
 - **Ovum:** the egg that is the female gamete
 - **Follicles:** a mature ovarian follicle that ruptures during ovulation to release the ovum
 - **Corpus luteum:** the remains of a follicle after ovulation. Secretes oestrogen and progesterone during the later parts of the oestrous cycle in mammals
 - **Embryo:** a stage of development between fertilisation and the development of characteristics that lead to the fetus stage
 - **Fetus:** following the embryonic stage and the organism has all the characteristics of an adult mammal- for humans this occurs from the 8th week of gestation until birth
 - **Gestation:** the length of time a pregnancy lasts for- includes the embryonic and fetal development
 - **Ovulation:** the release of an ovum (egg) by a mature follicle in an ovary
- *Stages of the ovarian cycle*
 1. **Follicular phase:**
 - Slow rising levels of FSH and LH causes the growth of follicles on the surface of the ovary
 - Prepares the egg for ovulation
 - As the follicles grow, they start to release oestrogen and a low level of progesterone
 - These cells then secrete fluids to move the egg to one side of the follicle, creating a bulge
 - Makes the follicle a mature one (Graafian follicle)
 - The cells that line the follicle will secrete the oestrogen which when stimulated, increases the production of LH
Results in ovulation and initiates the next phase
 2. **Ovulation:**
 - The Graafian follicle bursts to release the egg that moves from the ovary to the fallopian tubes
 - Occurs mid cycle and if a sperm is present, then fertilisation can occur
 - Ovum is only viable for 12-14 hours
 3. **Luteal phase:**
 - Lasts for 14 days
 - The corpus luteum is formed from the cells that remain from the burst Graafian follicles and it produces oestrogen and progesterone
 - Progesterone facilitates the growth of the uterine lining and inhibits the release of further FSH and LH

- The uterus is prepared to accept a fertilised egg
 - The inhibition of FSH and LH prevents any further eggs and follicles from developing, while progesterone is elevated
 - The level of oestrogen produced by the corpus luteum increases to a steady level for the next few days
 - If no fertilised egg is implanted into the uterus, the corpus luteum degenerates the levels of oestrogen and progesterone decreases
 - The decrease in progesterone also allows the hypothalamus to send GnRH to the anterior pituitary gland, releasing GnRH to the anterior pituitary gland, releasing FSH and LH and starting the cycles again
- **Menstrual cycle**
 - The cyclic preparation of the endometrium of the uterus for pregnancy in females with discharge of blood and tissue through the cervix if implantation does not occur
 - Key terms:
 - **Endometrium:** the inner lining of the uterus in mammals; the site of implantation of the embryo after fertilisation
 - **Menstruation:** the shedding of the blood and tissues from the endometrium of the uterus when an ovum is not fertilised
 - Stages of the menstrual cycle:
 1. **Menstrual flow:**
 - Lasts for 4-8 days on average but normally lasts 5-6 days
 - The endometrium breaks down and tears away, causing bleeding from the vagina
 2. **Proliferative phase:**
 - Occurs from the end of the period until ovulation occurs and involves building up the inner lining of the uterus as stimulated by the oestrogen
 - The endometrium becomes thicker
 - The endometrium is thinnest during this period, and thickens throughout this phase until ovulation occurs
 - Creates a place for a potential fertilised egg to be implanted and grown
 3. **Secretory phase:**
 - Occurs from ovulation until the start of the next period
 - Rising levels of progesterone causes the endometrium to stop thickening and to start preparing for the potential attachment of a fertilised egg
 - The endometrium secretes many types of chemical messengers
 - The most important chemical is prostaglandins which are secreted by endometrial cells and cause changes to other cells nearby
 - Two types of prostaglandins: PGF₂α and PGE₂ (cause the uterine muscles to contract “cramp”)

- These amounts of prostaglandins rise after ovulation and reach their peak during menstruation
- The cramping triggers the period
- If a pregnancy occurs, the prostaglandin production is inhibited so that contraction don't impact early pregnancy
- If pregnancy doesn't occur, the corpus luteum stops producing oestrogen and progesterone
- The drop in hormones, along with the effects of prostaglandins, cause the blood vessels to constrict and tissue of the endometrium break down



IQ1: How does reproduction ensure the continuity of a species?

5.1.3 evaluate the impact of scientific knowledge on the manipulation of plant and animal reproduction in agriculture

5.1.3 Plant and Animal Reproduction in Agriculture

Selective breeding: the process to produce animals and plants with more useful or more attractive characteristics. Humans decide which individuals may breed and leave offspring to the next generation

- Artificial insemination

- Artificial pollination
- 4 basic steps:
 - Determine the desired trait
 - Interbreed parents who show the desired trait
 - Select the offspring with best form of the trait and interbreed these offspring
 - Continue this process until the population reliably reproduces the desired trait

Genetic modification: development of techniques that allow for the alteration of an organism's genome and for the transfer of genes from organism to another

- Plants- increase crop productivity, provide crop resistance to insects and prevent disease
- Animals- improve fertility, and the quality and yield of meat, milk, eggs and wool

Cloning: used as an artificial method of producing genetically identical organisms

- E.g. cuttings and grafts, tissue culture, embryo splitting and somatic cell nuclear transfer

IQ2: How important is it for genetic material to be replicated exactly?

5.2.1 model the processes involved in cell replication, including but not limited to:

1. mitosis and meiosis (ACSBL075)
2. DNA replication using the Watson and Crick DNA model, including nucleotide composition, pairing and bonding (ACSBL076, ACSBL077)

5.2.2 assess the effect of the cell replication processes on the continuity of species (ACSBL084)

5.2.1: Cell Replication

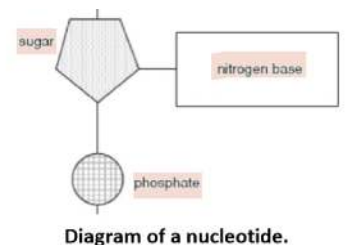
DNA transmits genes from one generation to the next generation.

- Codes for the production of proteins (transcription)
- Sexual reproduction
- The coding contained within the genetic material, can be altered by influences in the environment (mutations); can lead to new genes being formed

The DNA structure was discovered in March 1953 by Rosalind Franklin, Maurice Wilkins, James Watson, and Francis Crick

DNA: a nucleic acid that stands for deoxyribonucleic acid

- Double helix structure
- Each strand contains four different types of nucleotides which have a complementary base pair (adenine and thymine, guanine and cytosine)
 - Each nucleotide is made of deoxyribose sugar and a phosphate group

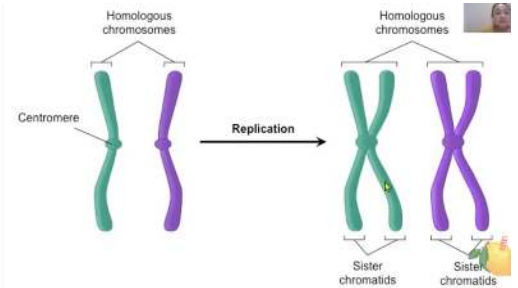


- Hydrogen bonds hold the base pairs together (2 hydrogen bonds between A and T, 3 hydrogen bonds between G and C)
- DNA contains genetic code for the synthesis of a polypeptide
- When a cell divides, genes are passed from one cell to the next. This is due to DNA replication that occurs in the interphase of meiosis.

Sister chromatids: identical copies of DNA created after DNA replication

Homologous pairs: chromosome pairs that carry the same genes but do not look identical

DNA is our genetic coding that determines the features of our body and its functions, whilst separating us from other human beings



Each of the cell's in the body are the results of an existing cell dividing into two- copying its DNA allowing for the new cell to have a complete set of instructions. If an error occurs during the replication stage, mutations can occur due to the change in DNA sequence.

Heterozygous	Two different alleles of a particular gene
Homozygous	Identical alleles of a particular gene in a diploid cell

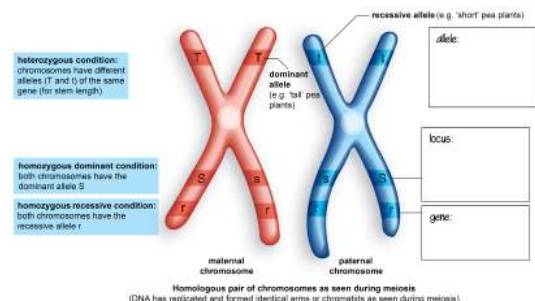
Alleles: different variation of the same gene

- Occurs in pairs in a diploid cell; two or more alleles for each gene may be present in a population
- Segregate during meiosis
- Occur individually in each haploid gamete
- Pair in fertilisation when the diploid condition is restored during meiosis

Gene: a segment of DNA on a chromosome

- Codes for a specific trait
- Has two alleles in an individual and two or more alternative alleles in a population

Histones: proteins that chromosomes are organised around



DNA is transported easily when it is compacted into a chromosome.

Steps of DNA replication:

1. The DNA double helix unwinds

- The enzyme helicase causes the DNA helix to slowly unwind and the strands separate from each other. This is due to the breaking of the hydrogen bonds between complementary base pairs.

2. DNA unzips

- The helicase breaks the weak hydrogen bonds between each of the complementary base pairs. The SSBs (single-stranded DNA binding proteins) then attach to the original strands of DNA after the helicase breaks the hydrogen bonds between complementary base pairs and hold the newly separated single-stranded DNA, preventing them from forming the double helix structure again.

3. Nucleotides are added against each single strand

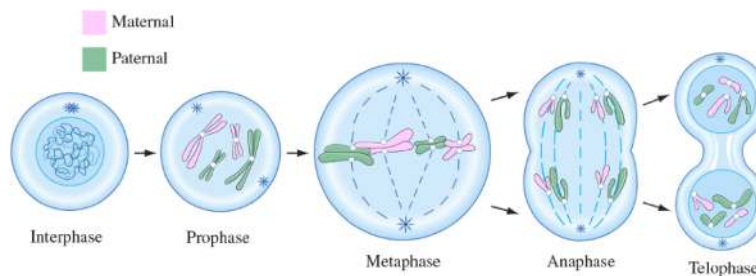
- DNA polymerase adds DNA molecules to continue the synthesis of the new strand, by taking free nucleotides floating in the nucleus, and then adding them to their opposite complementary base pair. Nucleotides are then added to the leading strand. On the complementary strand, nucleotides are added in Okazaki fragments. The ligase then joins the fragments to the strand forming a continuous strand.

4. Replication errors are identified and corrected

- DNA polymerase 'edits' and 'proof-reads' the strand checking for errors in the nucleotide sequence. Incorrect bases are 'spliced' out and then replaced with the correct bases. The ligase then seals the two new strands.
- The DNA molecules are **anti-parallel** as they run in opposite directions. Each DNA strand has a 3' and a 5' end. **Nucleotides are always added from the 3' end of a DNA strand.** This means that when nucleotides are inserted during replication, they are also anti-parallel along the two template strands.

Mitosis and Meiosis:

Mitosis: the formation of two identical daughter cells that promote growth and repair of an organism; occurs through asexual reproduction



Steps of mitosis:

1. Interphase

- DNA is replicated before the start of prophase
- Chromatin are not recognisable as they aren't chromosomes yet

2. Prophase

- Chromatin are coiled and two copies of DNA (sister chromatid) separates into chromosomes
- The nuclear membrane breaks down and disappears
- Chromosomes condense and become visible

3. Metaphase

- Spindle fibres attach to the centromere and are pulled to the centre of the cell
- Each chromosome consists of two identical sister chromatids

4. Anaphase

- Spindle fibres go through the centromere and separates the chromatids from each other, pulling them apart
- Sister chromatids then become daughter chromosomes

5. Telophase

- The nuclear membrane begins to form allowing the nucleolus to reappear
- Daughter chromosomes gather at opposite ends of the cell
- Results in two daughter cells with nuclei and identical chromosomes (identical to parents)

6. Cytokinesis

- Division of the cytoplasm occurs, creating two daughter cells
- **Animals:** splits between the two daughter cells and 'pinches off'
- **Plants:** cell plate forms while the nucleus is still in telophase. Cellulose is deposited to the other side, forming a cell wall to separate the daughter cells
- Splits evenly

Meiosis: the production of genetically different gametes for sexual reproduction; results in the number of chromosomes being halved (23)

Crossing over	The sister chromatids directly next to each other cross over each other, resulting in the swapping of genes; genetic variation
Independent assortment	Each pair of alleles segregates independently of all other pairs of alleles on non-homologous chromosomes
Random segregation	Two alleles separate during the formation of gametes and end up in different gametes

Steps of meiosis I:

1. Interphase

- DNA replication occurs
- Creates two chromosomes with identical copies (sister chromatid) and are held together by a centromere

2. Prophase I

- Spindle fibres begin to form, and chromosomes thicken and condense
- The chromosomes then line up in homologous pairs
- Crossing over occurs
- Nuclear membrane begins to break down, disappearing

- Chromosomes then become chromatids

3. Metaphase I

- Chromosomes line up at the centre of the cell in various combinations of paternal/maternal (independent assortment)
- Spindle fibres attach to the centromeres
- Random segregation (only 1 of the 2 genes copied are present in the organism is distributed to each gamete)

4. Anaphase I

- Spindle fibres pull chromosomes to opposite ends of the cell
- Cleavage begins to form

5. Telophase I

- Two daughter cells form creating two nuclei
- Chromosome combinations in the two cells differ from each other

6. Cytokinesis I

- Daughter cells are not identical and have half of the original number of chromosomes (23)
- Cytoplasm splits

Steps of meiosis II:

1. Prophase II

- Spindle fibres form in each daughter cell
- The nuclear membrane breaks down, disappearing

2. Metaphase II

- Chromosomes line up in the centre of each daughter cell
- Spindle fibres attach to the centromere

3. Anaphase II

- Spindle fibres pull chromatids to opposite ends of the cell
- Cleavage begins to form again

4. Telophase II

- Two daughter cells continue to divide
- Nuclei starts to form, containing the chromosomes

5. Cytokinesis II

- The cytoplasm divides
- 4 haploid daughter cells form

Nondisjunction: when a cell receives too many or too few chromosomes during separation; results in genetic disorders

Feature	Meiosis	Mitosis
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Location in organism	Ovaries and testes	All over the body to create new cells
Describe what happens to homologous chromosomes during process	Homologous pairs crossover and go through random segregation and independent assortment in meiosis 1. They then get separated into different cells.	No splitting or crossing over in homologous chromosomes
Comparison of chromosomes to parent cell	Not identical and half the number of chromosomes	Identical and same number of chromosomes
Genetic variability produced.	High variability due to cross over and random segregation	No genetic variability

IQ3: Why is polypeptide synthesis important?

5.3.2 model the process of polypeptide synthesis, including: (ACSBL079)

1. transcription and translation
2. assessing the importance of mRNA and tRNA in transcription and translation (ACSBL079)
3. analysing the function and importance of polypeptide synthesis (ACSBL080)
4. assessing how genes and environment affect phenotypic expression (ACSBL081)

5.3.2: Polypeptide Synthesis

Polypeptide synthesis: the production of proteins from DNA. Involves two processes, transcription and translation

RNA: a single-stranded sequence of nucleotide sequence due to the unzipping of a DNA strand

- Contains ribose sugar and uracil (replaces thymine)
- **mRNA:** carries genetic information from the nucleus to the cytoplasm
- **rRNA:** forms a structural part of the ribosomes
- **tRNA:** a link between mRNA and the growing chain of amino acids that make up a protein

Non-coding strand/ template strand (sense strand)	A strand of DNA that serves as a template during transcription; its sequence of bases is the same as anticodons on tRNA
Coding strand (antisense strand)	Not used as a template, but its base sequence corresponds to the base sequence of the mRNA; contains codons

Anticodons	Three unpaired bases at one end of tRNA that attach the tRNA to its complementary bases on the mRNA strand
Codons	A triplet or sequence of three bases on DNA or mRNA that codes for a single amino acid
Exons	Coding regions of genes
Introns	Non-coding regions of genes (spliced out)
Promoter region	A region of DNA that starts transcription of a gene
RNA polymerase	Binds to a part of the DNA called the promoter

Steps of transcription:

1. The Helicase unzips the DNA, and the RNA polymerase binds to the promoter. Only one strand of DNA contains the genetic information to make a protein (template strand). The coding strand is the one not being copied because the strand of RNA being made is carrying the RNA version of the code
2. The template strand acts as a template and the RNA nucleotides are assembled, forming a complementary single stranded mRNA molecule.
3. The mRNA moves out of the nucleus through the nuclear pores, into the cytoplasm. One mRNA molecule reads a large number of ribosomes, allowing multiple chains of the same polypeptide product to be produced from one mRNA template molecule.

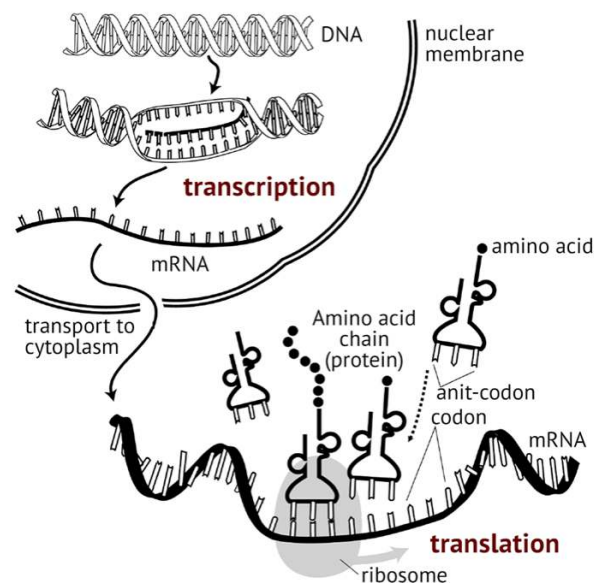
Steps of translation:

1. Initiation

- The mRNA attaches to the small subunit of the ribosome
- The start codon calls for the tRNA with the amino acid and the anticodon (initiates the translation process)
- The next tRNA will then enter through the A site of the ribosome attaching to the next set of codons
- A polypeptide bond is then formed between the two amino acids

2. Elongation

- tRNA(1) will leave its amino acid behind leaving via the E site of the ribosome
- The current tRNA is now holding two amino acids, it will then move along into the P site.



- According to the next set of codons it will then call for the next tRNA (enters through the A site)
 - A bond is now formed between one that is being held in the P site
- The old tRNA will drop the amino acid and exit via the E site
- This process is repeated until the end of the mRNA strand

3. Termination

- The ribosome will then reach a stop codon on the mRNA strand, meaning translation of this specific protein is complete
 - UAG, UGA, UAA
- These codons don't call any amino acids but instead calls for a 'release factor' (similar to a tRNA but no amino acid only the anticodons)
 - The release factor (RF) helps to release the newly made polypeptide chain and separates the mRNA from the ribosomal units
 - The newly formed polypeptide chain will then fold into a specific shape to then be sent to a specific parts of the organism to fulfil its function

IQ4: How can the genetic similarities and differences within and between species be compared?

5.4.1 conduct practical investigations to predict variations in the genotype of offspring by modelling meiosis, including the crossing over of homologous chromosomes, fertilisation and mutations (ACSBLO84)

5.4.1 Predicting Variation in Sexual Reproduction

Genetic variation in populations is due to mutation and sexual reproduction

Mutation: a change in the DNA of an individual

Nucleotide polymorphism: a change in a single nucleotide in a section of DNA (responsible for alternative alleles)

An organism's genotype is the combination of alleles that make up its genetic information

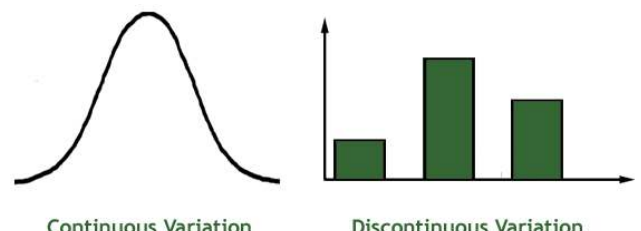
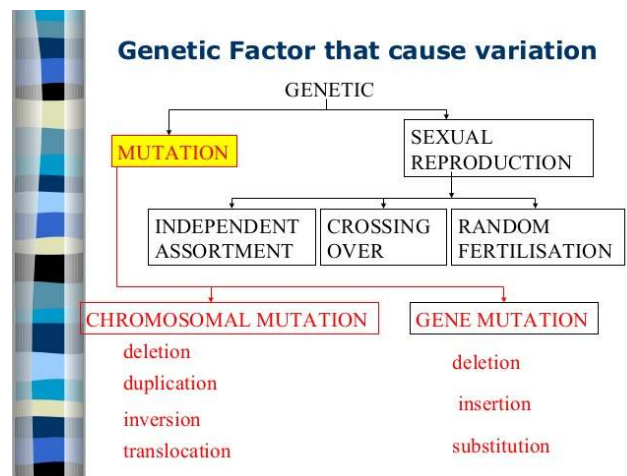
phenotype= genotype + environment

Gene: a sequence of DNA that is the unit of heredity

Allele: alternative form of a gene

Homozygous: an individual that holds the same alleles for a particular gene on both chromosomes of a homologous pair

Heterozygous: an individual that carries alternative alleles for a given gene



Continuous and discontinuous variation

Polygenic inheritance: characteristics that under the control of more than one gene

Continuous variation: traits are controlled by polygenes and characterised by a range of phenotypes; their distribution can be represented in a bell curve graph

Discontinuous variation: traits are typically controlled by a single gene, usually with two allelic forms and characterised by distinct phenotypes

IQ4: How can the genetic similarities and differences within and between species be compared?

5.4.2 model the formation of new combinations of genotypes produced during meiosis, including but not limited to:

1. interpreting examples of autosomal, sex-linkage, codominance, incomplete dominance and multiple alleles
2. constructing and interpreting information and data from pedigrees and Punnett squares

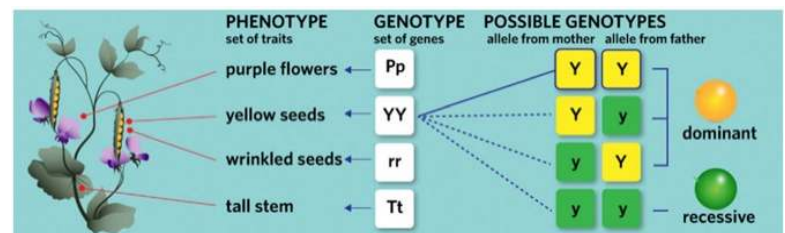
5.4.2 Modelling New Combinations of Genotypes

Genotype: the genetic information that produces the trait in an organism

Phenotype: the appearance of a trait in an organism

Dominant: refers to the gene that is expressed over the recessive gene in a phenotype.

Recessive: the gene that is masked by the dominant allele unless with another recessive gene



Autosomal punnett squares (Mendelian crosses)

Problem Solving Strategy

Step 1:

Designate letters that will represent the genes/traits according to the genetics.

Step 2:

Write down the genotypes (genes) of each parent. These may be stated in the outline of the problem or you may be expected to determine the genotypes by logical deduction from other information that has been given.

Step 3:

List the genes that each parent can contribute to the gametes.

Step 4:

Draw a Punnett square to show all the possible combinations of genes when gametes fuse. A Punnett square consists of nine cells in a table (three cells in each row and each column). Write the possible genes in the gametes of one parent across the top and the genes from the gametes of the other parent along the side of the Punnett square.

Step 5:

To show how the genes recombine during fertilisation, complete each cell of the Punnett square by combining the letters above and beside the appropriate cell. As a general rule, the capital letter goes first and a lowercase letter follows.

Step 6:

Write out the genotypic ratio and the phenotypic ratios for the offspring.

Co dominant and incomplete punnett squares (non mendelian)

Codominance: where both alleles are expressed and neither is dominant but are not blended e.g. speckled chicken

- Both expressed in the heterozygous phenotype
- E.g. a red cow and a white cow cross, producing a white cow with red speckles

Incomplete dominance: one allele is not completely dominant over the other, producing a heterozygous condition that is a blend in the phenotype of the parents to produce offspring that is somewhere in between

- E.g. a red and white flower cross, creating a pink flower

Epistasis: when one gene depends on another gene in order for it to be expressed

Sex linkage punnett squares (non mendelian)

Haemophilia: affects the platelets that help heal and stop bleeding in a wound

- Males are more likely to inherit haemophilia as they only have 1 X chromosome

Sex-linked traits: specifically on the sex chromosomes (only on X chromosome)

- The X chromosome is larger than the Y chromosome and contains thousands of more genes

Multiple alleles: focus on blood type

Blood groups are classified as either A,B,AB or O and by Rhesus positive (Rh+) or negative (Rh-)

Antigen: a molecule capable of inducing an immune response

The antigens in blood serve various purposes, including:

- Transporting other molecules into and out of the cell
- Maintaining the structure of red blood cells
- Detecting unwanted cells that could cause illness

1. ABO blood type (non mendelian)

- Blood type is controlled by 3 alleles: I^A , I^B and i . I^A and I^B are dominant over i , but are codominant with each other. Therefore, creating 3 blood types (A,B, AB,O) which exhibit different antigens
 - $I^A I^A$ and $I^A i$ code for type A, exhibiting the A antigen
 - $I^B I^B$ and $I^B i$ code for type B, exhibiting the B antigen
 - $I^A I^B$ codes for blood type AB, exhibiting the A and B antigen (codominance)
 - ii codes for type O exhibiting no antigens
- Punnett square e.g.

Parents Phenotype	AB X O														
Parents Genotype	$I^A I^B$ X ii														
Punnett Square	<table border="1"> <tr> <td></td><td>x</td><td>i</td><td>i</td></tr> <tr> <td>I^A</td><td>$I^A i$</td><td>$I^A i$</td><td>$I^A i$</td></tr> <tr> <td>I^B</td><td>$I^B i$</td><td>$I^B i$</td><td>$I^B i$</td></tr> </table>				x	i	i	I^A	$I^A i$	$I^A i$	$I^A i$	I^B	$I^B i$	$I^B i$	$I^B i$
	x	i	i												
I^A	$I^A i$	$I^A i$	$I^A i$												
I^B	$I^B i$	$I^B i$	$I^B i$												
Offspring Genotype ratio	$I^A i : I^B i$														
Offspring Phenotype ratio	1 type A to 1 type B														

2. Rh blood type

- Rh+: homozygous dominant (RR), heterozygous (Rr)
- Rh-: homozygous recessive (rr)
- Mother fetus incompatibility: when the mother is Rh- and her fetus is Rh+. maternal antibodies can cross the placenta and destroy fetal red blood cells
 - Can be prevented through treatment
- Rh type mother-fetus incompatibility occurs when an Rh+ man fathers a child with an Rh- mother
- Punnett square e.g.

Parents Phenotype	Rh+ X Rh-									
Parents Genotype	Rr X <u>rr</u>									
Punnett Square	<table><tr><td>x</td><td>r</td><td>r</td></tr><tr><td>R</td><td>Rr</td><td>Rr</td></tr><tr><td>r</td><td><u>rr</u></td><td><u>rr</u></td></tr></table>	x	r	r	R	Rr	Rr	r	<u>rr</u>	<u>rr</u>
x	r	r								
R	Rr	Rr								
r	<u>rr</u>	<u>rr</u>								
Offspring Genotype ratio	Rr: <u>rr</u>									
Offspring Phenotype ratio	1 Rh+ to 1 Rh-									

IQ4: How can the genetic similarities and differences within and between species be compared?

5.4.2 model the formation of new combinations of genotypes produced during meiosis, including but not limited to:

- interpreting examples of autosomal, sex-linkage, codominance, incomplete dominance and multiple alleles
- constructing and interpreting information and data from pedigrees and Punnett squares

5.4.2 Pedigrees-Modelling New Combinations of Genotypes

Pedigree: shows inherited traits passed across generations

- circles= females
- squares= males

Autosomal: non sex chromosomes (first 22 pairs)

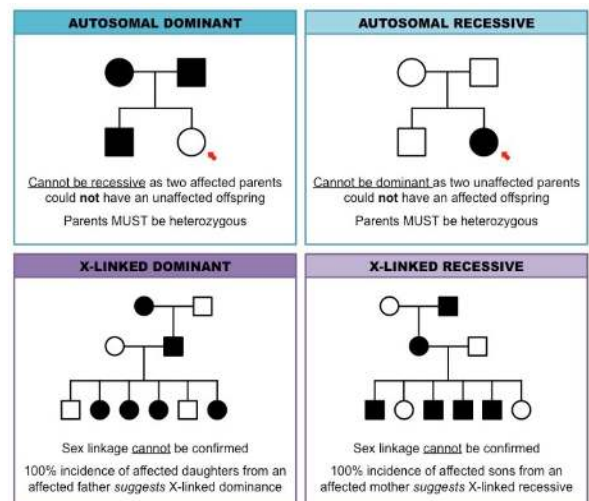
Pedigree rules:

Autosomal dominant inheritance

- Trait **should not skip** generation
- An affected person married to a 'normal' person should have approximately 50% of the offspring being affected (also indicated that the **affected individual is heterozygous**)
- Distribution of the trait should be close to equal distribution among the sexes

Autosomal recessive inheritance

- Trait often **skips** generations
- Distribution of the trait should be close to equal distribution among the sexes
- Traits are often found in pedigrees with consanguineous marriages



- If both of the parents are affected, all of the children should be affected
- Most affected individuals have 'normal' parents
- When a 'normal' person is married to an affected individual, all of the children are normal (indicating the normal parent is heterozygous dominant)
- If a 'normal' person is married to an unaffected individual and one or more of the children is affected then about half of the children should be affected

X linked recessive inheritance

- Most of the affected individuals are males
- For a female child to be affected, the father must be affected and the mother must be affected or a carrier
- All of the sons of an affected mother must be affected or a carrier (many times this can be determined by studying males in the mother's family line)
- About half of the sons of carrier females should be affected

The tricks to getting it right in less time

- Anyone in the pedigree with a different phenotype than their parents, means that the trait is recessive. (skipped a generation)
- Are there any girls with the recessive phenotype who have a father or a son with the dominant phenotype, then autosomal
- If sex linked then the female recessive will have a father and son with the recessive trait

IQ4: How can the genetic similarities and differences within and between species be compared?

5.4.3 collect, record and present data to represent frequencies of characteristics in a population, in order to identify trends, patterns, relationships and limitations in data, for example:

1. examining frequency data
2. analysing single nucleotide polymorphism (SNP)

5.4.3: Frequency of Characteristics in a population

Population genetics combines Mendelian genetics and Darwinian evolution (evolution through natural selection)

- Leads to microevolution and macroevolution

Allele frequency: measure of how common an allele is in a population

$$\text{Frequency of allele } G = \frac{\text{\# of copies of allele (G) in the population}}{\text{Total \# of copies of the gene (G+g) in the population}}$$

SNPs

Single nucleotide polymorphism (SNPs): most common genetic variation among people

- Represents a difference in a single nucleotide

- Key points:
 - SNPs are **common variation** in DNA sequences
 - Variation in a single nucleotide
 - To be classified as a SNP, the change must occur in more than 1% of the population
 - SNPs are identified by comparing many genomes
 - Millions of common variations among genomes and their locations of chromosomes can be identified
 - Specific locations are denoted by the chromosome number followed by the nucleotide number along the chromosome
 - SNPs are used to find the location of genes associated with particular traits
 - Once the location of a SNP is located in an organism's genome, they can be used to home in on the genes of interest
 - They then identify SNPs that are found to occur significantly more frequently in people with one version of the trait than the other
 - Those SNPs serve as 'markers' for the region of the human genome where a gene responsible for determining PTC tasting
 - Why are certain SNPs correlated, or associated, with certain traits?
 - SNPs that occur within a gene or in a regulatory area near a gene may directly affect that gene's function and the resulting trait
 - A SNP does not have to be responsible for causing a trait in order for it to be correlated with that specific trait
 - If a SNP is close enough to a trait-causing allele, they may be inherited together
- May associate with genetic change
- SNPs mostly occur in non-coding DNA and do not lead to any observable differences

IQ5: Can population genetic patterns be predicted with any accuracy?

5.5.1 investigate the use of technologies to determine inheritance patterns in a population using, for example: (ACSBL064, ACSBL085)

- DNA sequencing and profiling (ACSBL086)

5.5.1: Technologies

DNA sequencing

Patterns of inheritance: the observations of the way traits, or characteristics are passed from one generation to the next to form identifiable phenotypes

1. Polymerase chain reaction (PCR)
 - Exponentially amplify a single copy or a few copies of a specific segment of DNA to generate thousands to millions of copies of a particular DNA sequence

2. DNA sequencing

- Determines the exact nucleotide sequence
- Can be conducted using the Sanger method, Maxam-Gilbert method or next generation technologies

Human genome project (HGP)

- A project where scientists worked collaboratively and shared their data to determine the sequence of the human genome
 - Allows scientists to identify genetic disorders or illnesses that may be present in an individual
- 3 sad surprises about the HGP:
 - Doesn't have a lot of genes that code for proteins
 - The genes only make up 2% of our genome
 - Sequencing DNA hasn't helped with curing genetic diseases

DNA profiling

- Also referred to as DNA fingerprinting
- Used to identify and compare individuals by characteristics in their DNA
 - Determines the identity of an individual but doesn't determine the genome or DNA sequence
- **Short tandem repeats (STRs):** sections of DNA unique to an individual
- **Loci:** predictable inheritance patterns at a certain location
 - Useful for determining identity and biological relationships
- Involves:
 1. Isolating the DNA to be profiled from blood to other cells found on the scene
 2. The DNA is cut into fragments using the same restriction enzyme
 3. The polymerase chain reaction (PCR) is used to increase (amplify) the amount of DNA of the sequence
 4. Gel electrophoresis (agarose sugar) is used to separate the segments according to length
 5. Smaller fragments will migrate further on the gel compared to larger fragments
 6. The fragments are then made visible using a radioactive probe

IQ5: Can population genetic patterns be predicted with any accuracy?

5.5.2 investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example: (ACSBL064, ACSBL073)

- a) the use of population genetics data in conservation management
- b) population genetics studies used to determine the inheritance of a disease or disorder
- c) population genetics relating to human evolution

5.5.2: Conservation Management

Genetic diversity: the range of genetic material present in a gene pool or population of a species

Species diversity: all the different possible phenotypes of a species

Ecosystem diversity: the measure of the amount of organisms or species that live in one area

The use of population genetics in conservation

Population genetics: the study of genetic variation within a population and the changes in the frequency of genes and alleles within a population and among populations over time

- Involves the understanding of allele frequency, gene flow, genetic drift to predict the outcome for a population under particular circumstances

Conservation genetics: the application of genetics to preserve species through maintaining variation within population so that they are capable of coping with environmental change.

- Organisms that are studied are typically endangered or threatened due to:
 - Habitat destruction
 - Environmental change
 - Change in population size

Minimum viable population (MVP): the ecological threshold that indicates the smallest number of individuals in a species or population capable of surviving in the wild for a period of time

- Small populations are more open to genetic drift (a change in allele frequency due to random chance) than large populations.
 - Occurs when differential breeding leads to the loss of alleles from the gene pool (all the alleles of all the genes in a breeding population) due to the small number of individuals carrying particular alleles
 - Genetic drift can cause a problem as it occurs through chance compared to natural selection where the favourable alleles in individuals survive and pass on good genes

IQ5: Can population genetic patterns be predicted with any accuracy?

5.5.2 investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example: (ACSBL064, ACSBL073)

- a) the use of population genetics data in conservation management
- b) population genetics studies used to determine the inheritance of a disease or disorder
- c) population genetics relating to human evolution

5.5.2: Population Genetics and Inheritance of disease

Using the Human Genome Project (HGP) we are able to compare DNA sequences of people with genetic disorders to the genome sequence made by the HGP.

Breast cancer

- BRCA1 is a tumour suppressor gene found on chromosome 17

- BRCA2 is a tumour suppressor gene found on chromosome 13
- Both genes are responsible for proteins that repair genes involved in regulating the process of cell division
 - Mutation in BRCA1/2 gene leads to uncontrolled cell division which is associated with cancer

IQ5: Can population genetic patterns be predicted with any accuracy?

5.5.2 investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example: (ACSBL064, ACSBL073)

- a) the use of population genetics data in conservation management
- b) population genetics studies used to determine the inheritance of a disease or disorder
- c) population genetics relating to human evolution

5.5.2: Population Genetics and Evolution of Humans

The HGP allows us to compare the sequences of DNA and work out the relatedness of different organisms

- Used to compare the DNA sequence of humans and gorillas
 - Although later found through DNA Hybridisation (a technique that separates the double-stranded DNA of two species into single strands. They are then reheated and the temperature of separation is recorded) that humans were more closely related to chimpanzees

DNA hybridisation

- The more closely related the species, the stronger the hybridisation, the higher the temperature needed to separate the hybrid DNA = a more recent common ancestor
- Humans' closest living relatives are the chimpanzees and bonobos
- 98.8% of human DNA is shared with chimpanzees, and 98.4% with gorillas
- Humans and the African great apes (gorillas, chimpanzees, bonobos) all differ from the Asian great ape, the Orangutans by 3.1%
- Humans and the African great apes all differ by only 7% in their DNA with old world monkeys e.g. rhesus monkey
- **Mitochondrial DNA:** only comes from mothers
 - The egg holds organelles (e.g. mitochondria)

Out of Africa Theory

- Proposes that a common ancestor of all humans evolved in East Africa between 200,000 and 100,000 years ago.
- Determined by comparing mitochondrial DNA (mtDNA)

- **mtDNA:** DNA contained in the cell mitochondria from the mother to her children. The mitochondria separates into each gamete during meiosis in the ovary.
 - SNPs can cause changes to the mtDNA and helps track migration routes of our ancestors

Answering the syllabus outcome:

The large collaborative project, the human genome project, allows us to sequence all DNA including mitochondrial DNA by comparing our nucleus DNA to that of different monkeys, we know we are more closely related to the chimp (98%) than the gorilla. Therefore we must have evolved from a **common ancestor** to the chimp (not from the chimp).

The evolution of the human race since leaving Africa to form the different ethnic groups can be determined by comparing mitochondrial DNA. From doing this we can see that the Africans are the original homo sapiens and a small group that left Africa has spread throughout the globe.

Mod 6: Genetic Change

IQ1: How does mutation introduce new alleles into a population?

6.1.1 explain how a range of mutagens operate, including but not limited to

- electromagnetic radiation sources
- chemicals
- naturally occurring mutagens

6.1.1: Mutagens

Mutations can occur due to the result of errors in DNA replication, or they can be induced by mutagens such as:

- Ionising radiation (e.g. UV, X-rays)
- Viruses and microorganisms
- Alcohol, fatty and processed foods
- Environmental poisons and irritants

Mutation: a permanent change in the base sequence of DNA

- Addition, deletion, substitution

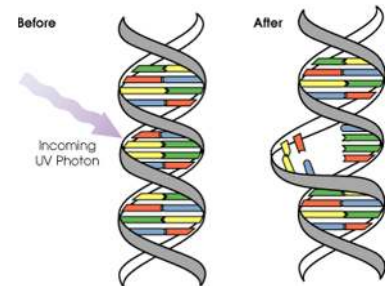
Non coding DNA does not code for a gene but code for **genes** that are **turned off**

Mutagen: any environmental factor that alters the genetic sequence of DNA

- When the repair system is compromised mutations may lead to diseases e.g. cancer
- Can be physical, chemical, or biological

Physical mutagens: refers to heat and ionising radiation (transfer of energy)

- E.g. leukaemia, melanoma, thyroid cancer



Chemical mutagens: when cells are exposed to chemicals for too long or in large amounts of time e.g. alcohol, nicotine, benzene, pesticides. It can cause a change in DNA and alters the function of proteins which impairs cellular processes.

→ E.g. lung cancer

Biological mutagens: natural environmental factors which can create mutations in DNA

- **Viruses:** replicate by inserting their DNA into host cells, creating a disruption in normal cell function and may lead to lasting changes
- **Bacteria:** can induce inflammation, reducing the efficiency of DNA repair systems and increasing the rate of mutation
- E.g. hepatitis B, liver cancer

IQ1: How does mutation introduce new alleles into a population?

6.1.2 compare the causes, processes and effects of different types of mutation, including but not limited to:

- point mutation
- chromosomal mutation

6.1.2: Point Mutations

Mutations can be:

- Harmful mutations can negatively change protein function and decrease the chance of survival
- Neutral mutations have no effect on protein function
- Beneficial mutations can positively change protein function and increase the chance of survival

Gene mutations are small localised changes in the structure of a DNA strand. Involves a change in a single nucleotide (point mutation) or the change to a triplet (deletion or repeat).

Substitution (nonsense)	Replaces one base pair only possibly impacting one amino acid code
Deletion (missense)	Makes a frameshift as it impacts all amino acids that come after it
Insertion (nonsense)	Makes a frameshift as it impacts all amino acids that come after it

Point mutation: change of only one nucleotide in the DNA sequence

- Results in a change in the mRNA; changes the amino acid sequence due to this change which results in a change in the enzyme being non-functional as the shape of the protein is changed (substrate won't be able to bind to the substrate in the active site)
- **Frameshift:** all codons after mutation are affected (different amino acids)
- **Missense:** different amino acids
- **Nonsense:** unfinished protein (results in stop codon)
- **Silent:** no effect on amino acid produced

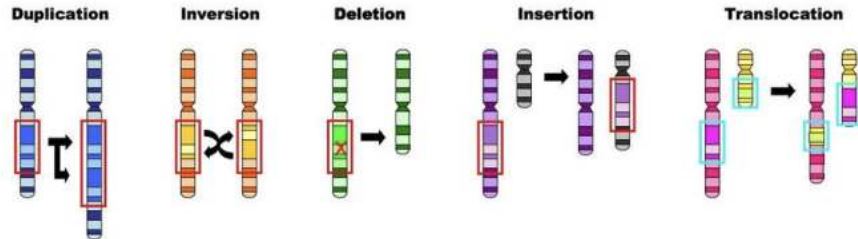
6.1.2: Chromosomal Mutations

- The number of chromosomes in the cell is changed or large sections of the chromosome are changed; nondisjunction

Chromosomal mutation: change in structure of chromosomes

- **Duplication:** results in gametes having missing genes or double sets of genes
- **Inversion:** the middle piece of the chromosome falls out and rotates 180° and then rejoins

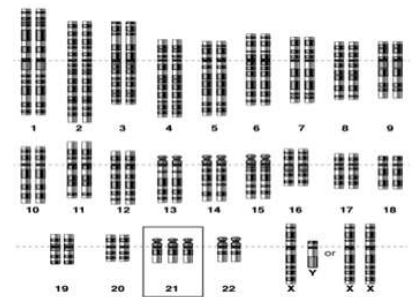
- **Deletion:** the middle piece of the chromosome falls out and the chromosome rejoins or the end of the chromosome falls out
- **Insertion:** a section of one chromosome breaks off and attaches to a different chromosome
- **Translocation:** involves the movement of a group of genes between different chromosomes



Polyploidy: more than 2 sets of chromosomes per nucleus

Aneuploidy: an abnormal number of chromosomes per nucleus

Trisomy: three copies instead of a pair of chromosomes. To the side is a diagram of trisomy 21 or down syndrome



IQ1: How does mutation introduce new alleles into a population?

6.1.3 distinguish between somatic mutations and germ-line mutations and their effect on an organism (ACSBLO82, ACSBLO83)

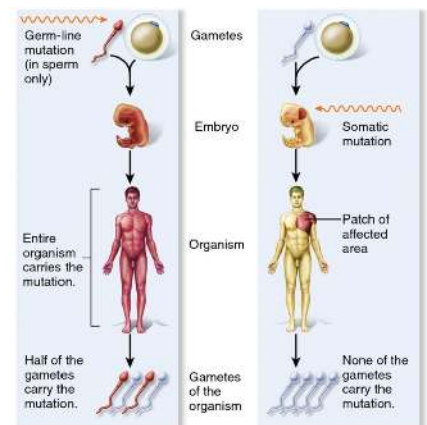
6.1.3: Somatic vs Germline Mutations

Germline mutations: occurs in the gamete of an organism, forming a new allele; can be passed onto the next generation

- Occurs in sexual reproduction and can happen in both meiosis in prophase I and anaphase I and II, and DNA replication
- Half of the gametes in the organism carry the mutation
- E.g. sickle cell anaemia

Somatic mutation: forms a new allele in a body cell, which may affect the individual; can not be passed onto the next generation

- Sometimes this mutation can be silent resulting in the coding for the same amino acid, therefore there is no change to the amino acid sequence
- **Oncogenes:** regulate cell division; if a mutation occurs, it can lead to skin cancer due to the uncontrollable cell division
- E.g. skin cancer



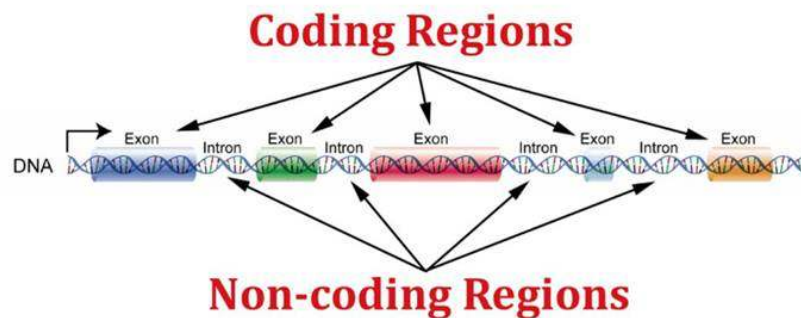
IQ1: How does mutation introduce new alleles into a population?

6.1.4 assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation (ACSBL078)

6.1.4: Coding and Non-Coding DNA

The human genome is comprised of coding DNA (2%) and non-coding DNA (98%)

Genome	A haploid set of chromosomes in a gamete in an organism
Genotype	The genetic makeup of an organism
Phenotype	The observable traits of an individual



Mutations in coding DNA:

- When the amino acid sequence is changed the protein produced or the effectiveness of the protein may change.
- A change in the genetic code can lead to a change in the expression of the gene which changes its phenotype of an individual, resulting in a genetic disease

If the errors to eukaryotic genes occur to products that involve DNA repair, there is an increase in mutations during DNA replication.

Prokaryotes have very little non-coding DNA. when parts of the coding DNA is inactive or unable to be maintained due to continuous changes to the environment, there is an increase rate of mutation

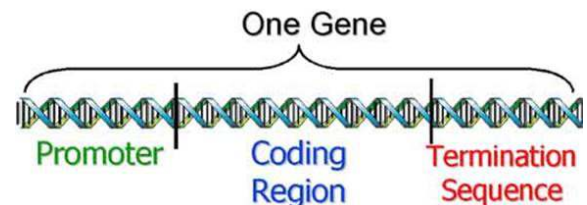
Mutations in non-coding DNA:

- Termed as 'junk DNA' - repeated sequences with no identified purpose
- Portions of non-coding DNA that have important functions
 - Introns
 - Non-coding RNA genes

- Promoter terminator regions
- The overall functioning of the cell or organism may be changed due to mutations in these portions

Introns are removed during transcription to create a functional mRNA strand. Once the introns are removed, the exons can join together in a neat, continuous coding sequence. If introns aren't removed, they can be read as coding sequences when mRNA codes for a polypeptide; pseudo-exon.

rRNA and tRNA are produced from non-coding sections of DNA called 'non-coding RNA genes'. Mutations in these sections can cause cell death.



Mutation within the promoter or terminator region can result in over or under expression of a gene, leading to either an excess or shortage of the gene product

IQ1: How does mutation introduce new alleles into a population?

6.1.5 investigate the causes of genetic variation relating to the processes of fertilisation, meiosis and mutation (ACSBLO78)

6.1.5: Causes of Variation

Genetic variation: the variation in alleles both within and among populations

3 ways of variation in a population through meiosis

1. Fertilisation
2. Crossing over
3. Independent assortment

IQ1: How does mutation introduce new alleles into a population?

6.1.6 evaluate the effect of mutation, gene flow and genetic drift on the gene pool of populations

6.1.6 Effect of Mutation, Gene Flow, and Genetic Drift on the Gene Pool of a Population

Gene flow effects on gene pool

Gene pool: the combination of all the genes present in a population

- When the allele frequencies change, we know the population has evolved

Gene flow (migration): the movement of genes from one population to another population.

- Source of genetic variation in a population who doesn't hold the gene
- Assisted gene flow and targeted gene flow are being used to increase gene variation

Genetic drift and mutation effects on gene pool

- A small population is more influenced by genetic drift than a large population

Genetic drift	Change in alleles frequencies due to chance in a population • Unlike natural selection, genetic drift is random • E.g. koala: their genetic gene pool has been reduced due to bushfires (random event)
Bottle neck	A type of genetic drift caused by a random event (fire) that wipes off the majority of the population, leaving the remaining population with limited alleles
Founder effect	A type of genetic drift where an organism moves to a new area that is physically different from its original area, its alleles don't match the allele frequency of the population (random sample)

- The founder effect occurs when a portion of the population separates from the old population to start a new population with different allele frequencies

IQ2: How do genetic techniques affect Earth's biodiversity?

- 6.2.1 investigate the uses and applications of biotechnology (past, present and future), including: (ACSBL087)
1. analysing the social implications and ethical uses of biotechnology, including plant and animal examples
 2. researching future directions of the use of biotechnology
 3. evaluating the potential benefits for society of research using genetic technologies
 4. evaluating the changes to the Earth's biodiversity due to genetic techniques

6.2.1 Applications of Biotechnology Content

Biotechnology: the exploitation of living systems and biological processes to develop tools for technological use for the benefit of mankind. Applications may be used in areas such as industry (e.g. food, energy), medicine, environmental sciences and computational design

1. Medical biotechnologies

- Makes use of biotechnology understanding to improve human health and needs
 - **Vaccines:** make use of the understanding of the human immune system and development of molecules to strengthen this response for future infections

- **Antibiotics:** use the understanding of cell biology. With this understanding cells make chemicals to help them survive synthetically designed molecules, we call them drugs, can be used to target specific proteins in bacteria, inhibiting their function with increased efficiency and fewer patient side effects
- **Stem cell treatments:** use the technology of therapeutic cloning to create skin grafts and treat certain cancers and autoimmune diseases
- **Insulin:** insulin from pork and beef is very similar to human insulin which is what makes up the insulin shots that people with type 1 diabetes take

Insulin manufacturing

- The gene that makes human insulin is extracted from a person's DNA and is stuck onto a bacteria, manufacturing the insulin
- Process:
 - Remove the plasmid
 - Cut open with restriction enzymes
 - Locate and remove human insulin gene using the same restriction enzymes
 - The DNA is combined with the plasmid using ligase enzymes
 - Bacteria are then incubated with new recombinant plasmid
 - Bacteria will take up plasmid and start to produce insulin
- *Restriction enzymes and recombinant*
 - DNA ligase enzymes repair DNA by helping nucleotides 'glue' together during DNA replication.
 - Restriction enzymes are produced by bacteria and act like scissors, cutting up DNA at specific DNA sequences

2. Environmental biotechnologies

- Make use of biological understanding to improve ecosystems and primarily agriculture
 - **Bioremediation:** an application of biotechnology to manage the pollution introduced to an ecosystem by the use of microorganisms. Bacteria are able to metabolise pollutants
 - Agriculture utilises transgenesis to produce more durable and nutrient rich crops
- **Biodegradable plastics:** create bio polymers derived from plant and bacterial system which increases their biocompatibility which allows for more efficient breakdown of plastics

3. Biotechnology in farming

- **Selective breeding:** selecting the best organism in a species to mate with the rest of the species

Human Mandate

- Genetic engineering techniques allows for species to be crossed (e.g. strawberry and salmon), and has helped with creating tomatoes that have a longer shelf life. Plants have also been engineered for fast yield and pest and disease resistance
- Agriculture crops and livestock have limited biodiversity
- Animal and plant agricultural practices can reduce a natural ecosystem rich in diversity
- Limited diversity is at serious risk of destruction by pest or disease
- Selective breeding techniques produce superior crops at the expense of diversity

IQ3: Does artificial manipulation of DNA have the potential to change populations forever?

6.3.1 investigate the uses and advantages of current genetic technologies that induce genetic change

6.3.2 compare the processes and outcomes of reproductive technologies, including but not limited to:

- artificial insemination
- artificial pollination

6.3.1 & 6.3.2: Reproductive Technologies

Artificial insemination: the process of collecting sperm cells and manually depositing it into the reproductive tract of a female

- **Process in humans**
 - Performed on couples who can't get pregnant the usual way and require an alternative fertility treatment (IVF)
- **Process in agriculture**
 - Allow for high quality semen from high quality males to be used to impregnate large amounts of females in a very fast amount of time (resulting in stronger and better animals)
- **Advantages:**
 - Increases variety

Artificial pollination: pollen is selected from the desired plant and is then used to dust the stigma of the chosen female flower

- Used to produce offspring with favourable characteristics e.g. higher yield, larger fruit or disease resistance
- **Advantages:**
 - Useful in conditions where natural carriers of pollen grains such as insects are fewer in number
 - Facilitates hybridisation because in this pollination method, interspecific pollen transfer can be done

Mechanical pollination: pollen grains are spread by a small aircraft or drone on a large area (not efficient as the probes dust the pollen grains in the air and some pollen grains may not reach flowers)

Hand pollination: human beings transfer pollen grains from one flower to another by the means of forceps or fine brushes (more efficient compared to mechanical)

Technology-assisted pollination: humans use one or more technologies to transfer pollen from one flower to another

IQ3: Does artificial manipulation of DNA have the potential to change populations forever?

6.3.3 investigate and assess the effectiveness of cloning, including but not limited to:

- whole organism cloning
- gene cloning

6.3.3: Cloning

Cloning: a process where an exact genetic copy of an organism can be made

- The DNA from a nucleus of a male organism is added into the somatic cell of another organism of the same species
- Done through 2 methods, **artificial embryo twinning** and **somatic nuclear transfer**
 - *Artificial embryo twinning*
 - Mimics the natural process that creates identical twins
 - A very early embryo is separated into individual cells, which are allowed to divide and develop for a short time in the Petri dish. The embryos are then placed into a surrogate mother, where they finish developing. Since all embryos are from the same fertilised egg, they are genetically identical
 - *Somatic cell nuclear transfer*
 - Produces an exact genetic copy or clone of an individual
 - **Nuclear:** the nucleus is a compartment that holds the cell's DNA. The DNA is divided into packages called chromosomes, and it contains all the information needed to form an organism
 - **Transfer:** researchers isolate the somatic cell from the adult organism then remove the nucleus and all of its DNA from an egg cell. The nucleus from the somatic cell is then transferred to an egg cell
- Cloning to make stem cells
 - Stem cells can be cloned to be genetically identical to an individual. These cells will be used for medical processes and may be able to grow whole organisms. The growing of stem cells could also help with assisting a person with a disease to help researchers understand the disease and develop treatments

IQ3: Does artificial manipulation of DNA have the potential to change populations forever?

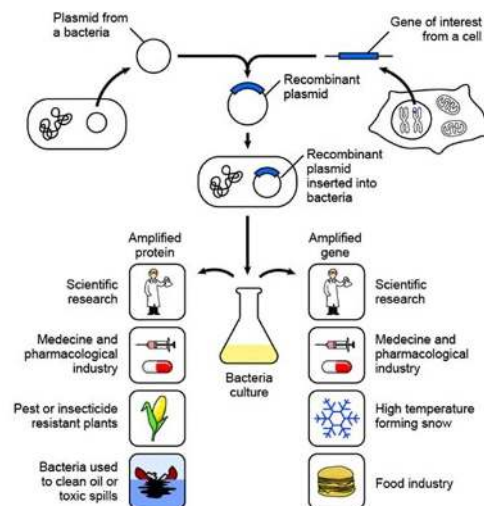
6.3.3 investigate and assess the effectiveness of cloning, including but not limited to:

- whole organism cloning
- gene cloning

6.3.3: Transgenic Species

Recombinant DNA technology: the joining together of DNA molecules from two different species

- The DNA molecule is inserted into a host organism to produce new genetic combinations
- Cloning is undertaken in order to obtain the clone of one particular gene or DNA
- Bacteria created through recombinant DNA techniques that are capable of synthesising
 - Human insulin
 - Human growth hormone
 - Alpha interferon
 - Hepatitis B vaccine
- **Gene therapy:** a normal gene is introduced into an individual's genome in order to repair a mutation that causes a genetic disease (e.g. cystic fibrosis)
- Process of DNA recombinant:
 1. The target gene from the human DNA is **isolated**
 2. A **bacterial plasmid** is extracted from a bacterial cell
 3. The **restriction enzyme** then cuts the DNA at a specific sequence. The cuts from the enzymes create sticky ends or overhangs which fit with the sticky ends on the human gene and the sticky ends on the plasmid. This is because both sticky ends were cut by the same enzyme
 4. The sticky ends of the **plasmid and human gene** come together and form a complete plasmid (recombinant DNA)
 5. The **recombinant DNA plasmid** is then **inserted** into a **bacterial cell**. The bacterial cell will clone itself many times over, therefore making lots of copies of the recombinant DNA plasmid



	Gene Technology	Benefit
Agricultural Examples		
Bt cotton	Genetic modification (transgenic)	allows for increase of productivity of marginalised land and reduces post-harvest losses.

Golden rice	Genetic modification (transgenic) - incorporates vitamin A	allows for a higher intake of vitamin A
Atlantic salmon	Genetic modification (transgenic) • Takes DNA from Chinook salmon	modified to grow faster and larger than the natural stock
Medical Examples		
Humulin - human insulin	Genetic engineering (Recombinant DNA technology)	produced for people with diabetes to regulate insulin levels. It is better tolerated by diabetes and can be produced quickly and efficiently
Monoclonal Antibodies (MABs)	Genetic technology to artificially clone antibody-producing cells	used more for cancer research. Used in cancer treatment to assist the natural immune system to produce specific antibodies that target cancer cells.
Industrial Examples		
Production of purer enzymes	manufacturing enzymes	quickly clones produce in large quantities- produces purer enzymes as traditional enzymes contain additional chemical substances
E.coli	genetical engineering	allows for heavy metals to be absorbed (e.g. mercury) from contaminated sites. Can aid in the remediation of mine sites and other heavily polluted areas, reducing harm to local ecosystems

Mod 7: Infectious Disease

IQ1: How are diseases transmitted?

7.1.1 describe a variety of infectious diseases caused by pathogens, including microorganisms, microorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including: (ACSBL097, ACSBL098, ACSBL116, ACSBL117)

1. classifying different pathogens that cause disease in plants and animals (ACSBL117)
2. investigating the transmission of a disease during an epidemic
3. design and conduct a practical investigation relating to the microbial testing of water or food samples
4. investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission

7.1.1: Causes of Infectious Disease

Disease: a state of **complete** physical, mental and spiritual well-being, not merely the absence of disease or infirmity

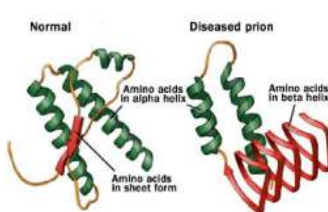
- Any condition that interferes with how an organism, or any part of it, functions
 - Usually associated with certain symptoms and signs that are different from physical injury
- **Non-infectious diseases:** those that are **not** caused by pathogens and **can't** be transmitted from one organism to another
- **Infectious diseases:** disease caused to an organisms by **pathogens** transmitted from **one organism to another** or from the **environment** to an **organism**

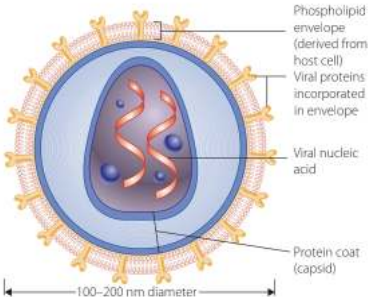
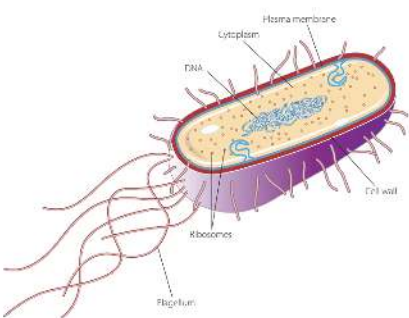
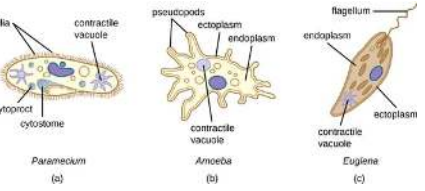
Pathogen: an organism or any other agent that can **cause disease** in an organism

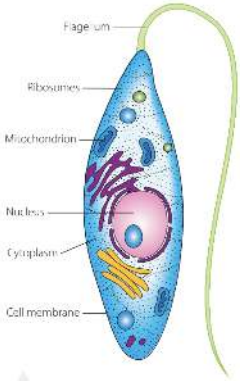
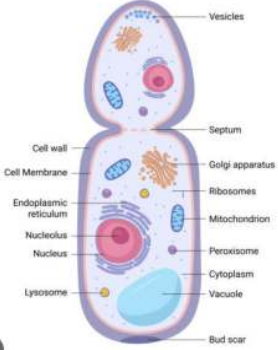
- **Disrupts** the **normal** functions of the infected organism
- Can be unicellular or non-cellular
 - **Unicellular pathogens:** bacteria, fungi, protozoa, macro-parasites
 - **Non-cellular pathogens:** viruses, prions and viroids

Microorganisms: bacteria, protists, prion, virus

Macroorganisms: ectoparasites (ticks, lice), endoparasite (tapeworm, flocks)

Pathogen	Features	Examples of diseases
Prion 	An abnormal protein that doesn't contain genetic material (DNA/RNA) that is capable of causing degenerative diseases to the nervous system.	Kuru: • First found in Papua New Guinea in the 1950s in the Fore tribe. • <u>Symptoms include:</u> shivering, trembling, uncontrolled laughter, then progressing to death • Found the cause of disease was from cannibalism of the dead ◦ The brain was consumed by worms carrying the

From smallest to largest in size			pathogen, therefore, causing anyone eating the deceased to contract the disease
	Virus 	A small particle containing genetic material (RNA/DNA) inside a protein coat (nucleic acid) that can be passed on genetically. As they are not free living, they can only reproduce and metabolise in a host - <u>Virion</u>: single viral particle	Influenza • Caused by the influenza virus (A or B) • Airborne • <u>Symptoms include</u>: fever, coughing, sneezing, sore throat, stuffy nose, muscle or body ache, headaches, fatigue and sometimes diarrhoea and vomiting • Spread through droplet transmission
	Bacteria 	A unicellular, single living and prokaryotic cell, that doesn't contain any membrane bound organelles. Reproduced by binary fission	Salmonella • Causes food poisoning • <u>Symptoms include</u>: abdominal cramps, vomiting and diarrhoea • Found in unhygienic kitchens, undercooked meat, egg and poultry, or the same food that hasn't been reheated properly
	Protozoa 	A single living eukaryotic animal like cell	Malaria • Spread through a mosquito (vector) carrying the Plasmodium protist • Often found in areas with higher temperatures (e.g. Africa, Asia etc.) • The mosquitoes suck on the blood of an infected person, then passing the protist to other people by sucking their blood, allowing the protist to enter the bloodstream • <u>Symptoms include</u>: fever, sweats and chills, headaches, vomiting and diarrhoea

			
	Fungi 	A eukaryotic non photosynthetic cell with cell walls, that may be single living or form more complex structures	Athlete's foot • A rash caused by a fungus usually found between people's toes • <u>Symptoms include</u>: dry, red and flaky or white, wet and cracked skin • Often found in communal areas (e.g. swimming pool change rooms) • <u>Transmission</u>: touching infected skin or contaminated surfaces • Treated through antifungal medication
	Macroparasites	Multicellular and eukaryotic pathogens that are visible to the naked eye • Some directly cause disease, others are vectors	Parasitic worms • Their eggs are taken into the body (usually through swallowing) • The eggs hatch and release the worms to live in the body • Larvae may burrow their way into the skin • Sickness is caused when they live in the body as they consume food before it's digested, removing the nutrients needed by the body

Epidemic Transmission

Epidemics: nationally spread disease

- The number of affected by a particular disease is **higher** than usual
- **Endemic level:** where the **level** of a particular disease is **steady** and its **occurrence** is **not increasing/decreasing** in a population
- Factors to be investigated for a reason for an epidemic:
 - The cause of the disease
 - How it is transmitted

- Who is at risk
- Method of treatment and control
- Long term measures of prevention
- Virulence of the strain
- Whether the pathogen changes over time

Pandemics: disease spread **globally**

Antigenic shift: the process by which the flu virus can become lethal

Transmission of infectious diseases

Transmissible disease: a disease which a pathogen can be passed from one host to another (can spread)

- Methods of transmission:
 - **Direct:** contact between people by touching, mixing of bodily fluids etc.
 - **Indirect:** via contamination of food, animals, the air, water etc.
 - **Vectors:** an organism that carries a pathogen from one host to another

Reducing transmission

1. Hygienic food preparation
 - a. Wash hands before touching food/wear gloves
 - b. Keep your hair out of food
 - c. Don't cough or sneeze around the food
 - d. Keep animals (including flies) away
 - e. Keep food cold (keeps bacteria and microorganisms from growing)
 - f. Keep raw meat separate
2. Personal hygiene
 - a. Oil and dirt (and pathogens) accumulates on skin
 - b. Sweating encourages bacteria
 - c. Washing regularly reduces transmission
3. Waste disposal
 - a. Rats, dogs, flies etc. are attracted to food waste
 - b. Bacteria multiply on food waste
 - c. Well-designed landfill sites are essential
4. Sewage treatment
 - a. Toilet and kitchen waste contains a lot of pathogens
 - b. Sewage that enters drinking water provides a pathway for transmission
 - c. Correct treatment of sewage is essential

Indirect transmission methods

Airborne: pathogen leaves the host (sneezing, coughing etc.), and **remains** in the air before it comes in contact with the **respiratory membranes** of another person

Droplets: pathogens contained in droplets of **body fluid** (saliva, blood, etc.) come in contact with another person or contaminate a surface.

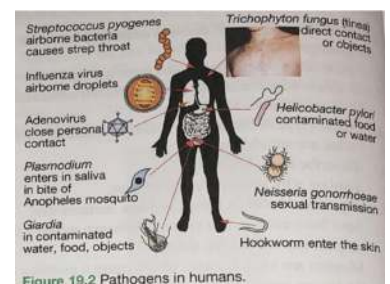


Figure 19.2 Pathogens in humans.

Foodborne: transmission occurs through **eating contaminated food** or by **improper cleaning** habits after handling contaminated food

Waterborne: pathogen is spread by consumption or contact with contaminated water

Zoonotic: pathogen is spread from animals to humans (vectors)

Reservoir	Where the pathogen lives and reproduces, can be a living organism or environment
Portal of exit	Pathway a pathogen uses to exit a host and be transmitted to another
Portal of entry	Is often the same as the portal of exit from the previous host. The pathogen must be able to access the type of tissues where it can grow and reproduce
Virulence	The ability of the pathogen to infect and cause harm to its host
Infectious agent	The pathogen that is capable of infection and causes disease
Susceptible host	An organism that will become infected. Depends on genetics, immunity to the pathogen and overall health
Virulence of a pathogen	An increase in ability to infect a host can occur due to mutations, creating new strains of a disease
Pathogen introduction	Migration of individuals into a new population due to bringing a new disease that the new population is not immune to
Host exposure and susceptibility	Loss of genetic variation or selection pressures can cause a population to lose genetic resistance to certain pathogens

Mode of transmission: the movement or the transmission of pathogens from a reservoir to a susceptible host; once the pathogen leaves the reservoir it needs a mode of transmission to the host through a portal of entry (indirect, direct or vector)

<u>Mode 1: Types of</u>	<u>Direct contact</u>		Examples
<u>Type</u>	<u>Description</u>	<u>Examples</u>	
1.person-to-person contact 	Occurs when an infected individual touches or exchanges body fluids with another individual. This includes touching, kissing, sexual contact, biting and contact with oral secretions or body lesions,	Ebola can be spread through saliva, mucus, blood, sweat, urine or from mother to baby during birth. Other diseases transmission by direct contact person to person contact include: herpes, gonorrhoea, HIV, chickenpox and measles	
2.droplet spread 	The result of droplets spray caused by talking, coughing and sneezing. Travels a short distance before settling	Other diseases transmission by direct contact droplet spread include. influenza, chickenpox, measles, tuberculosis	
<u>Mode 2: Types of indirect contact</u>			
<u>Type</u>	<u>Description</u>		<u>Examples</u>
1.airborne transmission 	Some pathogens can remain suspended in the air for many hours outside of the body. They can travel long distances and infect an individual long after the original host has left the area		cold and measles viruses. influenza, chickenpox, measles, tuberculosis.
2.contaminated objects 	Transmission occurs when a surface or object containing the pathogen is touched and the pathogen is transferred to the mouth, nose or eyes before washing the hands		slap cheek disease 

3.food and drinking water	Several pathogens are transmitted as a result of improper treatment, handling or storage of food and water.		Vibrio cholerae (cholera), escherichia coli (food poisoning, diarrhoea)
4.animal-to-person contact	Usually occurs from a scratch or bite from an infected animal or from handling of waste.		toxoplasma gondii (toxoplasmosis) is a protozoan parasite found in cat faeces, rabies virus
5. vector-borne 	A vector is an organism that is not the pathogen but can spread the disease from one host to another (carrier) (fleas, ticks etc), they carry the disease (they aren't affected) and spread the pathogen between hosts)		Malaria, Dengue fever,

Microbiology

- Microbes that are present in the air, water and food may cause human diseases, may be beneficial, or some may have no effect on humans at all
- As microbes reproduce they can form colonies that are visible to the unaided eye.
 - Colonies are distinguished by their size, shape, surface profile and colour.
 - **Bacterial colonies** are small, shiny, and have colour
 - **Fungal colonies** are fluffy and large.

Methods of treating drinking water

- **Filtration:** fine mesh screens, coagulation, removal of flocs
- **Disinfection:** chlorination, chlorination, ozone, UV light, kills pathogens, disrupting the life cycle and future transmission
- **Fluoridation:** prevents dental decay under the advice of NSW Health

Quantitative data: these observations are objective containing information which focuses on numbers of measurements.

Qualitative data: an observation that describes or provides information about the quality of something such as is it hot or cold, blue or green, without expressing numerical values.

Pure culture	A population of cells or multicellular organisms growing in the absence of other species
Colony	A visible mass of microorganisms all originating from a single mother cell
Streaking	Techniques used to isolate a pure strain from a single species of microorganism
Aseptic technique	Using practices and procedures to prevent contamination from pathogens Involves applying strict rules to minimise the risk of infection

IQ1: How are diseases transmitted?

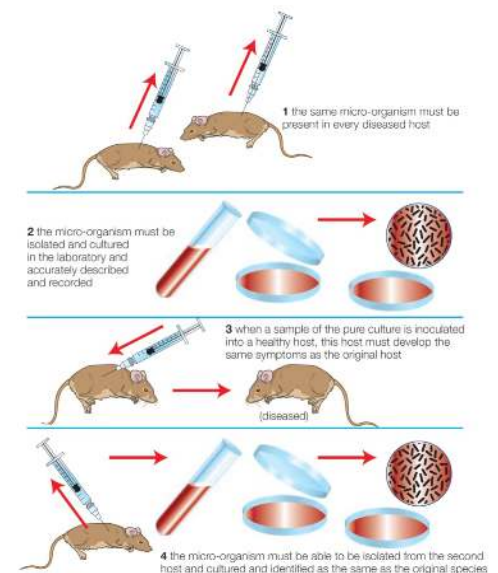
7.1.2 investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including:

1. Koch's postulates
2. Pasteur's experiments on microbial contamination

7.1.2: Robert Koch and Louis Pasteur

Robert Koch

- Developed the **agar plate technique** used for growing microorganisms
- Koch's extensive study of the **anthrax bacillus** is where he examined the blood of sheep that had died from anthrax and identified rod-shaped bacteria that he then isolated and grew in cultures. His experiment added to the **germ theory of disease** as they showed that a microorganism grown outside the body caused a disease
- Through his study he was able to develop a series of steps called **Koch's Postulates**
- He used **methyl** indicator to show the **bacterial growth** on the agar plates



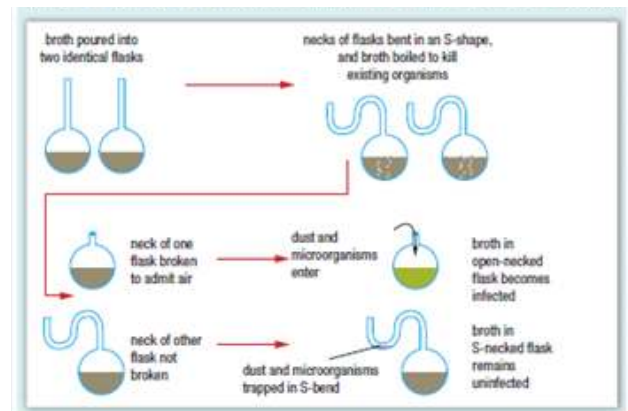
Koch's Postulates

- Determines if a particular microorganism is responsible for causing a disease
- The steps of Koch's Postulates includes:
 1. The **same microorganism** must be present in every diseased host

- Aim: to establish a constant association between the presence of the pathogen and occurrence of the disease
- 2. The microorganism must be **isolated** and **cultured** in the laboratory and accurately described and recorded
 - Aim: isolate the pathogen from diseased tissues into pure culture free of all living organisms
- 3. When a **sample** of the **pure culture** is inoculated into a **healthy host** this host must develop the **same symptoms** as the original host
 - Aim: inoculate an organism of the same species or variety and reproduce the same symptoms of the disease
- 4. The microorganism must be able to be **isolated** from the second host and **cultured and identified** as the **same** as the original host
 - Aim: re-isolate the pathogen from the inoculated plant into pure culture and compare it to the original isolation; it must be the same
- Links a specific bacteria with a specific disease
- His postulates helped with the identification of the pathogen for typhus, tetanus, and plague

Louis Pasteur

- Credited for creating the science of microbiology
- Pasteur studied **fermentation** of beet juice and found that the process was because of yeast (microbe)
- Helped in finding the bacteria that caused wine, beer and vinegar spoilage
- He found that **heating** these solutions long enough to **kill the contaminating bacteria** would help with keeping the wine and vinegar from going off
- **Pasteurisation** was therefore created and used today to ensure products like milk are **free of disease** causing microorganisms
- He also found that the rotting of foodstuffs was due to the activity of living organisms
- After further study he proposed the **germ theory of disease**, disregarding the spontaneous generation
- Through his investigation he found that using a '**swan-necked**' flask trapped the microorganisms present in the air from entering the broth in the flask
 - Therefore proved that organisms that contaminated the broth caused **decay** and must be **carried in the air** and isn't spontaneously generated
- Found that **specific bacteria** are linked to a **specific disease** (supported germ theory of disease)
- Investigated the cause of anthrax and found that the animals contracting the disease had **no known contact** with animals suffering from the disease. The carcasses that had been buried in the fields were **grazed by the healthy animals** therefore, causing infection.
- Pasteur found a way to introduce a bacteria into a host to recognise the infection (vaccine)



- **Principle of immunity:** ability to detect and fight an infection

Comparative summary table

	Pasteur	Koch
Similarities	● Vaccinations (Pasteur- anthrax, Koch- cholera) ● Both were credited for the germ theory of disease ○ Microbes caused disease	
Differences	Pasteur 'swan-neck' experiment to disprove the microorganisms were the cause of decay ● Both flasks boiled for 15 mins and left to grow for a couple of weeks	● Koch's postulates- series of steps to determine the specific microorganism that caused a particular disease 1. Same pathogen is present in every organism with microorganism 2. Isolate and grow pure culture 3. Inoculate healthy host --> must show the same symptoms 4. Isolate microorganisms --> pure culture. Compare to original culture to see if they are the same

Assessment of articles

Designing an investigation (first hand investigation)

Validity	● Efficiently manage independent variables (what you will change) ● Clearly state dependent variable (what you will measure) ● Keep controlled variables the same (e.g. control) ● Keep all other variables constant (e.g. temperature) ● Ensure method answers hypothesis Discussions about validity must: - identify what validity is - identify factors that affect the validity of the experiment (variables, equipment, range of values, ect) - assess the overall validity of the equipment used
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<i>Reliability</i>	The experiment must be repeated with consistent results obtained. Discussions about reliability must: - identify what reliability is - identify that reliability is assessed by repeating the experiment and averaging the results - identify that repetition minimises the effect of random errors/ outliers - gives examples of possible random errors - assess the overall reliability of the experiment
<i>Accuracy</i>	• When an experiment is as valid and reliable as possible. • Whether the results are close to the true value of the quantity being measured • Whether the results can be substantiated in secondary sources.

Evaluating an investigation (secondary data)

<i>Validity</i>	Evaluate: • Whether the information relates to the problem being investigated • Efficiently manage independent variables (what you will change) • Clearly state dependent variable (what you will measure) • Keep controlled variables the same (e.g. control) • Keep all other variables constant (e.g. temperature) • Ensure method answers hypothesis
<i>Reliability</i>	Evaluate: • The author's credentials • Whether the purpose of the article is not biased • Whether the article is reputable (ie textbook or .gov.edu) • Whether the data was gathered using an appropriate method and measuring devices • Whether the data is current
<i>Accuracy</i>	Evaluate: Whether the information can be substantiated in more than one reliable source (i.e. whether the information is consistent with information from other reputable sources.)

IQ1: How are diseases transmitted?

7.1.3 assess the causes and effects of diseases on agricultural production, including but not limited to:

1. plant diseases
2. animal diseases

7.1.3: Agricultural Disease and its Effects

Plants

Agriculture: a form of primary industry that involves the cultivation of crops and pastures and the rearing of animals to provide meat, milk, fibres, and other products for humans

- The wool industry is one of Australia's biggest economic successes

The importance of agriculture in Australia

- Australia's agriculture is relatively free of many infectious diseases that affect animals and plants in other countries as it's fairly secluded from the rest of the world
- **Stringent biodiversity** (strict border security) measures are put in place to reduce likelihood of disease transmission from pathogens

Infectious diseases in agriculture

- Plant disease affect the yield and quality of crops
- Biodiversity must be kept in crops to minimise the transmission of disease as crops are usually genetically similar
- **Intensive farming (many animals in a small area):** creates higher densities of livestock, increasing direct contact between animals and the risk of infectious disease transmission
 - E.g. mad cow disease
 - High usage of antibiotics results in resistance in bacteria producing **superbugs** (antibiotic resistance)
 - **Zoonotic diseases** (from animal to human) can be spread through the consumption of and contact of infected animals
- Impacts on travel
 - Increase of travel (plants being imported), imports, exports and global connection increases the chances of introducing new diseases
 - All plants and animals are to be quarantined for 4 months before entering Australia

Plant diseases: Panama disease on Cavendish Bananas

- There is currently no way of eliminating the fungi causing panama from the soil in which bananas are grown on

Cause (name the pathogen)	Effects on the Banana	Effects on agriculture production

Fusarium oxysporum f	This pathogen causes a fungus to grow in the soil and travels up to the stem of the banana plant. The TR4 causes the banana to wilt (change in colour of the leaves, splitting of stem). The banana plant eventually dies off.	Panama causes a large decrease in the production of bananas
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Animals

- **Epizootic:** the part of the animal population
- Any exotic animal disease must be reported and detected within the borders the disease has been identified to minimise risk of **zoonosis** (animal to human)
- Varroa mites
 - Slip into hives by attacking themselves unto the back of the adult bees
 - The mites feed off the bees blood within a few minutes of clinging unto the bees
 - Once the varroa mites enter the hive, the slip into the vulnerable uncapped brood cells
 - They wait until the bees cape the brood
 - Once the cell is capped, the mother mite climbs into the cocoon of the developing bee and tears open a hole, feeding on its blood
 - Within 3 days, the mother mite lays the first egg which always develops into a male
 - A female egg is then laid every 30 hours over the next week or so under the brood cap
 - As the female mites mature, they mate with their brother
 - By the time the baby develops and leaves its infected cell, as many as 3 fertilised mites will emerge with it and the cycle continues
 - This process allows the population of the varroa mite to grow as fast as the bee population it feeds on, but when summer ends and the bee population decreases, the hive is left with a huge population of mites that is dangerous
 - Too many mites can overwhelm the bee population and cause it to die
 - Honey bee populations with large mite infestation can't effectively pollinate or produce honey, decreasing the availability of honey to consumers

Cause (name the pathogen)	Effects on the Honey bee	Effects on agriculture production
Varroa mite	Causes mass infection to the population of honey bees, therefore causing the population to die	A decrease in honey production will occur, which increases the cost of honey due to the lack of honey available to the public

IQ1: How are diseases transmitted?

7.1.4 compare the **adaptations** of different pathogens that facilitate their entry into and transmission between hosts

7.1.4: Pathogen Adaptations for Entry and Transmission

Adaptation: a characteristic/trait that allows an organism to survive in its environment/past environment

- E.g. special digestive enzymes in a koala to digest the poison held in the eucalyptus leaves
- Those with better adaptations will survive to reproduce and will pass on the adaptation to the next generation (natural selection)

For a pathogen to successfully affect the host, the following must occur:

1. Entry into **host** (bypassing **first line of defence** e.g. nose hairs)
2. **Adhesion** to host cell (some need to enter cells as well e.g. virus)
3. **Resist/not stimulate** secondary immune responses (**white blood cells or lymphocytes**)
4. **Survive to reproduce** (natural selection), **colonise** other tissues, and **multiply** (takes control of cell replication processes)
5. **Damage** to host in order to leave the host and be **transmitted** to a **new host**

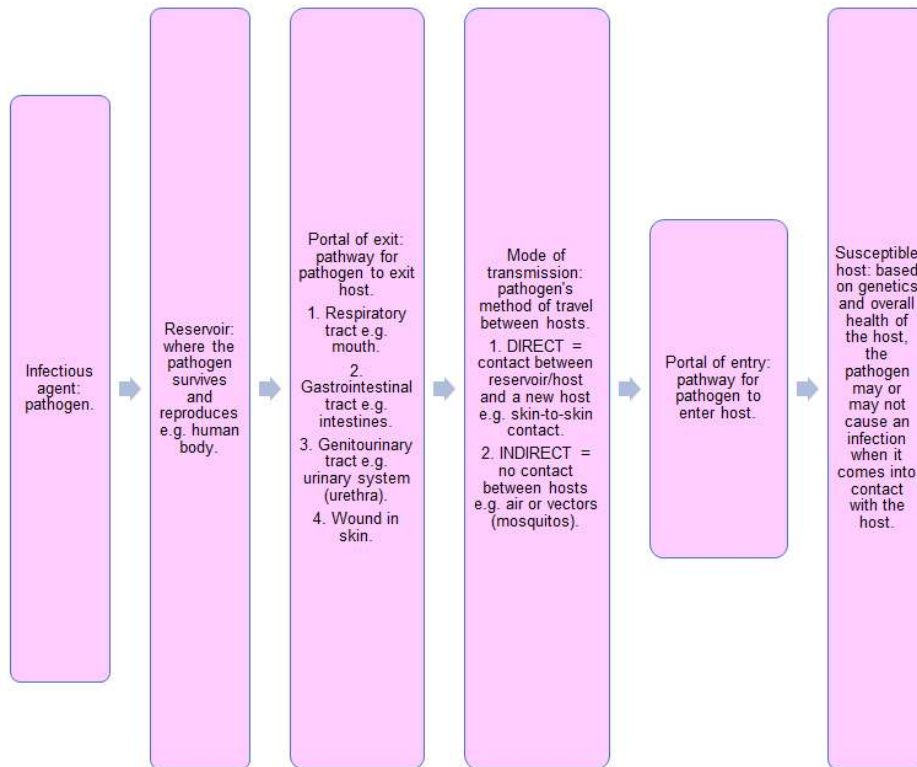
Adaptations of pathogens cause **virulence** (high virulence= highly contagious). Highly virulent pathogens possess many adaptations to cause disease

Virulence factors: molecule produced by a pathogen that increases its effectiveness in invading the host and causing disease

- Evolutionary strategies of pathogens are slightly ahead of development of host resistance strategies

Once pathogens have caused infection, many are able to **hide** within the host or **combat** the **host's immune system**

- Host symptoms often aid in pathogen reproduction and increase the chance of future infection in a new host



General adaptations of different pathogens

- Vectors
 - The outer layer of skin is impenetrable to most microorganisms but pathogens that use biting insects (mosquitos) as vectors are able to penetrate the skin
 - Pathogens can also replicate in the vector before entering the host → increases the number of pathogens entering the host e.g. malaria
- Eggs (macroparasite)
 - Tapeworms use hooks or suckers to attach to their host's intestines → makes it hard to remove
 - When they mature they become hermaphrodites and release thousands of fertile eggs in faecal matter
 - Releasing high numbers of eggs chance of survival is increased
- Prions
 - Can be easily absorbed from the soil by plants → they remain in the soil till animals consume the plant when grazing e.g. mad cow disease

Pathogen	Adaption
Bacteria	● Adhesions (cell-surface components- fimbria, pili) facilitate adhesion to

	other cells • Long pili enable DNA exchange between bacteria • Protective capsule prevents macrophage invasion • Resistance to washing effects secretions e.g. mucus, urine • Produce enzymes to dissolve cell walls
Virus	• Surface proteins connect to white blood cell receptor sites, immobilising immune system • Enter cell nucleus to replace host genome with viral DNA (e.g. polio) • Invades using receptor-mediated endocytosis, binding with cell surface to be engulfed
Prion	• Misfolded proteins produced by host body, change neighbouring proteins to prions • White blood cells produce tumour necrosis factor, helping prions invade dendrites
Protozoa	• Toxoplasma Gondii causes toxoplasmosis, transmitted through cat faeces and contaminated food • Possess microtubules, form structure enabling invasion of host cells
Fungi	• Cell wall contains mannoproteins, shields from immune system detection • Thermotolerance (production of heat shock proteins) resists heat • Capsule protects from being engulfed by macrophage • Produce hydrolytic enzymes to damage cell walls
Microparasites	• Hookworm secrete proteins reducing immune system response, teeth anchor worm to gut lining • Ticks use vasoconstriction inhibitors to prevent blood clotting

IQ2: How does a plant or animal respond to infection?

7.2.1 investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-source investigation, for example:

- fungal pathogens
- viral pathogens

7.2.1: Response of a Named Au Plant

	Fungi	Viruses
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Similarities	- Both can be pathogenic and cause decay - Both have a form of nucleic acid (DNA or RNA) - Both inhibit plant cellular processes like photosynthesis, water and nutrient uptake	
Differences	Eukaryotic heterotrophic (feeds off herself)	Non-cellular
1. cellular or non-cellular	cell structure.	Requires a host to reproduce inside of
2. size	Bigger: 2-10μm -mm in size	Smaller: 0.02-0.05μm
3. Adaptation to penetrate cell wall	Secrete enzymes that are able to digest plant cell wall .	Spread by vectors as they cannot pass through cell walls.
4. Relationship with plant	Necrotrophic – in plants results in death of the host.	Biotroph – a plant pathogen that does not kill the host
5. Symptoms	Symptoms of fungal plant infection: - Chlorosis - Leaf and stem wilting - Leaf and stem rusting - Leaf blotching - Leaf mildew	Symptoms of viral plant infection: - Chlorosis - Yellowed leaves - Mosaic leaf pattern - Crinkled leaves - Growth stunting

Ways that plants defend themselves

- Physical barriers**
 - Bark made of lignan (protects the plant from pathogenic attacks)
 - Waxy cuticle (prevents transpirations- water from escaping)
 - Thorns, spines and prickles (discourages big herbivores)
 - Trichomes (small hair like structures) - prevents pathogenic attacks --> releases a chemical that leave a stinging feeling
- Mimosa:** folds away and doesn't let insects touch it at all
- Plant responses to prevent further infection/damage from pathogens
 - Each single cell has the capacity to protect itself
 - The waxy cuticle will start to thicken
 - The cell walls get thicker especially in areas of attack
 - Guard cells stop invading pathogens from entering the plant through the spores

5. Cell death around the pathogen will prevent the pathogen from spreading
6. Hormones- an area of a plant under attack can release hormones to warn surrounding similar or the same plants and itself of the pathogen

Fungal infection	
Named pathogen	Phytophthora cinnamomi in eucalyptus trees
Symptoms	• Yellowing of foliage • Dieback of the limbs and branches • Splits in the trunks of infected trees • Darkening of the larger roots • Presence of mycelial fans (large sheets of fungal growth), behind the bark
Transmission	• Infected roots coming into contact with uninfected roots (direct contact) • Fungal spores are dispersed and land on dead wood surfaces, initiating infection (rare) • Physical transportation of the contaminated roots can also cause uninfected species to be infected
Response	• Pathogen acids physical barriers (thick cuticles, cellulose, small stomata) • Plant launches a defending response (involving phytohormones) once the passive barriers are breached • When the host is penetrated, a group of cell degrading enzymes that include cellulases are released, causing significant damage to the host • The cellulose and lignin of the sapwood are both broken down (creating stringy wood) ○ Results over time in the dieback (causes limbs to drop due to weakness)

Viral infection	
Named pathogen	Cymbidium mosaic virus
Plant species affected	Cymbidium (orchid) species
Symptoms	• A mosaic of irregularly shaped lesions • Sunken patches on the leaves • Stunted growth
Transmission	• The wound on the plant is often caused by insects and enables the virus to bypass the usual physical barriers of the plant (e.g. waxy cuticle)
Response	• The host is infected through a wound on the plant ○ The virus will move through each of the orchid's cells'

	endoplasmic reticulum • The virus will activate the active defences (e.g. the release of hydrogen peroxide that can destroy some pathogens) • Apoptosis
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IQ2: How does a plant or animal respond to infection?
7.2.2 analyse responses to the presence of pathogens by assessing the <u>physical and chemical changes</u> that occur in the host animals cells and tissues
IQ3: How does the human immune system respond to exposure to a pathogen?
7.3.1 investigate and model the <u>innate</u> and <u>adaptive</u> immune systems in the human body 7.3.2 explain how the immune system <u>responds</u> after primary exposure to a pathogen, including <u>innate</u> and <u>acquired</u> immunity

7.2&3: Animal, Plant and Human Response to Infection

Normal flora: refers to the microorganisms that are normally found in or on the human body

Effects of pathogens on plant tissue

- Death of plant
- Abnormal growth
- Destruction of tissues (necrosis)
- Discolouration of tissues
- Wilting
- Reduced yield of grains, pastures, fruits and vegetables
- Loss of trading opportunities
- Economic loss for the farmer

Animal responses to pathogens

- Internal compartments are closed off to the environment- helps to remain sterile
- **Antigen:** any molecule that the body recognises as foreign and that triggers an immune response
- The body recognises **marker molecules** on the pathogen as non-self and causes the immune response to be activated

Non-specific immunity	The response is the same for every foreign object
Specific immunity	There is a specific response for every antigen
Antigen	Any molecule the body recognises as foreign and that triggers the immune response (produces antibodies)

Antibodies	Proteins produced by B plasma cells that bind to and neutralise antigens
Innate immunity	Defence mechanisms that an individual is born with (non specific)
Adaptive immunity	After initial exposure the immune system retains a “memory” of an antigen and is able to produce a specific response to that antigen
Acquired immunity	Specific immunity obtained by the immune system after exposure to an antigen
Pathogen	Disease causing organism
Active immunity	The body produces antibodies in response to being challenged by an antigen
Passive immunity	The body is supplied with antibodies from another source

First line of defence- innate/nonspecific immunity

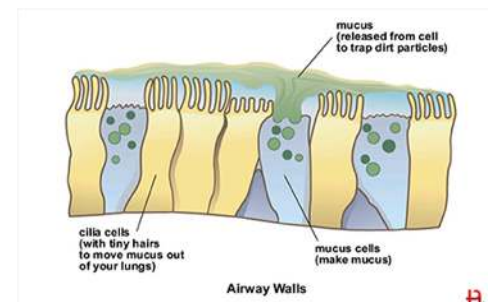
- On the surface of cells in the body, there are **‘marker’ molecules** that identify the cells as **belonging to the body**; protects the body from attacking its own immune system
- The body recognises the **antigens** attached to the pathogens entering the body as **foreign**, **triggering the immune response**
- Any foreign cell, cell fragment, proteins debris or toxin produced by bacteria can also contain antigens

Biological barrier

- Skin contains **harmless bacteria** that help stop the invading microbes from **multiplying**
- Done by competing for food and shelter with the pathogenic microorganisms therefore reducing the available requirements resulting in a reduction of the number of pathogenic microbes

Physical barriers

- Skin**
 - The largest organ of the human body and forms a tough outer barrier that covers the body and **prevents penetration by microbes**
 - Dry, and has a layer of dead cells which helps to **prevent the growth of pathogens**
- Mucous membranes**
 - Thick layer of mucus which **traps the entering pathogens**
 - The pathogens are held in the mucus until they are **removed by coughing and sneezing**



- The mucus contains an **antibody** that prevents **bacteria** and **viruses** from **attaching to the surface**
- Provides a **moist, nutritious layer** where harmless microbes (biological barrier) live and produce substances that **inhibit the growth and entry of pathogens**
- **Cilia**
 - Tiny 'hairs' that line the respiratory surfaces of the trachea and bronchial tubes
 - Constantly beating in an upwards direction to **move the mucus containing the trapped pathogens toward the throat**, where they are **removed by coughing, sneezing or swallowing**

Chemical barriers

- **Oil and sweat glands on skin**
 - Produce **antibacterial** and **antifungal** substances that **stop the growth** of invading pathogens
 - If the skin is cut, the blood clots to produce a temporary patch (scab) to maintain the barrier until new skin forms
- **Alimentary canal**
 - Pathogens entering with **food or drink**, or swallowed with **mucus** are destroyed by the **stomach acid**
 - The **urinary** and **vaginal** openings and the surface of the skin are also **acidic**; stopping the growth of pathogens
- **Urine**
 - Sterile and slightly acidic, flushing and cleaning the ureters, bladder and urethra; **prevents bacterial growth**
- **Tears**
 - Contains **lysosomes** that destroy the cell walls of some bacteria
 - The surface of the eye is cleaned and pathogens are washed away through blinking of the eye
- **Saliva**
 - Contains **lysosomes** and washes microorganisms from the teeth and the lining of the mouth

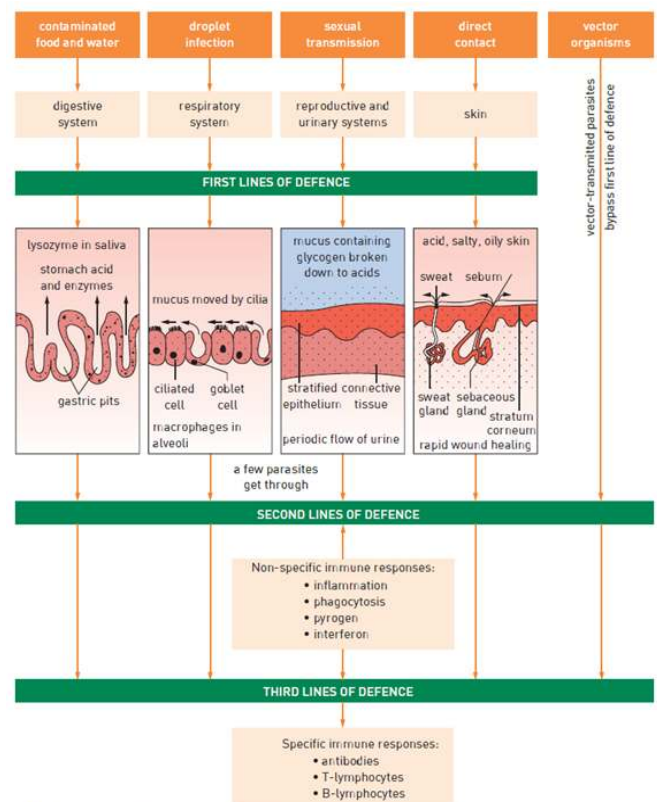
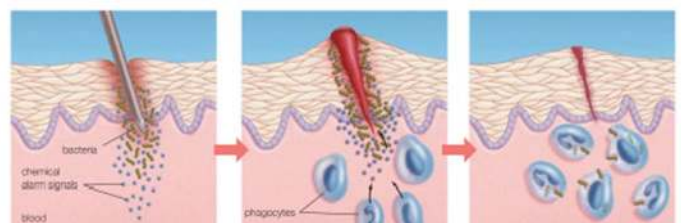


FIGURE 1 The body's defences against disease

Second line of defence- innate/nonspecific immunity

Inflammation response (swelling, redness, fever, pain, heat)

- Allows the body to optimise disease fighting in areas where injury has occurred to body tissue



- Chemicals such as **histamine** are released, causing the **capillaries** to **dilate** so fluid containing **phagocytes** can enter the tissues and **destroy the invaders**
- Plasma also moves into the tissues, carrying more **phagocytes** and causing **swelling in the area of infection**, forcing tissue fluid into the lymph and taking waste and pathogens with it
- The temperature **increases** as cytokines are released, stopping the growth of pathogens
- **Fever** is caused by infection as **cytokines** are released by **phagocytes**
 - **Increases** metabolism and activates heat shock proteins
- Chemicals secreted by mast cells:
 - **Histamine:** causes vasodilation, increased vascular permeability, smooth muscle contraction and itching
 - **Serotonin:** vasoconstriction and increases vascular permeability
 - **Heparin:** cause fever, vasodilation, vasoconstriction and increased vascular permeability
 - **Interferon:** a natural antiviral protein. Taken up by neighbouring cells to start the production of antiviral enzymes

Phagocytosis

- The process where phagocytes **change** their shape to **surround** a foreign bacteria and **engulf** it
- Enzymes are then released to destroy the bacteria
- Main types of phagocytes:
 - *Neutrophils*
 - **Phagocytosis:** Quickly enter tissues and phagocytose pathogens in acute infection
 - **Antimicrobial compounds:** Release hydrogen peroxide, disrupting bacterial and fungal cell membranes killing them
 - Release **cytokines**, attracting other immune cells to **promote inflammation**
 - *Macrophages*
 - **Phagocytosis:** Live longer than other phagocytes, help with fighting **chronic diseases**
 - **Antigenic presentation:** Presents **antigenic fragments** on their surface, then meeting with the **T lymphocytes** and activating them
 - Release **cytokines**
 - *Dendritic cells*
 - Have a role in **phagocytosis** and antigen presentation
 - *Natural killer cells*
 - Constantly patrol the body, and are important in defence against virus-infected and cancerous cells
 - Release **cytotoxic chemicals** (kills cells directly) when they're in close proximity to the target cell
- After the macrophage destroy the foreign particle, parts of the antigen are **displayed** on the surface of the **macrophage**
- Not always successful as the pathogens can sometimes **repel** the **phagocytes** and escape before being completely destroyed

- **Leukocytes** are white blood cells
 - Manufactured in the **bone marrow**

Lymphatic system

- Removes **unwanted toxins, waste** and **excess water** from the body
- 3 main duties:
 - Contains **lymphocytes** (white blood cells) that fight infection
 - Acts as a **one way drainage system**, transporting fluid from body tissues into the blood circulation
 - Removes **waste products** from cells
- The debris move with the tissue fluid into the **lymph vessels**
- Once they reach a **lymph node**, the waste particles are filtered off and any foreign particles are destroyed by **macrophages**
- *Swollen lymph nodes indicate infection*

Apoptosis (cell death)

- A wall of dead cells forms a capsule (granuloma) or cyst
 - Cells inside die, causing the pathogen to be destroyed
 - The debris inside are destroyed by **macrophages** surrounding the walled-off area
 - Granulomas form when a person is suffering tuberculosis or leprosy

Complement proteins

- Specifically **causes death** to **pathogenic cells**
- The **cascade** is composed of many **plasma proteins**, synthesised in the liver. The functions of proteins include:
 - **Triggering** the recruitment of **inflammatory cells**
 - **Selecting pathogens** from destruction by other cells by opsonizing (process where foreign particles are marked for phagocytosis), or coating, the surface of the pathogen
 - **Form holes in the plasma membrane** of the pathogen, resulting in **cytolysis** of the pathogen cell, causing the **death** of the pathogen
 - Remove the “corpse” of neutralised antigen-antibody complexes

Natural killer cells

- **Nonspecific** lymphocytes that can **target** and **destroy** infected body cells or tumour cells
 - Infected cells release chemicals called **interferons**, which function to promote the **activation** of natural killer cell
 - **Induce apoptosis** (cell death) in the infected cell

Fever

- **Macrophages** release chemicals which reset the body's thermostat in the hypothalamus; the **high** temperature can **kill** some pathogens, **increase** the activity of some **lymphocytes** and **phagocytes** and **speed up** the **death** of virus infected cells

Third line of defence- adaptive/acquired/specific immunity

- Antigens infect cells are 'possessed' by phagocytes before **specific** defences are triggered by the presence of **antigens or antigenic fragments** in **plasma membranes** (antigen presentation)
- When a foreign pathogen successfully penetrates the first line of defence (e.g. skin), and survives the **nonspecific** responses of the second line of defence (e.g. inflammation), the pathogen then faces the third line of defence (adaptive immune/specific response)
 - Involves the production of 2 different lymphocytes T and B cells, specific to antigen
 - **Acquired immunity:** memory T and B cells are produced to remember the specific antigen so that they can be destroyed quickly as soon as they enter the body again
 - **Cell mediated** immunity (T cells) and **antibody-mediated** (or humoral) immunity (B cells) are major ways of **destroying** invading **foreign** material

Very important to know the difference between antibodies & antigens for this dot point

	Antibodies	Antigen
Similarities	• Both are involved in the immune response • Both made by proteins	
Differences	• Y shaped protein made by B cells in response to an invading pathogen • Made in response to the antigen entering the body	• Found on the surface of the pathogen (identity marker) • Causes the immune response to occur

1. Antigen presenting cell

- Defence is initiated by the **macrophages** that carry out phagocytosis and dead or dying cells
 - **Dendritic cells** are a specific type of phagocyte that initiates the third line of defence
 - The dendritic cell **engulfs** the **antigen** present on the foreign pathogen through **phagocytosis**. As the foreign material is being destroyed, the antigen moves to the surface of the macrophage, which is then transported to the **lymph nodes**
 - The dendritic cell (or antigen-presenting macrophage) is presented to the **helper T cell** that has a **T cell receptor** that corresponds to the particular antigen
 - The **helper T cell** can be **activated** by a **B cell**
 - When a B cell comes across the antigen that corresponds to its surface antibodies, it **binds** to the antigens, then processing the antigen and attaching to its surface molecules. It is then presented to the **helper T** cells that have **matching T cell receptors**

- Much of the interaction between the two different cells involved in the immune response is regulated by the **secretion of cytokines**.

2. Cell mediated immunity (T cells)

- **Cytokines** are secreted by the **helper T cell** to activate (initiate process of cell replication through mitosis) more of the **same helper T cells and macrophages**
- A specific cytokine (**interleukin**) produced by the **helper T cells**, activate the **production of clone** of the **B cells** that are **specific** to the **antigen**
- **Interleukin** also **activates** the production of **proliferation of clones** of the **cytotoxic T cells** that have the particular antigen receptor on their surface
- The **suppressor T cells** are then responsible for **suppressing** the activity of the **B cells** and **cytotoxic T cells** once the immune response has defeated the infection
- **T cells** are lymphocytes produced in the **bone marrow** and mature in the thymus gland, found in the thoracic (chest) cavity
 - Once they mature, the **T cells** are **released** into the **blood, spleen, tonsils and lymph nodes**
- Each **T helper cell** has a specific receptor protein that can recognise a specific antigen
- When the matching T helper cell comes across the antigen in the body that matches their receptor protein, they are **activated** and **produce** many **clones of cytotoxic (killer) T cells** specific to that antigen
- **T helper cells** are triggered by an antigen presenting dendritic cell
 - They then move to the site of the infection and release chemicals that destroy the infected cells
- **T cells defend the body against cellular diseases like:**
 - Bacteria and viruses that are inside cells, protozoa, fungi, flatworms and roundworms, cancerous cells and transplanted foreign tissue

Type of T Cell	Role in Acquired or Adaptive Immunity (3 rd Line of Defence)
T Helper Cell	Releases interleukins and cytokines that activate the cloning of cytotoxic T cells and B cells and increase macrophage activity after the antigen has been recognised
T Killer Cell (Cytotoxic T cell)	move to the site of infection and release chemicals that destroy infected cells
T Suppressor Cell	suppress the immune response when the infection has been defeated
T Memory cell	remain in the body to respond to future infections by the same antigen

3. Humoral immunity (B cells)

- B cells are lymphocytes are produced in the **bone marrow**

- After they **mature** they are **released** into the **blood, spleen, tonsils and lymph nodes**
- On the **surface** of B cells has different **antibodies** that only respond to **specific** antigens
- B cells only **live for a few days** and if they come cross their antigen they become activated; if not they **die**
- When they become activated they clone themselves and each clone forms plasma cells that produce antibodies specific to the particular antigen
 - *Neutralisation*
 - **Deactivating** a pathogen or toxin by **blocking** its **active site**
 - *Precipitation*
 - Antibodies **bind** to **soluble** antigens causing them to form **insoluble** clumps
 - *Agglutination*
 - Antibodies **bind** to antigens on the **surface** of cells to form **clumps** of cells
 - *Activation of the complement system*
 - Help **disarm** pathogens, **enhancing** phagocytosis, inflammation and pathogen **removal** by cell lysis
- These antibodies then move to the **site of infection** and combine with the antigen to form the **antigen-antibody complex**, deactivating the antigen and the pathogen
- **B memory cells** have a **specific antigenic receptor** located on their surface
 - Therefore when the pathogen enters the body again, the B memory cell is able to **quickly** recognise the antigen on the pathogen due to the receptor on the cell
 - This causes B cells to be quickly **proliferated** to defeat the pathogen

Type of B Cell	Role in Acquired or Adaptive Immunity (3 rd Line of Defence)
B Cell	defends the body against bacteria and viruses that are outside the cell. By producing antibodies that are specific to the antigen , which deactivated the antigen/pathogen
B Memory	designed to live a lot longer and remain in the body, up to many years. They allow for a more faster and stronger response if the pathogen were to re-enter

IQ4: How can the spread of infectious diseases be controlled?

7.4.1 investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease

7.4.1: Limiting Spread

Local, regional and global spread of a named infectious disease (SARS-COV2)

- COVID only kills 3% of people who get it but it spreads at a fast rate

<i>Local</i>	<i>Regional</i>	<i>Global</i>
Covering the mouth and nose when coughing, washing hands with soap and water, use of PPE (e.g. masks, gloves), social distancing. Cheap however requires education, campaigns and widespread cooperation.	Limiting person to person transmission by closing borders and entry screening. Requires extensive resources.	Travel bands and restrictions, screening travellers exiting areas with the disease.
E.g. in COVID: wearing masks, remote learning, quarantine.	E.g. in COVID: each LGA having different policies.	E.g. in COVID: country borders being closed
Public health policy for tracing close contacts, early detection, cleaning contaminated facilities and closing down schools/affected workplaces. Can be costly to businesses/families who may be affected by shutdowns.	Widespread use of antiviral medications, surveillance, quarantining, restrictions on "hotspots", common infection control guidelines.	Surveillance, international networking, liaising with governments of other countries to determine travel guidelines, redistributing antiviral medications to areas of demand, emergency preparedness for new strains.

IQ4: How can the spread of infectious diseases be controlled?

7.4.2 investigate procedures that can be employed to prevent the spread of disease, including but not limited to:

1. hygiene practices
2. quarantine
3. vaccination, including passive and active immunity (done in previous section because it flows nicely from immunity)
4. public health campaigns
5. use of pesticides
6. genetic engineering

7.4.2: Procedures to Prevent the Spread of Disease

Hygiene

- The need for hygiene
 - Food, water and people can hold pathogenic organisms
 - Cholera, dysentery and tapeworms can infect humans who ingest poorly prepared food
 - Dirty water is one of the main causes of disease and death in the world
 - Waterborne pathogens use water as a path to enter a new host
 - Giardia, typhus, botulism, E.coli, salmonella and hepatitis A can be transmitted through poorly sanitised water supplies
- Hygiene practices can range from personal hygiene practices e.g. hand washing after using the bathroom, showing frequently and covering your mouth when coughing, to community hygiene practices e.g. effective sewage systems and garbage removal

Cleanliness in food	
Freezing foods	Pathogens die in cold temperatures • Chicken should be frozen unless used immediately
Dehydrating foods	Limits any impacts of pathogens on food
Washing vegetables with clean water	Removes pesticides and germs from the hands of the pickers
Washing hands before food preparation	Removes bacteria off the surface of your hands
Avoiding the time foods are kept above 5°C and below 60°C	Pathogens will die between the 2 temperatures
Cleanliness in water	
Membrane filtration	Prevents the pathogen from moving with the water (blocks the pathogens path)

Chlorination	Kills pathogens
UV sterilisation	Kills pathogens

Quarantine

- A public health measure based on restricting the movement of people, plants and/or animals through ports of entry to a country, to prevent or limit the spread of pests and contagious diseases
 - Minimise the risk of exotic pests and diseases from entering Australia (protects native flora and fauna, our agricultural industries, our environment and health)
- Types of quarantine
 - Animal quarantine
 - Involves all animals that come into Australia to spend time in quarantine stations to ensure they're free from disease before they're released
 - For pets to enter Australia, they must be quarantined for a number of weeks
 - Plant quarantine
 - Involves examining all plants, parts of plants and plant products (seeds, fruits, cuttings, bulbs and wood) brought into Australia for pests/diseases
 - Many of these items are refused entry, although in some cases, items are allowed after treatment by quarantine
 - Live plants must be kept in quarantine until any diseases that may be present have had time to develop
 - Human quarantine
 - Captains of aircraft and ships are required to notify Australia Quarantine and Inspection Services (AQIS) if any passengers or crew show any symptoms of prohibited diseases (rabies, yellow fever, malaria, SARS or avian influenza)
 - Aircraft are also sprayed with insecticide to kill any pests that have Australia with the aircraft
 - Australian international airports have mosquito-trapping programs to detect any entering the country as they could be acting as vectors
 - Northern Australia quarantine strategy
 - Northern part of Australia is only a short distance from countries that have exotic pests and diseases not present in Australia
 - Established as early warning system
 - System involves the use of 'sentinel' animals such as cattle and pigs that are regularly checked for diseases such as Japanese encephalitis and blue tongue
 - Insect traps are also set up and checked regularly for insects such as screw-worm flies, Asian Honeybees and papaya fruit-flies

Vaccinations

- Vaccines help in protecting those with a **reduced** immune response e.g. elderly, babies
- Main types of vaccines:

- **Whole agent (live attenuated) vaccine:** made of weakened or damaged version of the pathogen
 - Pathogens are completely killed
- **Subunit vaccines:** made of specific antigenic fragments from pathogens e.g. proteins, polysaccharides
 - Triggers an immune response
- **Toxoid:** contain antigenic toxins from pathogens
- **(EXTRA) DNA vaccines (safer- no original pathogen is present):** the genes of the specific antigen is isolated and injected into the body triggering the production of antigens to protect the body
- Induce **long-term immunity** to specific pathogenic infections by triggering the production of memory cells
 - **Vaccine:** weakened or attenuated form of the pathogen that contains antigens but can't trigger a disease
 - Antigenic determinants in a vaccine may be conjugated to an adjuvant (functions to boost the immune response)
 - Injecting a vaccine initiates a **primary immune response**, creating memory cells
 - When exposed to the original pathogen, a more strong **secondary immune response** is triggered by the memory cells
 - **No disease symptoms** develop in the secondary immune response
- The immunity of an individual to a disease depends on **how long** the **memory cells survive** for
 - Individuals may need a **booster shot** to maintain immunity
- *Immune response*
 - First encounter (primary response)
 - Time taken to fight the infection is quite long
 - The **activation** of T and B cells and time to **build up clones** of these cells is **longer**
 - **Cytotoxic T cells** (to kill infected cells) and **B cells** to produce plasma cells that then secrete antibodies and bind with the antigen to neutralise it, takes **longer**
 - Second encounter (secondary response)
 - If the same antigen re-enters the body in the future, the response is much **quicker**
 - After the antigen has been identified, memory cells will activate the production of cytotoxic T and B cells
 - A large number of B cells will form plasma cells, secreting **larger** amounts of antibodies than the primary response
 - **Effect:** destroy invading antigens before their numbers are large enough to cause any symptoms

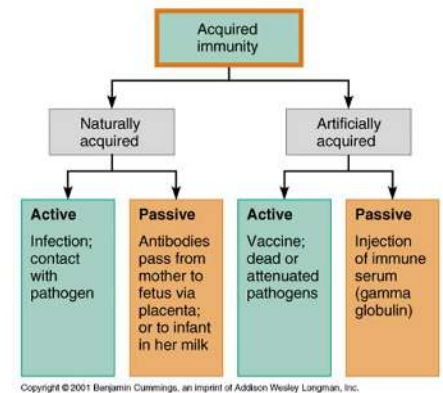
	Primary response	Secondary response
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Similarities	Respond due to its exposure to an antigen	
Differences	14 days (longer response) - Weaker (not as much B and T cells produced) - Immunity drops at a faster rate (protection is lost easily)	9 less days (shorter response) - A lot stronger (more B and T cells produced) - The body is protected a lot longer after secondary exposure

Types of immunity

- Natural active immunity**

- When a person is infected with a pathogen, **specific antibodies** are produced to combat the pathogen
- These antibodies stay in the blood long after the infection has been fought
- If the pathogen re-enters, the antibodies immediately react to attack and destroy it
- Once the body learns to create a specific antibody, it is more likely to be protected from reinfection in the future
- If a person is **infected** but fights off the infection without showing symptoms, the person is said to be **immune** to the disease



- Natural passive immunity**

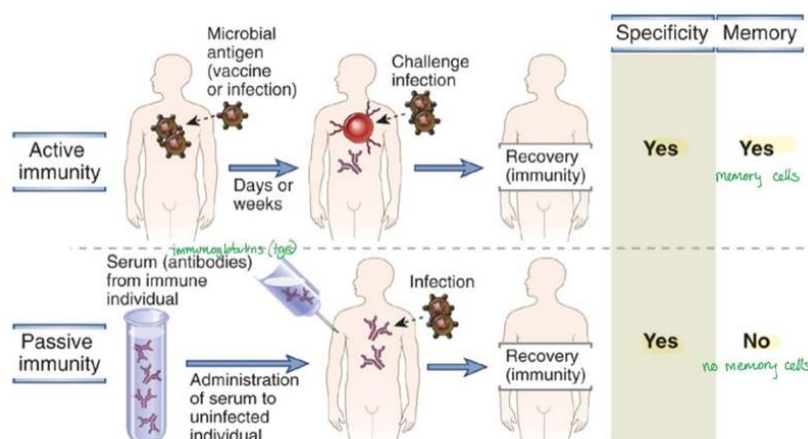
- Occurs during **pregnancy** when antibodies are passed from the **maternal blood** into the **fetal bloodstream**

- Acquired (artificial) active immunity**

- Through **vaccination**, a person is prompted to produce antibodies without the disadvantage of becoming ill, which usually leads to immunity
- Use of vaccinations has resulted in many diseases becoming rare or eradicated

- Acquired (artificial) passive immunity**

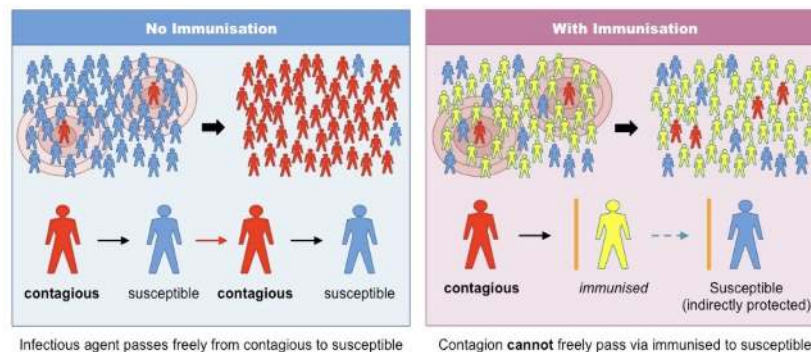
- Normally done through injection and is used in a **recent outbreak** of a particular disease or as an **emergency treatment** for toxicity
- Antibodies can be produced in animals, called “serum therapy”, although there is a high chance of anaphylactic shock because of immunity against animal serum itself



	Active immunity	Passive immunity
Similarities	- Both include antibodies - Both are apart of the acquired immunity	
Differences	Made by our body and produces long lasting T and B memory cells	Not made by the body but antibodies are injected in the body, without producing T and B memory cells

Herd immunity

- When individuals who are not immune to a pathogen are protected from exposure by the large amounts of immune individuals within the community
 - Vaccination grants immunity to vaccinated individuals but also protects non-vaccinated individuals (herd immunity)



Applications of herd immunity

Smallpox	Before Vaccination Program:
Occurrence of disease	The disease killed billions of people across the world ~300-500 million people were killed in the 20th century alone
Spread of disease	Smallpox was spread by the Egyptian prisoners held captive by the Hittites in the Egypt-Hittite war. The disease was further spread by Egyptian merchants who came in contact with the Crusades and the Spanish and Portuguese, spreading the disease to the Americas

Date first vaccine produced/used	(1022 A.D.) Crushed up scabs from those who survived the Smallpox disease (May, 1796) Edward Jenner took matter from the pustules of a milk maid who had freshly contracted Cowpox
Description of vaccination program / when introduced	The 'vaccination' was called Variolation, and was discovered when a Buddhist nun living in O Mei Shan, in the southern province of Sichuan, China, crushed the smallpox scabs and would blow the power into the nose of healthy people after finding that those who survived the disease, never got the disease again Jenner inoculated James Philip with the matter from the pustules, causing him to contract the Cowpox disease. 2 months after he had recovered from the disease, he inoculated Philip once again but with the matter of a fresh pustulate from a fresh smallpox lesion, and no disease developed. This study led to the development of the Smallpox vaccination leading to the eradication of the disease in 1979.
	After Vaccination Program:
Occurrence of disease	Smallpox disease was eradicated in 1980 and became a rare disease to contract
Spread of disease	Transmission through face-to-face contact through droplets from the mouth or nose, usually by coughing
Judgement of effectiveness of vaccination program	Therefore, the smallpox vaccination is highly effective in eradicating the smallpox disease (1980) and building the global immunity to the disease

The eradication of smallpox by vaccination was successful for a number of reasons:

- Smallpox was **easily identifiable** due to overt clinical symptoms, which helped to limit potential transmission
- Transmission only occurred via **direct contact** and there were no animal vectors or reservoirs to sustain the infectious agent
- The infection period was **short lived (3 – 4 weeks)** and the virus was stable and **didn't mutate** into alternate strains
- There was global cooperation and immunity was long-term so **repeated booster shots were unnecessary**

Polio	Before Vaccination Program:
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Occurrence of disease	Polio can come in a few different forms • Non-paralytic polio (infected however no symptoms apart from headaches, vomiting, fatigue, fever, sore throat) • Less than 5% of cases can develop paralytic polio (affects spine and potentially brain stem) Some forms of polio may affect brain stem only, or spine only, or both • Both infected - develop muscle spasms, complete paralysis or one side of body or other
Spread of disease	1894 America - 132 cases were found in Vermont 1916 New York - first large polio epidemic (6000 + people died)
Date first vaccine produced/used	1955 - created by Jonas Salk developed an injectable and inactive polio vaccine first administered in US and Canada
Description of vaccination program / when introduced	Jonas Salk developed an injectable and inactive polio vaccine first administered in US and Canada 1957 - incidents of polio began to fall
	After Vaccination Program:
Occurrence of disease	The amount incidents with polio decreased by a significant amount, not many cases were reported, countries were beginning to see an end to the epidemic Polio may be eradicated worldwide by 2018
Spread of disease	1957 - incidents of polio began to fall to 90% 1988 - US did not see any more epidemics however other countries were still infected by the disease After all the humanitarian efforts, the amount of countries affected (in 2013) stands at 3 (Afghanistan, Nigeria and Pakistan)
Judgement of effectiveness of vaccination program	The vaccination program introduced by Jonas Salk in 1957 had a significant impact and effect on the spread of polio, as it was able to reduce cases by 90% in the first two years it came out. The US started to see a fall in the epidemic, and many incidents were decreasing. By 2013, only 3 countries were affected by the disease due to the significant humanitarian efforts of stopping the spread with the vaccine. Overall, the vaccination program was significantly effective in reducing the spread of polio.

Measles	Before Vaccination Program:
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Occurrence of disease	In the lead up to the measles vaccine it was said that it caused an estimated 2.6 million deaths each year. ● In the U.S nearly all children get measles by the time they are 15 years of age. ● Estimates 3-4 million people in the united states were infected each year More widespread as global exploration increased in the 16th century.
Spread of disease	○ An airborne virus ○ Starts with flu-like symptoms and a raised rash on the face, followed by inflammation ○ In the very young can cause brain damage, hearing loss and even death ○ One infected person will infect, avrg. 12 to 18 other people ○ In 1775, Scottish doctor Francis Home discovered measles was caused by a pathogen, found in the blood of infected patients
Date first vaccine produced/used	○ 1954 Thomas Peebles, cultured the pathogen from an 11-year old school boy (David Edmonton) - allowing doctors to create the first vaccine against measles ○ John Franklin Enders, Peebles' boss developed the measles vaccine from the Edmonton-B strain and is now used as a basis for most live-attenuated vaccines to this day. ○ Enders tested measles vaccine on small groups of children from 1958 to 1960 before beginning trials on thousands of children in New York City and Nigeria ■ In 1961 it was hailed as 100% effective and the first measles vaccine was licensed for public use in 1963
Description of vaccination program / when introduced	○ The first internationally focused measles immunisation program took place in Africa from 1996 ○ By may 1967 The Gambia became the first country in the world in which transmission of the virus was interrupted (stopped) ○ In 1974 measles was among the first diseases targeted by WHO and it established the Expanded Programme on Immunization
	After Vaccination Program:

Occurrence of disease	○ In the EPI vaccination against measles drastically reduced the disease rates in children ■ WHO now recommends vaccination at 9 months for babies in areas where measles are common, and at 12-15 months for those in other areas. ■ Second dose is recommended for all children, essential to protect approx. 15% of children who don't develop protective immunity after their first dose.
Spread of disease	○ Can still be spread through coughing, sneezing, close personal contact or direct contact with infected nasal or throat secretions
Judgement of effectiveness of vaccination program	The measles is still around though with the introduction of vaccines the mortality rate and the amount of infections have decreased drastically.

Public health campaigns

- Government regulations
 - Ensure that standardised procedures are put in place and followed when handling, cooking, and storing food
 - Strict guidelines must be followed in hospitals, surgeries and clinics when sterilising equipment and when health workers move from one patient to another
 - These regulations stop the spread of pathogens and possible diseases e.g. food poisoning
 - Also ensure that garbage disposal is done correctly, drinking water is treated effectively and sewage is removed and treated
 - Prevents the occurrence of a disease in individuals and communities
- The law
 - Requires certain disease to be reported to authorities if detected e.g. COVID
 - Allows for early detection of the diseases in order to create strategies to control the spread of the disease in a community
- Public health programs
 - Encourage regular screening for certain diseases are in place
 - Women are encouraged to regularly check their breasts for presence of any unusual lumps and have regular mammograms between the ages 50 and 65 years
 - Aids in early detection
 - Regular 'pap' tests (every 2 years) for the early detection of cervical cancer is recommended

- Men over the age of 50 are encouraged to take a PSA blood test every 2-3 years to aid in the early detection of prostate cancer
- These screening programs help in early detection of these diseases and increases the chance of survival for sufferers and help to control these diseases
- Child immunisation programs are put in place for diphtheria, tetanus whooping cough, measles, mumps and rubella (prevents the occurrence of these diseases)
- Public education programs
 - Aimed to increase public awareness of lifestyle-related risk factors
 - **Aim:** to change the lifestyle of individuals to prevent development of disease
 - E.g. 'Quit' (graphic images of people developing lung cancer on cigarette packets are designed to let the individual aware of the consequences of smoking)
 - E.g. slip, slop, slap

Use of pesticides

- **Pesticides:** chemicals that are used to kill the pests of plants and animals, pathogens and vectors that transmit pathogens from one organism to another
 - If pests and vectors are killed through pesticides, the occurrence of disease will be prevented and the spread of disease in a population will be controlled
 - Types of pesticides:
 1. Insecticides (kill insects)
 - *DDT*
 - Used widely in WWII to kill lice that transmitted pathogens that caused Typhus (controlled the spread)
 - Also used to kill the Anopheles mosquito (carries the plasmodium that causes malaria)
 - Effectiveness of DDT was reduced due to natural selection
 - Variation → resistant mosquitoes and nonresistant mosquitoes
 - Overpopulation → more mosquitoes are born than can survive
 - Selection pressure → DDT
 - Death → the less resistant die before they are able to reproduce
 - Survival of the fittest → the slightly resistant mosquitoes will survive to reproduce
 - Reproduction and passing on the favourable traits → the resistant genes are passed onto the next generation through reproduction in the population
 - Although DDT is useful in killing insects, it has a harmful effect on the environment
 - **Other insecticides:** pyrethrum (less harmful to the environment and more effective in control)

2. Fungicides (kill fungal pathogens)
3. Herbicides (kill weeds)

Genetic engineering

- **Genetic engineering:** a process that involves altering the genetic composition of an organism
 - By altering the genetic makeup of organisms, there is a chance to make them resistant to diseases
 - **Transgenic species:** organisms with genes from other organisms are inserted into their own genetic material
- Genetically modified pea:
 - Resistant to the pea weevil (pest)
 - No chemical pesticides are to be used on these genetically modified peas

IQ4: How can the spread of infectious diseases be controlled?

7.4.3 investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example:

- a. antivirals
- b. antibiotics

7.4.3: Pharmaceuticals

Antivirals

- Only affected on viruses
- Interfere with an important enzyme of the influenza virus (neuraminidase)
- Controls viral infections
- Inhibit their development inside infected cells
- Slow down progress of the disease to allow the body's natural defences to take over
- Usually used by individuals who have a higher risk of complications e.g. those over the age of 65
- How they work:
 1. The drugs keep the virus from escaping from one cell to infect the neighbouring cells
 2. They are the most effective within a day or two of the onset of symptoms
 3. Preventing a virus from entering a cell by binding to a cell receptors that allow the virus to enter
 4. They can inhibit enzymes that aid in the replication of the virus
 5. They block transcription and translation of viral proteins
 6. They prevent the virus from leaving the ce

Positives

Negatives

- Designed to target specific viral pathogens
- They can significantly improve the health outcomes and life expectancy of infected individuals → i.e. HIV used to be a death sentence, antivirals allowed for improved life outcomes

Antibiotics

- Antibiotics are drugs (natural or synthetic) used to fight bacterial infections. They are one of the most frequently prescribed medications in medicine
- Full dosage must be taken to fully eradicate the bacteria from the body
- Chemicals that are capable of destroying or inhibiting growth of bacteria that cause disease
- They target bacteria without destroying the host and are **not at all effective against viruses**.
- Alexander Fleming discovered the first antibiotic in 1928
 - Became available for medical use in 1941
- Types of antibiotics
 - Broad spectrum
 - Acts on a wide range of bacteria
 - Use when the identity of the bacteria causing the infection is not known
 - Narrow spectrum
 - **Acts on only one or two types of bacteria**
- Bacteriostatic antibiotics stop bacteria from multiplying by interfering with DNA replication, protein production or other cellular metabolic processes, e.g. amoxicillin
- **Antibiotic resistance** is being further accelerated by a number of common practices.
 - These include:
 1. The **overuse of antibiotics** for treatment of many diseases and not just bacterial diseases. Examples are the prescribing of antibiotics for the treatment of coughs, colds and 'the flu'. These are diseases caused by viruses, which are not affected by antibiotics. This practice just gives the bacteria more chances to build up populations of resistant strains.
 2. A very common practice is to only take the antibiotics provided until the symptoms of the particular disease disappear and **not finish the whole course**. This is also a dangerous practice as not all the bacteria present may be killed before the end of the course of tablets. This allows yet another chance for the bacteria that are more resistant to survive, reproduce and develop populations of resistant strains.
 3. Food-producing animals such as **chickens and pigs are fed antibiotics** as part of their diet to prevent infections and to promote growth. This widespread use of antibiotics further promotes the formation of antibiotic resistant populations of bacteria.
 4. The use of cleaning products that contain **antibacterial ingredients** are encouraging the development of antibiotic-resistant strains of bacteria. The quantities of these products used are not enough to kill all of the bacteria, so those that are more resistant will survive, reproduce and form populations that are resistant to antibiotics.

Positives	Negatives
• The discovery of penicillin rates as one of humanity's greatest discoveries. Estimates are that 200 million lives have been directly saved from antibiotics. • Due to different cell structures, e.g. cell walls, bacteria are more easily targeted than viruses.	• Antibiotics only work on bacterial infections. • Misuse and overuse have led to multiple examples of antibiotic resistance. • People can be allergic to antibiotics

IQ4: How can the spread of infectious diseases be controlled?

7.4.4 investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic

7.4.4: Environmental Management and Quarantine to Control a Epidemic or Pandemic

- seeks to **change the environment** in order to prevent **OR** minimise **vector propagation** and human contact with the vector-pathogen by destroying, altering, removing or recycling non-essential containers that provide egg/ larval/ pupal habitats"

3 main types of environmental management created by WHO to control dengue fever (similar for malaria with a vector):

1. **"Environmental modification – Long-lasting physical transformations to reduce vector mosquito larval habitats.** For example:
 - Installation of a reliable piped water supply to communities, including household connections.
2. **Environmental manipulation – Temporary changes to vector habitats involving the management of "essential" containers,** such as:
 - Frequent emptying and cleaning by scrubbing of water-storage vessels, flower vases and desert room coolers.
 - Cleaning of gutters.
 - Sheltering stored tyres from rainfall.
 - Recycling or proper disposal of discarded containers and tyres.
 - Management of plants close to homes that collect water in the leaf axils.
3. **Changes to human habitation or behaviour – Actions to reduce human coming into contact with vectors.** These include:
 - Installing mosquito screening on windows, doors and other entry points.
 - Using mosquito nets while sleeping during daytime."

Main vectors that are responsible for epidemics

- Mosquitoes
- Non-biting flies
- Lice
- Mites
- Rodents (Rats)

Evaluation

- What is evaluated:
 - The environmental management methods and strategies used to control vector propagation and human contact with vectors carrying the pathogens
 - The consideration place in the balance between effectiveness as well as economically acceptability in terms of cost

IQ4: How can the spread of infectious diseases be controlled?

7.4.5 interpret data relating to the incidence and prevalence of infectious disease in populations, for example:

1. mobility of individuals and the portion that are immune or immunised
2. Malaria or Dengue Fever in South East Asia

7.4.5: Effect of Mobility and immunised on these

Incidence and prevalence

- **Incidence:** emerging cases
 - Measure of disease that allows us to determine a person's probability of being diagnosed with a disease during a given period of time
$$\frac{\text{number of new cases during a specified time}}{\text{size of population at start of monitoring period}} \times 100$$
- **Prevalence:** existing cases
 - Measure of disease that allows us to determine a person's likelihood of having a disease
$$\frac{\text{all new and previous cases during a time period}}{\text{population during the time period}} \times 100$$
- Determines the pool of pathogens already present in a population
- **Mobility of the population:** determines how easy it is for the pathogen to spread
- **Percentage of the population that is immunised against an infectious disease:** determines what proportion of the population are vulnerable and may act as carriers for the pathogen

Malaria

	Identify or Describe
--	----------------------

Pathogen	Caused by a parasite - transmits ● P. Falciparum ○ The deadliest malaria parasite ● P.vivax ○ Dominant malaria parasite
Disease Name	Malaria
Mode of transmission	Vector - Female Mosquito
Adaptations of the pathogen to aid transmission (look at the life Cycle)	● The female mosquito (Anopheles) ● The Mosquito bites the individual and the pathogen is then transfused into the bloodstream of the individual
Mobility	It is spread through a vector, therefore depending on the amount of vectors there are can affect how quickly the disease can spread. ● Though it doesn't spread through person to person contact
Symptoms	● Fever ● Headache and chills ● Start within 10-15 days of getting bitten ● Infants, children under 5 years, pregnant women, travellers and people with HIV or AIDS are at higher risk More severe symptoms ● Extreme tiredness and fatigue ● Impaired consciousness ● Multiple convulsions ● Difficulty breathing ● Dark or bloody urine ● Jaundice (yellowing of eyes and skin) ● Abnormal bleeding

Treatment	It is important for those experiencing signs of malaria to be tested immediately in order to be treated properly. Most common medicines for malaria: • Artemisinin-based combination therapy medicines like artemether-lumefantrine are usually the most effective medicines. • Chloroquine is recommended for treatment of infection with the <i>P. vivax</i> parasite only in places where it is still sensitive to this medicine. • Primaquine should be added to the main treatment to prevent relapses of infection with the <i>P. vivax</i> and <i>P. ovale</i> parasites.
Prevention Some measures can be applied before the occurrence of a communicable disease to protect a community from getting it, and to reduce the number of cases locally in the future	Simple controls • Use mosquito nets when sleeping in places where malaria is present • Use mosquito repellents (containing DEET, IR3535 or Icaridin) after dusk • Use coils and vaporizers. • Wear protective clothing. • Use window screens. Other controls • Vector Controls • Chemoprophylaxis • Preventative chemotherapies • Vaccine
Control Once a communicable disease occurs and is identified in an individual, measures can be applied to reduce the severity of the disease in that person, and to prevent transmission of the infectious agent to other members of the community.	

Dengue Fever

	Identify or Describe
Pathogen	Dengue- part of the Flaviviridae family

Disease Name	Dengue fever (or Breakbone fever)
Mode of transmission	vector • Spread through a female mosquitos (Aedes aegypti) Airborne disease
Adaptations of the pathogen to aid transmission	• The pathogen is able to spread through a vector (mosquito) without affecting the vector • The pathogen is able to phagocytosed by macrophages when it enters into the host, changing the pH of the macrophage and forming a pore ◦ The formation of the pore allows the virus to enter into the host's cytoplasm ◦ The macrophage unintentionally duplicates the virus (budding) ◦ The antibodies aren't able to find as well to the antigens of the virus
Symptoms	• Fever • Rashes • Painful joint • Muscle aches • Vomiting • Haemorrhagic fever • Dengue shock fever
Mobility	Spreads through a vector • A single female mosquito can feed off 3-4 hosts in their lifetime --> disease can be easily spread
Treatment	• A vaccine must be effective against all four stereotypes simultaneously
Prevention Some measures can be applied before the occurrence of a communicable disease to protect a community from getting it, and to reduce the number of cases locally in the future	• Know your surroundings • Cover up • Insect repellent • Dump standing water

Control

Once a communicable disease occurs and is identified in an individual, measures can be applied to reduce the severity of the disease in that person, and to prevent transmission of the infectious agent to other members of the community.

IQ4: How can the spread of infectious diseases be controlled?

7.4.6 evaluate historical, culturally diverse and current strategies to predict and control the spread of disease

7.4.6: Historical, culturally diverse and current strategies

Mediaeval times

- Lack of proper diagnosis methods and techniques limited knowledge about the causes of disease leaving small chances of preventing it
- Most treatment methods included:
 - Herbal remedies
 - Sometimes astrology
- Some religions taught that illness was punishment for evil
- They were suspicious of anything scientific, as it challenged the authority of a religion
- **Mediaeval technique:** drilling of the skull to 'release' pressure in the head to stop headaches

Miasma theory was the prevailing 19th C dogma of the public and medical community alike.

- The theory of indirect and airborne transmission held that diseases like cholera were caused by the smell of the bad air, *miasmata*, a poisonous vapour with suspended particles of decaying matter and a foul smell.
- At the time miasma theory made sense to most as disease and epidemics were concentrated in poor, filthy and foul-smelling city neighbourhoods.
- Physicians tended to believe that cholera was a condition of the blood. (hence blood letting)
 - **Digestive system-** bacteria in the digestive system
- Some believed that cholera was related to altitude.
 - Cholera is pathogen caused
- Most believed it was not contagious.
 - Cholera is highly contagious

Some doctors did not even diagnose a patient with a fear of catching a disease himself. Thus, “doctors” had a very poor status during early times.

The Early 18th and 19th centuries:

- Though many doctors still based their treatment on the four humours, the works of Galen and other ancient writers gradually became less important.

Strategy	Description	Evaluation
Historical	In the Middle Ages infectious diseases and parasites, e.g. head lice, worms were common. Poor hygiene meant cuts on the skin easily became infected and developed tetanus or gangrene. Leprosy was common and leper colonies (or lazer houses) were widespread. It was believed the body had four 'humours' – blood (sanguine), yellow bile (choleric), black bile (melancholic) and phlegm (phlegmatic). These humours needed to be balanced, e.g. by bleeding with or without leeches and causing purging or vomiting. As many people believed that disease, e.g. Black Death was caused by horrible smells they believed that by removing the bad smell, e.g. by using the pleasant scent from a pomander they could repel the disease in the air. The pomander was a hollow container filled with spiced wax which people wore around their necks, at their waist or around their wrists. Malaria was named from the medieval Italian for 'bad air' (had formerly been called ague or marsh fever).	The isolation of lepers into leper colonies is a form of quarantine which is a suitable strategy to control the spread of the disease. However, the lack of proper hygiene and sanitation meant there were many possible routes for the transmission of pathogens, e.g. the Black Death killed around 40% of the population of Europe. Thus these historical strategies were ineffective.
Another culture	Traditional Chinese medicine (TCM) involves many strategies, e.g. acupuncture, herbal medicine, maintaining a specific diet, massage and tai chi and qigong practices. Classical Chinese medical books provide many herbal formulas to treat and control many types of illnesses. 'Warm' diseases which went from one person in a household to another then from street to street were treated with bitter and cold herb formulas and 'cold damage' diseases were treated with heat. The ancient Chinese doctors realised that the cause of these warm diseases had no smell, shape or shadow. Modern analyses of these formulas show some have notable antibiotic properties. Traditional Chinese herbs include ginseng, honeysuckle and <i>Forsythia suspensa</i> , <i>Gentiana</i> , <i>Astragalus propinquus</i> . The ancient Chinese also advocated bathing, oral hygiene, washing hands before eating using steam heated cloths and advised not to drink contaminated water.	By careful observation the ancient Chinese doctors identified strategies that would help treat and control disease, e.g. using particular herbal formulas for particular diseases and understood there were different modes of transmission, e.g. 'warm' diseases spread from person to person within a household. Combined with better hygiene practices these strategies helped control the spread of disease.
Current	Many strategies are currently being used to predict and control the spread of disease, e.g. immunisation programs, public health campaigns, provision of fresh water, good sewage treatment and disposal, garbage disposal, applying good hygiene practices in all situations, quarantine and vector control measures. The Australian government has a National Framework for Communicable Disease Control that was developed with the assistance of the states and territories and provides strategies for communicable disease prevention, detection and response with improved organisation and delivery of communicable disease control. If a person shows symptoms of a notifiable disease the medical practitioner must register the case with the Notifiable Diseases Surveillance System (NNDSS) Once diagnosed and confirmed the case is investigated and the situation contained as much as possible to prevent further transmission.	Modern strategies cover a diverse range of activities that help predict and control the spread of disease. Thus if an outbreak does occur these strategies aid a fast and efficient response that helps contain the outbreak and prevent further illnesses and loss of life.

IQ4: How can the spread of infectious diseases be controlled?

7.4.7 investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example:

1. bush medicine
2. smoke bush in Western Australia

7.4.7: Aboriginal protocols and intellectual property

- Indigenous knowledge has been acquired over tens of thousands of years
 - Includes:
 - Use of flora and fauna for food and medicine
 - Seasons
 - Climate
 - Water and fire management

Bush medicine

- Bioactive compounds that are useful in managing certain infectious diseases
- The alkaloid compounds from the Modern Bay Chestnut is able to manage HIV/AIDS
- Specialised preparation techniques are used to minimise harmful effects

Aboriginal protocols in the development of medicines

- Substances from plants e.g. tannins, mucilage, oils, latex and alkaloids were used for medicine
- The plant material was crushed and used as a poultice or infused with water to drink
 - Increased fat solubility of the plant substance and increased absorption rates into the tissues

Smoke bush

- Grown on the coastal Western Australia, and is used for its healing properties held in the plant
- Contains Conocurovone, which has the potential to destroy low concentrations of HIV
- The Indigenous although first finding the healing properties of smoke bush, were not credited for their knowledge by the USA's National Cancer Institute
 - Poses a large ethical issue as they aren't being credited as being the original founders of the healing properties of smoke bush

Mod 8: Non-Infectious Disease

IQ1: How is an organism's internal environment maintained in response to a changing external environment?

8.1.1 construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited

- a) temperature
- b) Glucose

8.1.1: Homeostasis

Homeostasis: the maintenance of a stable, or constant, internal environment, despite external and internal environmental changes

- **Internal environment:** the fluid in which cells live
- **Set point:** the optimum point
- Involves maintaining blood pH, water/salt balance, glucose, pressure, CO₂ and O₂ levels, maintain body temperature at a set point or between limits
 - Maintained even in *changes in the external environment*

Importance of homeostasis

- Keeps the body's systems working under **optimal conditions**
- Changes to internal conditions can change enzyme reactions in the body
 - A decrease in enzyme activity can lead to a decrease in essential compounds

Detecting changes from a stable state

- Internal and external stimuli help the body *monitor changes to the body's internal and external environment*
 - **External changes may include:** light or oxygen levels
 - **Internal stimuli may include:** core body temperature (physical stimuli) or CO₂ levels (chemical stimuli) in the blood
 - External receptors:
 - Visceral receptors
 - Mechanoreceptors (skin and ear)
 - Chemoreceptors (tongue and nose)
 - Photoreceptors (eye)

2 ways that the body responds to external/internal stimuli

- Through the nervous system
- **Endocrine system:** hormones transported in the blood
 - Thyroid glands
 - Sweat glands
 - Any gland that produces a hormone

Counteracting changes from the stable state

- Responses to changes in the body's external environment are called **homeostatic responses**

- Occur *quickly* and follow the **stimulus response pathway**
 - Includes stimulus detection by a *receptor*, it then sends a signal to the *control centre*, then an *effector* carries out a desired response to **counteract** the change

How does homeostasis work

- Homeostatic regulation works by the **nervous** and **endocrine** system working closely together
 - The nervous system provides a rapid response to stimuli
- The endocrine system uses **hormones** to maintain the activity of organs and tissues
- The link between the nervous and endocrine systems is the **hypothalamus**, located in the brain
- **Negative feedback** in the process by which the body senses a change from the set point of the internal condition and activated processes to negate that change

Maintaining a stable state

- The nervous and endocrine system work together to ensure many aspects of your internal environment remain within the narrow range required to maintain your health
 - E.g. maintaining the temperature of 37°C or close to 37°C in the body
- Maintaining a consistent blood volume, blood pressure and blood solute concentrations is done by balancing fluid and ion levels
- E.g. excretion of CO₂ and nitrogenous wastes (urea through the kidney) to help ensure that the blood pH stays close to 7.4
 - Hormones are also used to help regulate blood glucose levels
- *Maintaining a constant internal environment through homeostasis ensures...*
 1. The efficient **transport** of substances around the body and **diffusion** of substances in and out of cells (glucose homeostasis)
 2. It also ensures that **enzymes** can function correctly, which in turn ensures that biological reactions can occur quickly enough to sustain life (temperature homeostasis)

Homeostasis- maintaining a stable internal environment

- Conditions in the body must be maintained at a **constant level** (homeostasis) in order to achieve optimal *metabolic efficiency*; **enzymes**, responsible for controlling all metabolic activities within the cell (enzymes are metabolic catalysts that lower the E_A), will only operate within a **limited range of temperature and acidity** (pH)
- Solute concentration must in turn be kept constant for these reactions to occur properly

Stimulus response

- *Flow chart*
 - In a '**feedback loop**' the receptors detect the response and send messages back to the *control centre* to stop further adjustment
 - Often referred to as '**negative feedback**' → results in a negative response by the *effector*

Term	Definition
Stimulus	(1) An environmental (internal or external) change that an organism can detect and respond to. (2) A chemical or physical change that activates a receptor molecule in a cell and generates a response , such as the production of a hormone.
Set point	This is the desired level or norm, at which the system operates.
Receptor	It detects any deviation from the set point (stable state) and informs the controller Eg: retina
Control Centre	Generally the hypothalamus detects any change from the stable state and sends out messages to counteract the change.
Messenger/ Impulse	chemical (endocrine system with hormones in the blood) and electrical signals (nervous system with neurons) which are being transferred through cells.
Effector	It brings about the changes needed to return the system to the set point. This return to normality creates a feedback loop. The glands that release a hormone or the muscles that carry out the response.
Response	Is the action of the muscle or gland (hormone) to counteract the initial stimulus and return the body to the initial stable state.
Neuron	A nerve cell, which is a key part of the nervous system in animals
Sensory Neuron	This is the neuron that detects the stimulus and carries the message to the control centre.
Interneuron	A neuron that transmits information from a sensory neuron to a motor neuron. Most of the nerve cells in the brain and spinal cord are interneurons.
Motor Neuron	This is the neuron that takes the message (impulse) from the control centre to the effector to carry out the response
Feedback loop	Once the response has been carried out, a message is sent to the brain to inform it that the system has returned to set point equilibrium or a larger response is needed to counteract the stimulus.

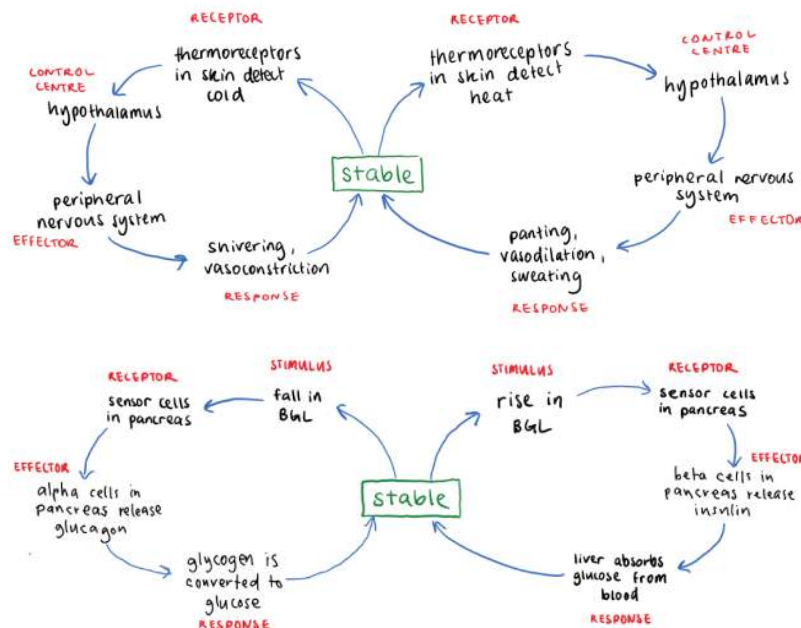
Role of the nervous system

- The nervous system consists of the **CNS** (brain and spinal cord) and the **peripheral nervous system** (motor and sensory nerves) and **receptor cells** (e.g. photoreceptors in the eye, chemoreceptor, mechanoreceptor)
- The **CNS** acts as the **control centre** to access variations from optimal levels and to send signals to effector organs in response

- The **hypothalamus** is vital in this role in detecting and responding to internal changes by measuring changes in the blood temperature and chemical levels
- The **PNS** acts the carrier of signals and in some cases will detect changes in stimuli

Negative and positive feedback

- **Positive feedback system:** the response of the system is to **increase or reinforce** the initial change and cause a *greater deviation from the set value*
 - E.g. the entry of Na^+ into the neuron
- **Negative feedback system:** the response of the system is to **counteract** the change and *return it to the set value*
 - The most system used
 - E.g. maintenance of blood glucose levels, water levels (osmoregulation) and body temperature
 - E.g. of negative feedback loop diagrams for temperature and glucose maintenance



IQ1: How is an organism's internal environment maintained in response to a changing external environment?

8.1.2 investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including:

- trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis
- internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways
- mechanisms in plants that allow water balance to be maintained

8.1.2: Mechanisms for Homeostasis

Neural pathway

- **Nerve impulses:** electrical signals which are produced by the plasma membrane (cell membrane) of the nerve cell
- The human body is electrically neutral, having almost the same number of positive charges as negative charges
- The side of the plasma membrane exposed to the cytoplasm is negative while the side exposed to the extracellular fluid (outside the cell) is positive
- The difference on either side of the membrane results in a cellular voltage (resting membrane potential), measures ~70 millivolts
 - Indicates that the inside of the membrane is negative (membrane is polarised)
- The cytoplasm of a cell and the extracellular fluid are both electrolyte solutions (differ from each other)
 - The cell membranes are impermeable to proteins and most organic phosphates (kept in the cytoplasm)
 - The membranes are selectively permeable to Na^+ , K^+ , and Cl^- because of ion channels
 - When ion channel pores are open, ions can move from one side of the membrane to the other
 - Each channel allows only a specific type of ion to diffuse through it
- *Depolarisation and action potential*
 - Changes in the environment of a neuron can affect the permeability of the plasma membrane to ions and therefore change the membrane's potential
 - Any environmental factor which causes such a change is called a stimulus and neurons are highly reactive to stimuli
 - A cell's membrane potential can change in response to appropriate stimulation
 - A positive shift in membrane potential, from -70mV to -40mV (depolarisation)
 - If the depolarisation is strong enough, this flow of ions causes the neuron to generate a nerve impulse or action potential
- *The threshold*
 - The amount of positive change in membrane potential which is required before an action potential is produced
 - The depolarisation must reach a threshold, which is at least 15 mV more positive than the resting potential of -70mV
 - No action potential is produced if the depolarisation is below this level
 - One of the reasons why not all stimuli generate an action potential
- *The synapse*
 - Each stimulus produces either a full action potential or none at all
 - Each action potential is a separate event
 - A cell cannot produce another action potential until the previous one is complete

- The action potentials are transmitted from neuron to neuron across small gaps (synapses)- the junctions between the end of one axon and the dendrites of the cell body of a receiving neuron
 - The movement is in one direction only
- At the synapse, chemicals (neurotransmitters) diffuse across the gap from one neuron to the membrane of the receiving neuron, causing an electrical impulse

Endocrine system

- The glands responsible for producing hormones that help internal communication and maintaining homeostasis
- **Hormones:** chemical signals that travel in blood and body fluids to act on specific target cells to cause specific responses
 - The targeted cells have receptors for the particular hormone (ensures the targeted cells are the only ones affected)

Gland	Hormone	Function	Regulated by
Pancreas	Insulin	Change glucose to glycogen to increase blood glucose levels	High [glucose] in blood
	Glucagon	Convert glycogen to glucose to increase blood glucose levels	Low [glucose] in blood
Adrenal glands (adrenal medulla)	Adrenaline	● Raise blood glucose levels ● Increase metabolic activities ● Rapid breathing and heartbeat	Nervous system
Testes	Testosterone	Support sperm production and development of male secondary sex characteristics	FSH & LH
Ovaries	Oestrogen	Stimulates uterus lining formation and promoter female secondary sex characteristics	FSH & LH
	Progesterone	Maintains uterine lining ready for implantation and pregnancy	FSH & LH
Pituitary (posterior)	FSH	In females, the combined action of FSH and LH stimulates growth of ovarian follicles	
Pituitary (posterior)	LH		
Pituitary (posterior)	Oxytocin	Contraction of the womb (uterus) during childbirth and lactation	

Hypothalamus

- Contains temperature sensors and control mechanisms
- Located in the central part of the brain (between the thalamus and pituitary gland)
- Responsible for the coordination of the nervous system and the endocrine system's response in maintaining homeostasis
- Once internal and external stimulus are detected by the receptor cells, a message is sent via neurons to the control centre (through the hypothalamus)
- Directly responsible for a large proportion of the body's regulatory response
- Controls an endocrine gland to release its hormone, which then affects and controls the response of the regulatory organ
 - Causes the change that maintains homeostasis

Endotherm- homeostasis

- **Ectotherm:** an animal that derives body temperature from an external source
- **Endotherm:** an animal that derives body temperature from internal heat production
- Heat is produced by metabolism (chemical reaction in the body that produce heat as a side product)
- **Adaptation:** structural, behavioural or physiological functions that make an organism well suited to its environment and which therefore increases its chances of survival and reproduction
 - **Physiological:** sweat, vasodilation/vasoconstriction
 - **Structural:** fur
 - **Behavioural:** shuttling in and out of the sun

Advantages of being an ectotherm	Disadvantages of being an ectotherm
Less food and energy required- don't have to maintain temperature	Less capable of regulating their internal body temperature in changing environmental conditions
Do not require as much food	Less capable of movement when cold- can't eat
Less affected by small environmental temperature variation	Cannot survive in extreme temperature changes

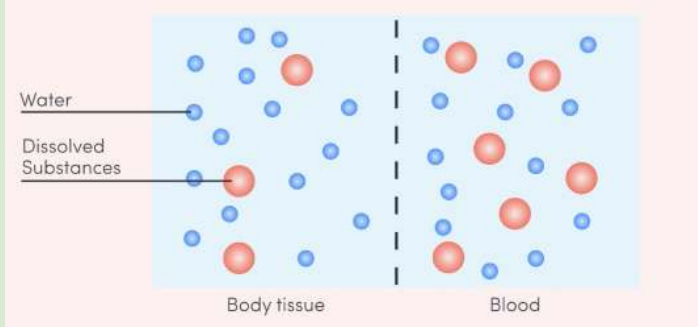
Thermoregulation in endotherms

- Endotherms generate most of their heat through metabolism
- As they are warmer than their environment they tend to lose body heat to their surroundings
- Adaptations for heat conservation include:
 - **Insulation:** insulating layers of fat, fur or feathers to prevent heat loss
 - **Metabolism:** metabolic rate can be varied to generate more or less heat

- Endotherms also use their skin to vary their heat loss and maintain a constant core body temperature
- Endotherms use similar behavioural adaptations of thermoregulation to ectotherms
 - The main difference between the two is that ectotherms rely on these behavioural mechanisms to maintain their constant body temperature, unlike endotherms who have a much wider range of physiological mechanisms to adjust their temperature
- *Mechanisms for thermoregulation for endotherms*
 - Insulation (helps keep the heat inside the body) examples include:
 - Fur
 - Blubber
 - Feathers
 - Evaporation, cooling
 - Shivering
 - Circulation adaptations
 - Vasodilation
 - Vasoconstriction
 - Behavioural responses
 -

Plant- homeostasis

Word	Definition
Stomata	● Pores found on the underside of a leaf which allow for gas exchange to occur between a plant's tissues and the environment ● Stomata also allow transpiration (water vapour escapes) to occur ○ Without a steady water supply, transpiration can lead to water loss ○ Without water, plants cells lose their turgor (pressure exerted by fluid in a cell that presses the cell membrane against the cell wall), wilt and eventually die
Mesophytes	Land plants which are adapted to neither particularly dry nor particularly wet environments ● E.g. roses, corn plants
Xerophytes	Plants that are specifically adapted to survive in arid regions ● No water found in surrounding environment ● E.g. eucalyptus tree, banksia, spinifix grass
Hydrophytes	Plants adapted to wet environments ● E.g. water lily

Osmosis	The movement of water from high to low concentration, across a semipermeable membrane  The diagram illustrates osmosis across a semipermeable membrane, represented by a vertical dashed line. On the left side, labeled 'Body tissue', there are many small blue circles representing water molecules and a few large red circles representing dissolved substances. On the right side, labeled 'Blood', there are fewer blue circles and more large red circles. Labels 'Water' and 'Dissolved Substances' point to the respective molecules on the left. The background is light blue on the left and light pink on the right.
Halophytes	Plants that can survive in salty/saline environments • E.g. estuaries, mangroves

Osmoregulation in plants: Xerophytes, Mesophytes and Hydrophytes

- **Dry environments**

- *Adaptations*

- Stomata can close themselves to minimise water loss
 - Achieved by decreasing the amount of water in the guard cells so that they close the gap
 - Controlled by abscisic acid
 - Plants will close their stomata when it is hottest (e.g. midday) and leave them open at all other times
 - Allows the leaf to maximise photosynthesis whilst minimising water loss
 - Waxy cuticle (lines the stems and leaves)
 - Provides a waterproof barrier, which helps to reduce water loss
 - Thicker cuticles = better at reducing water loss
 - Vacuoles
 - Organelles which store large amounts of water

- *Xerophytes (adaptations)*

- Adapted to arid regions
 - Reduced stomata
 - Fewer stomata = less potential for water vapour to evaporate out of the plant
 - Drooping leaves
 - Sunken stomata
 - Hairy leaves
 - All adaptations trap air around the stomata
 - Air becomes saturated with water vapour, creating a humid microclimate where water isn't pulled out of the plant as easily

- Both mesophytes and xerophytes are adapted to dry and moderately wet environments, so excess water isn't typically an issue

Osmoregulation in plants: Halophytes

- **Wet environments**

- *Hydrophytes (adaptations)*
 - Increased stomata
 - Many on upper surface help to optimise gas exchange
 - Large and flat leaves
 - Large SA:V further promotes water loss

- **Salty environments**

- Osmoregulation in plants
 - Regulating amounts of water and salt

- **Mechanisms used by halophytes**

- *Salt exclusion*
 - Special tissues in the roots and lower stem stop salt from entering the plant but allow water uptake
 - E.g. red and grey mangroves
- *Salt excretion*
 - Plant actively concentrates salt
 - Salt is then excreted through special glands on the leaves
 - Salt crystallises and is blown or washed away
 - E.g. grey and river mangroves
- *Salt accumulation*
 - Plant deposits salt in older tissues e.g. leaves
 - Leaves are shed by the plant
 - E.g. milky mangroves

IQ2: Do non-infectious diseases cause more deaths than infectious diseases?

8.2.1 investigate the causes and effects of non-infectious diseases in humans, including but not limited to:

1. genetic diseases
2. diseases caused by environmental exposure
3. nutritional diseases
4. cancer

8.2.2 collect and represent data to show the incidence, prevalence and mortality rates of non-infectious diseases, eg

1. nutritional diseases
2. diseases caused by environmental exposure

IQ3: Why are epidemiological studies used?

8.3.1 analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:

- a) nutritional diseases
- b) diseases caused by environmental exposure

8.3.2 investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above

8.2.1-2, 8.3.1-2: Causes and Effects of diseases

Non-infectious disease	Definition	Example	Cause	Effects
1.Genetic disease	Something that is inherited	Trisomy 21	Caused by trisomy 21 (3 chromosome 21's) caused by abnormal cell division.	Heart defects, hearing/vision problems, learning problems, and facial features.
		Cystic fibrosis	Mutation in a gene called the cystic fibrosis transmembrane conductance regulator (CFTR) • CFTR helps control the amount of salt and water in cells	Affects the mucus, sweat, and digestive juices

2.Environmental disease	Disease caused by factor in the environment	Hay fever	Allergic reaction to allergens in the environment.	cold-like signs and symptoms, such as a runny nose, itchy ey
		Asthma	dust	es,
	Chronic lung disease condition that is caused by prolonged exposure to high concentrations of asbestos fibres in the air	Asbestosis	Fibres in building products that have been distributed are small enough to penetrate the nucleus and cause mutations	Lungs are damaged, cancer forms in the lungs
3.Nutritional disease	Lack or excess of nutrient either micro or macro	Iron deficiency	Blood loss, lack of iron in diet, inability to absorb iron or pregnancy	Fatigue, pale skin, brittle nails and muscle pain
	Lack of vitamin c	Scurvy	Caused by insufficient intake of vitamin c	Skin and teeth fall out
4.Cancer	Uncontrollable cell division	Melanoma	Ultraviolet radiation Likely combination of environmental and genetic factors. • The lighter the skin, the higher the chance of developing melanoma	Illness and death

Genetic diseases

- Some diseases, particularly those caused by genetic factors, have limited treatments available so screening, early diagnosis and management of the symptoms are the best ways to deal with the disease
- **Genetic disease/dysfunction**
 - Are a result of an **altered or incorrect** expression of a gene or genes somewhere in the genome (all genetic material in an organism)
 - Can be a result of:
 - Being inherited

- Mutations to the genetic material during gamete formation- errors during cell division e.g. unseparated chromosomes
- Can result in disruption to normal gene expression and protein production, which results in the individual having a disease
- Down syndrome (trisomy 21)
 - Caused by an extra chromosome on the 21st chromosome pair and is a result of the chromosomes not separate during gamete formation
 - Statistics:
 - In 2007, there were 270 people in Australia with down syndrome
 - The rate of Australian babies born with down syndrome is approximately 1:1000 → lower than the worldwide rate of around 1 in 700 (due to high termination rates in Australia)
- *Albinism*
 - Results in the individual missing the pigment melanin from the skin, hair and eyes
 - Inherited disease that is the result of a mutation to the genes that produce the pigment
 - TYRP1 is the gene affected and results in an enzyme (tyrosinase), not being produce which stops the production of melanin in the body
 - Statistics:
 - Rate of occurrence: 1 in 17000 people
 - The effects on the individual include:
 - Sensitivity to UV light- easily burn in UV light
 - Nystagmus- rapid eye movement
 - Low visual acuity (clarity)
 - Photophobia- discomfort in bright lights

Nutritional diseases

- An individual with an unbalanced diet can result in developing non-infectious diseases
 - Types disease caused by unbalanced diets:
 - Insufficient intake of nutrients e.g. anorexia or malnutrition
 - Excessive intake of nutrients e.g. obesity or type II diabetes
 - Imbalanced diet for a particular vitamin or mineral e.g. scurvy (vitamin c) or rickets (vitamin D)
- *Scurvy*
 - Caused by insufficient intake of vitamin C
 - Most common disease on ships before the 1850s
 - Treatment:
 - Giving the sufferer regular doses of vitamin C and modifying diet to ensure that the minimum required daily intake is achieved
 - Effects:
 - Bleeding gums

- Loss of teeth
- Prevention:
 - Taking sufficient amounts of vitamin C
 - Educating school children
- Management:
 - Vitamin C supplements
- **Adult-onset (type 2) diabetes**
 - Risk of development increase with obesity, poor diet and lack of exercise (as well as genetic factors)
 - Results in reduced insulin (hormone that allows glucose to enter cells)
 - Without glucose, cellular respiration cannot occur
- **Obesity**
 - Statistics:
 - (Australia's Health report 2018) 63% of AUstralian aged 18 and over, and 28% of children aged 5-17 are overweight or obese
 - Compared with 1995, in 2014-15 a greater proportion of adults were in obese weight range, and the proportion in the severely obese range were nearly double
 - Caused by an excessive i take of nutrients and is contributed by a diet high in fatty and/or sugary food, high alcohol consumption and a lack of regular physical activity
 - Causes a high risk of developing a number of chronic diseases, higher rates of arthritis, back pain problems, diabetes and CVD than adults in the normal weight range
 - Overweight and obesity at the population level is measured by calculating BMI

Environmental disease

- Caused by factors that can cause harm
- **Lifestyle disease**
 - E.g. CVD and diseases caused by substance abuse (e.g. alcoholism) and smoking related diseases
- **Exposure to chemicals**
 - *Lead poisoning*
 - Caused by high exposure to high levels of lead in the atmosphere, soil and products used in everyday life
 - The lead accumulates in the body and causes lead poisoning
 - Most heavy metals are cumulative in organisms (cadmium and mercury) → once the heavy metal enters the body it does not leave
 - Children who have large levels of lead in their bodies often eat dirt that contains lead
 - Products including lead (includes banned products):
 - Lead based paints
 - Lead smelting plants
 - Lead pencils (no longer contains lead)

- Leaded petrol
- Lead attacks the CNS and can cause shaking and psychological problems
 - Causes a metallic taste in the mouth
 - In mild cases there may be vomiting and diarrhoea but in more extreme cases, coma or death may occur
- **Physical factors**
 - *Skin cancer*
 - Caused by prolonged exposure to the sun (UV radiation)
 - Melanoma is a type of skin cancer that develops when a mole becomes cancerous
 - The UV light (environmental mutagen) causes changes to the DNA of the skin cells → may cause continued abnormal cell division
 - Can lead to the formation of basal cell carcinomas, squamous cell carcinomas or malignant melanomas
 - Effects:
 - Most lethal form of skin cancer
 - Tumours start in the pigment producing melanocytes in the skin
 - New or unusual growths on the skin or changes to moles are often the first signs of the effect of melanoma

Cancer

- Refers to a group of diseases that involve the unregulated and abnormal cell growth leading to the growth of tumours
- Cancer starts when some cells in the body become abnormal, multiple out of control, and take over from normal cells in the area
- Cancerous cells do not differentiate (depending on which genes are switched on and off, determines what the cell becomes) from initial forms, and so they can't perform their specialised tasks in the tissue
 - E.g. lung cancers prevent gas exchanges in the lungs, leukaemia can prevent blood from transporting O₂, CO₂, and nutrients around the body
- Cancerous growth in tissue that is not vital for survival (e.g. skin or breast tissue) can become malignant (spreads to other parts of the body) or benign (does not move from origin area where tumour forms), where upon the uncontrolled growth spreads to more vital organs of the body
- *Melanoma treatment*
 - If melanoma is detected early, surgery is often the only treatment required
 - The surgery involves:
 - Removing the tumour and the skin around the tumour → ensures that none of the cancerous cells are left behind (reduces the risk of recurrence in the future)
 - Treatment used depends on various factors:
 - How far the melanoma has progressed

- The person's age and health
- Radiation- uses x-rays to damage DNA of cancerous cells
- **Chemotherapy:** drugs slows growth of cells (not as effective for melanoma as other types)
 - Results in all cells that divide quickly to be killed → hence why hair falls off as hair cells die
- Targeted therapies- drugs interrupt the pathways that cause uncontrolled cell division
 - The drugs target specific genes and proteins that are involved in the growth and survival of cancer cells
- Immunotherapy- uses the body's own immune system to fight the melanoma
 - Uses checkpoint inhibitors (drug) that causes the immune system to recognise and destroy melanoma cells
 - Cytotoxic T cells locate and destroy melanoma cells
 - Vaccinations can be used where an antigen is produced using melanoma cells so immune system will destroy them
- Personalised cancer vaccinations
 - Cancerous growths contain unique genetic sequences and biomarkers depending between individuals
 - By comparing DNA sequences of tumour and healthy cells, vaccine companies use recombinant DNA technology to create a vaccine for a specific tumour uniquely within an individual

IQ3: Why are epidemiological studies used?

- 8.3.1 analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:
- a) nutritional diseases
 - b) diseases caused by environmental exposure
- 8.3.2 investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above
- 8.3.3 evaluate the method used in an example of an epidemiological study (covered in module 7)
- 8.3.4 evaluate, using examples, the benefits of engaging in an epidemiological study (covered in module 7)

8.3: Epidemiology

Epidemiology	The study of distribution and determinants of health-related states or events, and the applications of this study to the control of diseases and other health problems
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Epidemiological population	Analysis is always carried out with reference to a population (group at risk for disease/condition) Group defined geographically but if only a small amount of individuals susceptible, additional criteria may be needed
Morbidity	State of being diseased Rate expressed as the number of diseased individuals out of a standard number of individuals in the population
Mortality and mortality rate	The incidence and prevalence of mortality may also be reported. Mortality rate can be expressed through a percentage of the population that has died from a disease or as the number of deaths per 100,000 persons
Sporadic diseases	Diseases that are seen occasionally, and usually without geographic concentration - Disease examples include: - Rabies - Plague - Tetanus
Epidemic disease	Diseases for which a larger than expected number of cases occurs in a short time within a geographic region are called epidemic diseases e.g. influenza
Endemic	(of a disease or condition) regularly found among particular people or in a certain area

Analysis of patterns of non-infectious disease

- An epidemiology study can:
 - Determine the cause of a disease and which populations are affected by the disease
 - Help to develop strategies to control disease and improve the public health
 - Evaluate the effectiveness of strategies in place to treat/control disease
- Identifies incidence, prevalence and mortality rates of a disease
 - Used to identify areas and ways where the overall health of the population can be improved

Methods used in epidemiological studies

- *Descriptive*
 - The first type of study conducted when investigating the cause of a disease
 - Provides patterns of the disease (frequency, demographics affected, geographical location of impact, and any time periods of impact)

- Data types determined include:
 - Age
 - Sex diet
 - Occupation
 - Drinking habits
 - Location of work and home
 - Places visited
- *Analytical*
 - Once a descriptive study is done, analytical studies are done to collect more data (used to test the hypotheses to the likely cause(s) of the disease)
 - Mortality and morbidity are two indicators used
 - Data about incidence and prevalence are put together in this study
- *Cohort*
 - Involves studying two or more similar groups of people who are free of the disease
 - The groups are separated on one factor; 1 group is exposed to the possible cause of the disease and the other group isn't
 - These groups are followed over a long period of time

IQ4: How can non-infectious diseases be prevented?

8.4.1 use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to:

- educational programs and campaigns
- genetic engineering

8.4.1: Prevention

- The preferred option
- Reduces the burden of disease, physical, financial and psychological, on the individual and the health care system
- Strategies to change behaviours include education programs and campaigns, legislation, support groups, and targeting those at risk
- Epidemiology studies provide data to identify areas of risk, to direct strategies for prevention

Education programs and campaigns

- Tobacco control measures became prevalent in Australia during the late 1980s and 1990s
- These measures included quit campaigns, ads advising of the dangers of smoking, banned tobacco advertising and a tobacco excise (e.g. government tax incorporated into the sales price of cigarettes)
- Epidemiology studies have led to a number of government investments in preventative measures
 - Measures include:
 - Educational programs and campaigns

- Free disease screening tests and investment in research for new treatments
- A successful public health campaign includes:
 - An evidence base for action (epidemiology studies)
 - A package of a limited number of high priority evidence-based interventions
 - Effective performance management with real-life monitoring, evaluation and program improvement
 - Partnership between the public and private sector
 - Communication of accurate information to health care workers and the public
 - Political commitment

Genetic engineering

- The ability to manipulate genes and has enabled us to devise a number of ways to prevent certain non-infectious diseases/ disorders but there are ethical issues associated with its use that need to be addressed
- **Preimplantation genetic testing (PGT):** involves testing 3-day-old embryos for specific gene mutations associated with a known disease before implantation
- Genetic disorders such as *cystic fibrosis, albinism, Huntington's and muscular dystrophy* result from mutations in the genetic code
 - The mutated gene gives out faulty instructions leading to the production of an abnormal protein
 - Gene therapy is a solution to fix genetic disorders
 1. Involves the transfer of a normal gene into a person to correct a faulty gene or genetic disorder
 - Tackles the cause of the disorder rather than the symptoms
- **Somatic gene therapy**
 - A normal gene copy of the gene is delivered to those cells so they can produce the normal protein
- **Germline gene therapy**
 - Involves modifying a gamete, embryo or fertilised egg before the germline has split off from the cells that make the rest of the body, so any change made will be passed onto future generations (more controversial)
 - Currently prohibited because of the ethical concerns it raises
- **Gene therapy**
 - In gene therapy the 'healthy' gene at the loci for cystic fibrosis is removed from the genes of a healthy person using restriction enzymes (scissors)
 1. The correct gene is inserted into a plasmid (circular piece of DNA) and then within a bacterial cell where it is replicated or cloned to produce vast numbers of identical copies of the gene for experimental use
 2. The next step is to find a safe way to deliver this normal gene into the epithelial cells of CF sufferers
 - 2 delivery systems, or vectors, have been developed:
 - **Method 1**

- The normal CFTR gene has been attached to the DNA in an adenovirus which has been treated to make it completely safe to use (e.g. of viral vector)
- **Method 2**
 - Liposomes can also be used as vectors for the CFTR gene
 - Liposomes are tiny hollow spheres made up of a positively-charged lipid bilayer
 - The correct gene is enclosed in a liposome and inhaled as an aerosol into the lungs of the person suffering cystic fibrosis
 - The liposomes fuse with the lung tissue and the correct gene moves into the genome and restores normal gene functioning to the person with cystic fibrosis
 - Gene therapy targets the cause of cystic fibrosis
 - Gene therapy targets the cause of cystic fibrosis not just the symptoms
- **CRISPR:** used to repair or replace the CFTR gene that has been changed, in order to restore normal mucus
 - Changes are permanent
 - 3 components of the CRISPR Cas9:
 1. An **RNA guide** that finds the location of the mutation
 2. The **Cas9 nuclease enzyme** that acts like scissors cuts the genetic sequence at the precise binding site
 3. A **repair template** containing the correct genetic sequence for the gene
 - Process:
 1. The **guide RNA** binds to the **Cas9** and activates it
 2. Once activated, **Cas9** finds the location of the mutation by searching the **DNA** for potential targets (mutation)
 3. Once Cas9 finds the mutation, it **unwinds the DNA** and ensures that the **guide RNA** sequence **matches**
 4. **Cas9 nuclease** cuts the **DNA** to create a break on the strands
 5. The DNA template that has the correct sequence is introduced into the cell. The cell uses the **fragment as a template**

	Gene therapy method	Gene editing (CRISPR Cas9)
similarities	● Both modify a gene sequence ● Both aim to cure cystic fibrosis ● Both use DNA manipulation ● Both unethical ● Can be both done on germline and somatic cells	

differences	• Must be repeated multiple times • Used after birth • Temporary • Treatment/management method • Can only be used in differentiated stem cells	• Only needs to be done once • Used before birth • Not temporary • Prevention method • Can be used more on the base cells (before cells are stem cells)
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IQ5: How can technologies be used to assist people who experience disorders?

8.5.1 explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example:

a) hearing loss

8.5.2 investigate technologies that are used to assist with the effects of a disorder, including but not limited to:

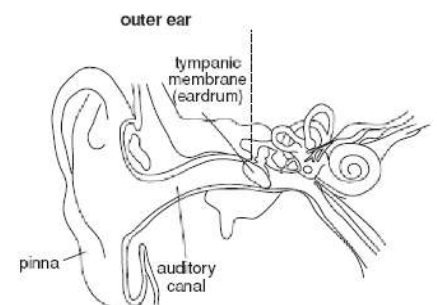
a) hearing loss: cochlear implants, bone conduction implants, hearing aids

8.5.3 evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder

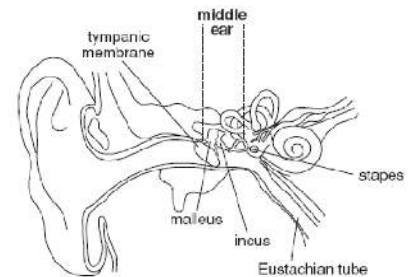
8.5.1-8.5.3: Technologies and Disorders

Ear

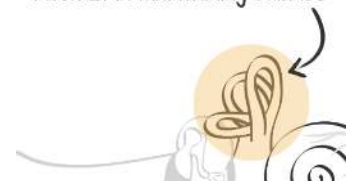
- **Pitch:** refers to the **highness** or the **lowness** of a sound and is determined by the frequency and wavelengths of the wave
 - Pitch **increases** as the number of waves that pass a point in one second (frequency) **increases**
 - Pitch **lowers** as the frequency **decreases**
- Amplitude does not affect pitch but affects **how loud or soft the sound is**
- Sounds are perceived as being loud or soft according to the amplitude of their waveform
 - *High amplitude=louder sounds*
 - *Small amplitude=softer sounds*
- **Outer ear:**
 - **Structure**
 - Consists of fleshy organs located on the side of your head (ears, ear canal, eardrum)
 - **Pinna (external part of the ear)**



- **Structure:** fleshy, external tissue which consists of a flap of cartilage and skin
 - **Function:** collects and funnels sound waves into the ear canal
 - **Auditory canal (ear canal)**
 - **Structure:** a passage comprised of bone and skin
 - **Function:** leads sound waves to the eardrum
 - **Tympanic membrane (ear drum)**
 - **Structure:** thin membrane between the outer and middle ear
 - **Functions:**
 - Vibrates in response to sound (vibrates at the same frequency as the sound wave)
 - Transfers sound vibrations to the **ossicles**
- **Middle ear:**
 - Ear-filled chamber that contains the ear **ossicles** and the **eustachian tube** (between the eardrum and the oval window)
 - **Ossicles**
 - **Structure:** 3 small bones- *malleus (hammer)*, *incus (anvil)*, *stapes (stirrup)*- smallest bones in the body
 - **Function:** amplify and transmit vibrations from the eardrum to the oval window
 - The **hammer** is attached to the **tympanic membrane** and vibrates at the same frequency as the vibrating tympanic membrane
 - The vibrations are then passed through the **anvil and stirrup**
 - **Stirrup:** pushes against the oval window allowing vibrations to be transmitted through to the fluid in the inner ear
 - **Eustachian tube**
 - **Structure:** narrow tube that connects the *middle ear* to the *pharynx* at the back of the throat
 - **When swallowing:** allows air into the *middle ear*, so that air pressure on either side of the *tympanic membrane* is the *same*
 - **Function:** equalises air pressure on both sides of the eardrum so that it is not overly stretched
 - Sudden changes to pressure causes the eardrum to *bulge or retract* and *less able to transmit vibrations* (causes temporary hearing problems)
 - Pressure can be equalised by *swallowing or 'popping' the ears*
 - **Round window**
 - **Structure:** thin membrane between *middle* and *inner ear*
 - **Function:** *bulge outwards* to allow for equalisation of pressure in the cochlea when the *oval window* vibrates
- **Inner ear:**
 - Located within the *bone of the skull*



The vestibular apparatus is involved in maintaining balance



- Consists of *fluid filled* chambers (semicircular canals and the cochlea)
- **Anatomy**
 - The semicircular canals are involved in balance
- **Oval window- separates the middle and inner ear**
 - **Structure:** thin, flexible membrane
 - **Function:** transmits vibrations from the stapes to the fluid in the cochlea
- **Cochlea**
 - **Structure:**
 - Snail-shaped (pea size), spiral tube filled with fluid
 - **Organ of corti:** the middle chamber of the cochlea consists of *hair cells* attached to the *basilar membrane*
 - Contains thousands of specialised *sensory hair cells* with *cilia*
 - The hair cells have cilia that are in touch with the *tectorial membrane*
 - Hair cells at the *larger end* of the cochlea responds to **very-pitched sounds**
 - Hair cells at the *smaller end* (and the rest of the cochlea) respond to **low-pitched sounds**
 - The vibrations transmitted from the *middle ear* cause *tiny waves* to form in the *inner ear fluid* (then interpreted as sound)
 - As the *stapes* push back and forth against the cochlea, it compresses the fluid to create *waves* in the fluid-filled compartments which make the *cilia* vibrate
 - Different characteristics in a wave create different impulses
 - The hair cells convert these vibrations into *nerve impulses*, or *signals* which are sent via the *auditory nerve* to the *base of the brain* where sound is interpreted
 - **Function:**
 - Contains ~30,000 hearing nerve endings in the hair cells
 - Hair cells act as *sound receptors* (when bent, they generate nerve impulses)
 - When the oval window vibrates it causes the fluids in the cochlea to vibrate
 - The basilar membrane then *flexes* in response to the movement of the fluid, bending the hair cells as they rub against the *upper membrane*
 - **Organ of corti:** responsible for converting the vibrations they experience into a *nerve impulse*
 - The nerve impulse is transmitted to the brain through the *auditory nerve* so that the sound can be processed and interpreted
- **Path of sound**
 - Sound enters the ear at the **pinna** as energy in the form of *sound waves* in a gas

- When the sound waves reach the *tympanic membrane*, the energy is changed into vibrations in a solid
 - Energy is then *transferred and amplified* by the **ossicles**
 - At the **oval window** the energy is transferred to vibrations in a fluid in the cochlea
 - Energy is then transduced into *electrochemical energy* by the **stereocilia** of the hair cells
 - The stereocilia are attached to neurons
 - The axons of the neurons join together to form the auditory nerve
- **Hearing loss**
 - **Conductive hearing loss**
 - Deafness caused by damage to the **outer or middle ear**
 - Results in *reduced or muffled* sound
 - Causes:
 - Malformed structure of birth
 - Aging
 - Noise strain
 - Injury
 - Infection
 - Blockage due to earwax, a tumour or foreign item
 - **Treatment:** medication or surgery
 - **Sensorineural hearing loss**
 - Deafness caused by damage to the **inner ear/nerve pathways** from the inner ear to the brain
 - Results in *faint or muffled* sound
 - Causes:
 - Malformed inner ear at birth
 - Exposure to damaging noise can cause the hair cells in the cochlea to become fatigued/kill them
 - Head injury
 - Can't be treated medically or surgically
 - **Auditory processing disorder**
 - Processing problems in the auditory areas of the brain
 - Results in inability to detect the direction of a sound
 - **Tinnitus**
 - When a sound is heard that isn't there
 - **Ring in the ears:** buzzing, hissing, may occur
 - Causes:
 - Exposure to loud sound
 - Injury
 - Aging
- **Treatments for hearing loss**
 - **Hearing aids**



- Battery operated electronic device that fits into the ***hollow*** outside of the ear canal

- **How does it work?**

- *Amplifies* sound waves within the ear canal
 - The ***eardrum, ossicles and cochlea*** must still have an *adequate* level of function so that sound can be registered and converted into an electrochemical signal
 - Can be used for people with *conductive or sensorineural hearing loss*



- **Parts of a hearing aid**

- **Microphone:** receives sound waves and changes them into electrical energy
 - **Amplifier:** makes the signal stronger
 - **Receiver:** changes electrical energy into sound energy
 - **Speaker:** directs sound energy into the ear canal

- **Advantages:**

- Relatively cheap
 - Easy to install (no surgery required)
 - Improved speech benefits
 - Customisable settings

- **Disadvantages:**

- Amplify all sounds in an environment (including background noise) and can cause pain
 - It doesn't help people with severe damage to the inner ear/auditory nerve

- **Cochlear implants**

- Used by people with *profound hearing loss* (profound damage to any region of the ear)

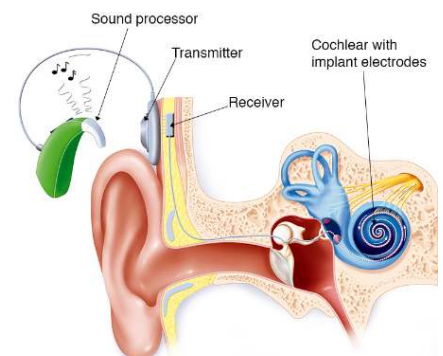
- **sensorineural hearing loss**

- **How does it work?**

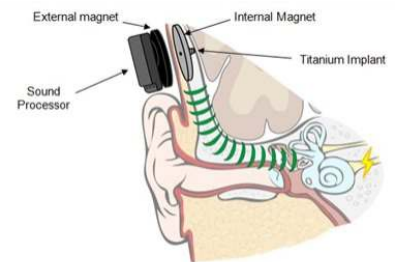
- Instead of transmitting sound waves, it converts sound into *electrical signals* and directly stimulates the auditory nerve

- **Parts of the cochlear implant**

- **Receiver:** surgically implanted into a patient's skull
 - Fine wires attach the receiver to the cochlear
 - **Microphone:** behind the patient's ear picks up sound
 - **Speech processor:** converts the sound to an electrical signal



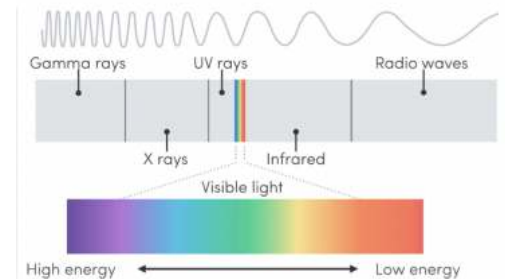
- **Transmitting coil:** sends the message as a radio signal to the receiver
 - The wires then stimulate the auditory nerve, stimulating the brain
- Advantages:
 - Provides hearing to profoundly deaf people
- Disadvantages:
 - Highly expensive
 - Operation side effects (e.g. infection, drooping face, numb tongue due to nerve damage)
 - The wearer must learn to interpret sounds again
 - Success of the implant system is limited if the auditory nerve or region of the brain is damaged
- **Bone conduction implants**
 - A medical device that transmits sound by **direct conduction** through bone to the **inner ear** (bypasses the outer and middle ear)
 - Consists of a small titanium implant, abutment and sound processor
 - Sound is transmitted as **vibrations** from the sound processor to the implant, through the **bone** to the **inner ear**
 - Similar to cochlear implants and have an **external** and **internal component**
 - Type of hearing loss it can assist with:
 - Conductive hearing loss
 - Mixed hearing loss
 - Single sided deafness
 - **How does it work?**
 1. Sound waves are detected by the **sound processor**, which is located on the side of the head behind the ear.
 - *These sounds are digitally analysed, amplified and converted into vibrations*
 2. The implanted part of the device then transmits the vibrations through the bone to the **inner ear** (bypasses the outer and middle ear)
 3. These sound vibrations reach the inner ear where they cause **movement** in the fluid filled **cochlea**
 - The movement stimulates the tiny hair cells, **generating electrical impulses**
 4. These electrical impulses are sent along the **hearing nerve to the brain**, where they are interpreted as sound
 - Advantages:
 - Free of hearing without an ear mould in the ear



- More consistent and reliable hearing, unaffected by chronic ear conditions
- Improved quality and awareness of sound
- Improved confidence in conversation
- Disadvantages:
 - Risk of infection and inflammation at the implant site after surgery
 - Small risk that the growth of new bone around the implant
 - Leading to the implant to possibly fall out
 - Loss of residual hearing
 - **Rare complications:**
 - Facial paralysis
 - Infection at surgery site
 - Balance problems
 - Dizziness
 - Taste problems

Eye

- **Visible light:** a kind of electromagnetic radiation
 - Energy which travels through the universe as a wave
 - Light waves can have *different* amounts of energy
 - We perceive these differences in energy as different colours
 - When light comes into contact with an object its either:
 - Absorbed
 - Reflected
- *Eye structure and functions*



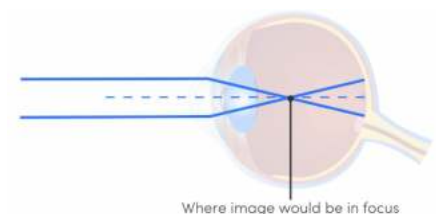
Structure	Function
Conjunctiva	• The conjunctiva lines the inside of the upper and lower eyelids, producing mucins and tears to keep the eye lubricated. Continuous with the cornea, the conjunctiva <i>lies on top of the sclera</i> • Mucins produced by the conjunctiva maintain a healthy tear film. The conjunctiva plays a vital role in the defence of the outside eye, as mucin secreted by the conjunctiva helps to trap and clear allergens
Cornea	• Protects the <i>iris and pupil</i>, while providing much of the eye's focusing power. Dome-shaped and clear, the cornea covers the <i>pupil, iris and anterior chamber</i> • The cornea is transparent in order to refract light correctly. Any <i>cloudiness or opaqueness</i> can <i>heavily</i> interfere with vision • <u>Serves as a filter</u>, screening out damaging UV wavelengths in sunlight. Without this protection, the lens and retina would be susceptible to injury from these UV rays

Sclera	- Protects and supports the structure of the eye - Commonly known as the "white of the eye" - The tough tissue of the sclera maintains the shape of the eyeball and protects from injury, and provides an attachment for 6 of the eye's extra ocular muscles
Retina Rods and cones	Retina - Converts light into signal that are passed on to the brain - Cells within the retina process and convert light energy into electrical signals. These electrical signals are then transmitted to the brain through the optic nerves and interpreted as the images we see Rod's Photoreceptor cells in the retina - Sensitive to light levels and help in providing good vision in low light Cones - Responsible for colour vision
Iris	- The coloured section at the front of the eye, which controls the size of the pupil. By changing the size and diameter of the pupil, the iris adjusts the amount of light that enters your eye - Surrounds the pupil, and is attached to the middle of the anterior surface of the ciliary body. It separates the anterior chamber (between the cornea and iris) from the posterior chamber (between the iris and lens)
Pupil	- Small black hole visible at the centre of the iris, and controls how much light enters your eye - Changes in size in order to control how much light hits the retina at the back of the eye. It is controlled by two groups of surrounding muscles, known as the <i>sphincter pupillae</i> and the <i>dilator pupillae</i>
Lens	Focuses the light that enters your eye unto the back of your eye - The lens is a transparent, biconvex structure located behind the iris. Ciliary muscles change the shape of the <i>crystalline lens</i>, focusing light so that images come to a sharp focus on the retina
Aqueous and Vitreous humour	Aqueous humour - Produced by the ciliary body - Flows from the <i>ciliary body</i> through the pupil, and into the <i>anterior chamber</i>. It diffuses into the trabecular meshwork and into the canal of Schlemm, then finally into the tiny blood vessels surrounding the eye Vitreous humour - A clear gel that exerts enough pressure to keep the retinal layers tightly pressed together - The pressure keeps the retinal layers tightly compressed together, and also maintains the round shape of the eye

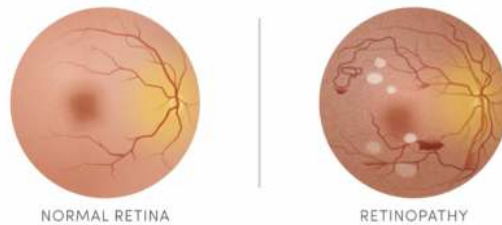
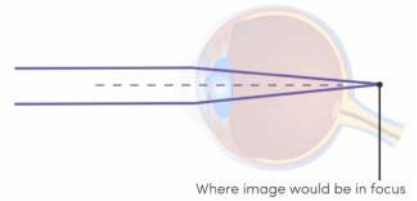
Ciliary body and Sensory ligaments	Suspensory ligaments - Connect the <i>ciliary muscles</i> to the <i>crystalline lens</i>, holding it in place and helping to change its size Ciliary muscle - Relaxes when your eye looks at a distant object, which pulls on the suspensory ligaments. This in turn pulls on the crystalline lens, causing it to become flatter and more able to focus light from the close object unto the <i>retina</i> Ciliary body - Controls the shape of the <i>crystalline lens</i> and produces a fluid called <i>aqueous humour</i>
Optic nerve	Transmits impulses received from the <i>retina</i> to the <i>visual cortex</i> in the brain, forming images in our minds - Light is converted into impulses via <i>photoreceptors</i> found within the retina. The information is sent through the optic nerve and past the optic chiasm - The <i>optic nerve</i> becomes the optic tract; where <i>impulses are sent to the brain</i> to be processed as a visual image
Macula	Highly sensitive circular area that is responsible for providing detail and colour to your vision in <i>bright light conditions</i>. Such detailed vision can be provided because of its high concentration of <i>photoreceptors</i> (particularly cone cells), found in its centre
Fovea	- Small part of the macula that provides the <i>sharpest and most accurate colour vision of the retina</i> - Makes up less than 1% of the retina, but information received here takes up over 50% of the Visual Cortex in the brain
Blind spot	Optic disc: marks the entrance of the optic nerve into the retina - The area of the retina that doesn't respond to light '<i>Blind spot</i>' is commonly used to describe the optic disc

- Visual disorders

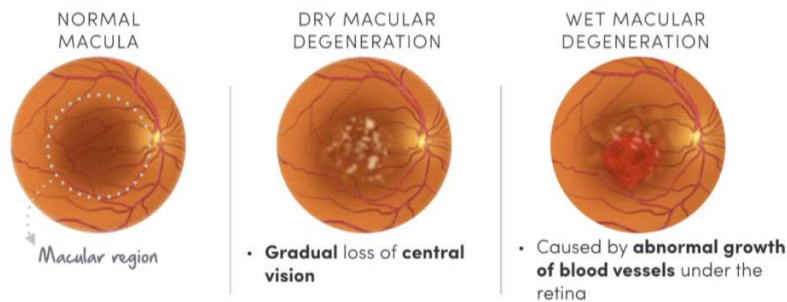
- Myopia and hyperopia are two common types of visual disorders
- Refractive errors**: results from when light is not being properly bent unto the retina so that vision is *blurred*
- Myopia**: a defect of the eye due to which the eye is **not able to see the distant object clearly**
 - Causes the image to form **in front** of the retina
 - Close objects are *clear*, but *distant* objects are *blurry*
 - The image is being focused *in front* of the retina
 - Causes:
 - The lens is **too thick** (bends light too much)



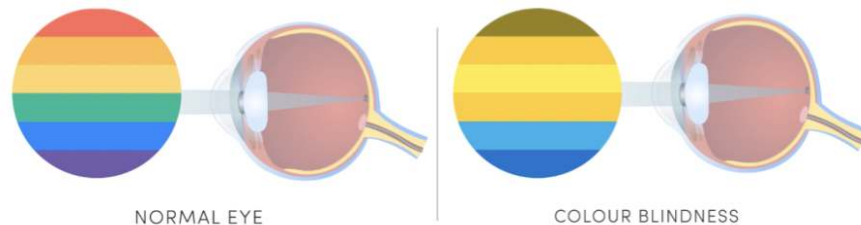
- The eyeball is **too long** (the retina is further away)
- **Hyperopia:** a defect of the eye due to which the eye is **not able to see the far object clearly**
 - Causes the image to form *behind* the retina
 - Distant objects are *clear*, but *close* objects are *blurry*
 - The image is being focused *behind* the retina
 - Causes:
 - The lens is **too thin** (doesn't bend light enough)
 - The eyeball is **too short**
- **Retina conditions**
 - **Retinopathy:** damage to the retina, often caused by *changes to the blood vessels* surrounding this region



- **Macular degeneration:** caused by a **distortion or absence** of the fovea

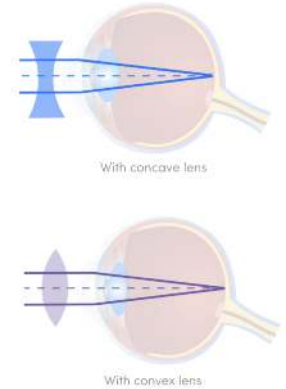


- **Colour blindness:** occurs when one of the three types of colour detecting cone cells is **absent or doesn't function properly**
 - Typically inherited
 - Affects both eyes



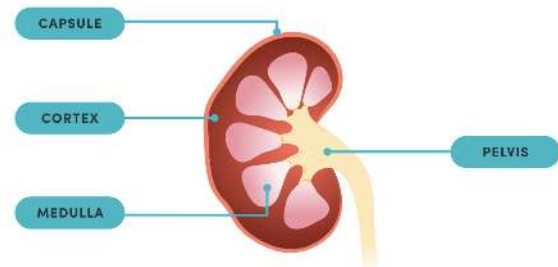
- **Treatments for visual disorders**
 - **Glasses**

- Glasses have lenses which are shaped in order to correct individual needs, **by compensating for misshapen eye lenses or eyeballs**
- Glasses cause *light rays to bend before entering the eye* so that the image focuses on the retina
- 2 different types of lenses:
 - *Concave (curved inwards)*
 - Used to correct **myopia**
 - Causes light rays to diverge so that the image forms further back, falling on the retina
 - *Convex (bulging outwards)*
 - Used to correct **hyperopia**
 - Causes light rays to converge so that the image forms further forward on the retina
- **Disadvantage:** Difficult to wear during active activities
- **Contact lenses (works the same way as glasses)**
- **Laser surgery/ lasik**
 1. A thin flap of the cornea is *cut and lifted up*
 2. A *laser* (high energy light beam) is used to *reshape the cornea*
 3. The flap of the cornea is *repositioned* and it rebonds to itself

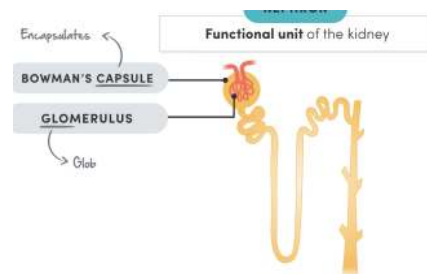


Kidney

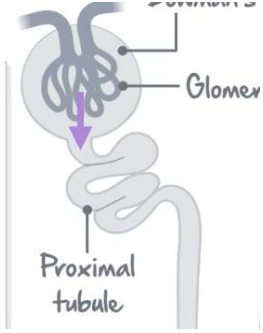
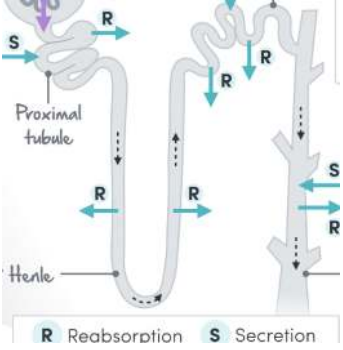
- **Kidney structure**
 - Responsible for *filtering* the blood
 - The kidney's **excrete excess wastes, salt and water** in order to maintain homeostasis (a constant internal environment)
 - **Capsule:** protective covering
 - **Nephron:** functional unit of the kidney that filters the blood
 - **Bowman's capsule:** surrounds a dense clump of capillaries (glomerulus)
 - **Glomerulus:** the main *filtering* unit of the kidney



- **Kidney function**
 - Blood enters the kidney via the **renal artery**
 - The blood then flows through a series of *smaller arteries* until it reaches the *glomerulus* in the *nephron*

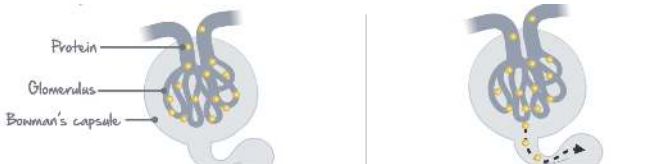
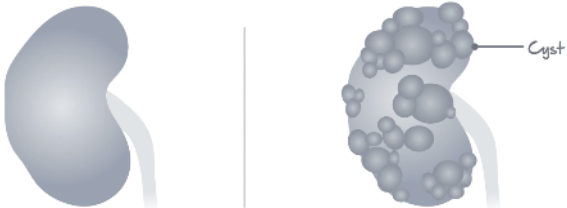


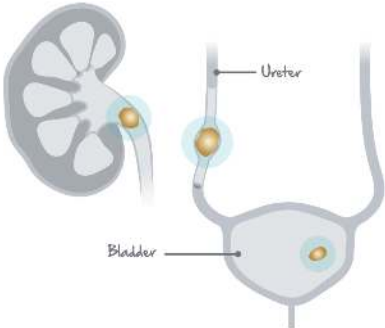
1. Filtration	● Blood pressure forces <i>plasma (water, salts and dissolved solutes)</i> out of the blood, through the glomerulus and into the Bowman's capsule
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	• Larger blood components (e.g. red blood cells) stay in the blood • Filtrate: the fluid that now enters the capsule ○ Flows into the <i>tubule</i> (semi-permeable: allows things to pass through its walls- permeability varies within the tube) 
2. Reabsorption and secretion	• Useful substances (sugars, nutrients and certain amounts of water) are passively reabsorbed back into the blood 
3. Storage	• Urine: fluid containing unwanted materials (e.g. urea, unwanted salt) • Urine is <i>removed</i> through the <i>ureter</i> • Urine is <i>stored</i> in the <i>bladder</i>
4. Urination	• Urine is <i>released</i> via the <i>urethra</i>

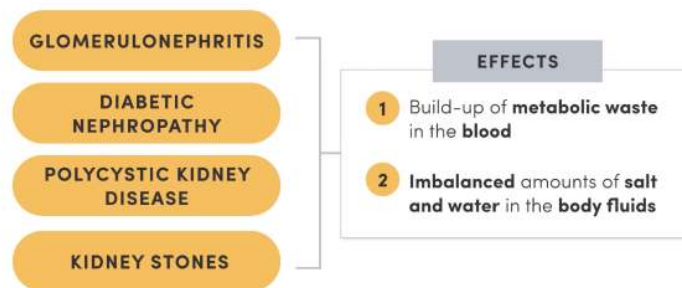
- **Kidney disorders**

- **Definition:** when the kidneys are damaged and don't function as they should
- Can vary from **temporary disorders to renal failure**

Glomerulonephritis	● Inflammation of the glomerulus ● Often occurs after a <i>bacterial infection</i>  NORMAL KIDNEY GLOMERULONEPHRITIS
Diabetic nephropathy	● Kidney damage caused by diabetes (the individual is unable to maintain healthy levels of sugar in the blood) ○ The kidney tries to remove all the <i>unwanted sugar</i>, causing it to become <i>fatigued and damaged</i> by extra strain ● Less efficient removal of waste ● Large substances (e.g. red blood cells and other proteins) can be filtered out of the blood when they shouldn't be able to  Protein Glomerulus Bowman's capsule NORMAL DIABETIC NEPHROPATHY
Polycystic kidney disease (inherited)	● Causes cysts to grow in the kidney ○ Cysts are filled with a watery fluid ● Cysts can enlarge and damage the kidney  Cyst NORMAL KIDNEY POLYCYSTIC KIDNEY DISEASE

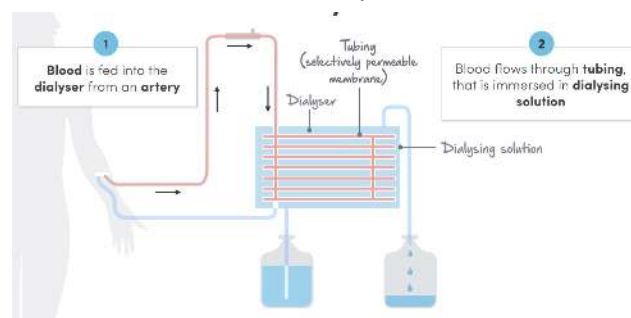
Kidney stones	- Form when waste products <i>crystallise</i> into lumps - Can be incredibly painful and can cause blockages <u>Factors which increase a person's risk:</u> - Being dehydrated - Poor diet - Hereditary factors 
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○ **Effects of kidney disorders**



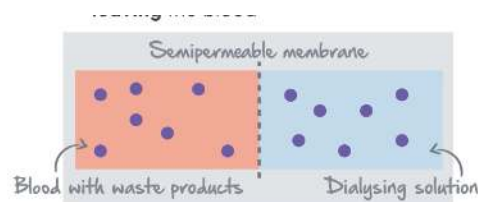
● **Treatments for kidney disorders**

- **Dialysis:** the process of removing waste products and excess fluid from blood by an external machine called a dialyser



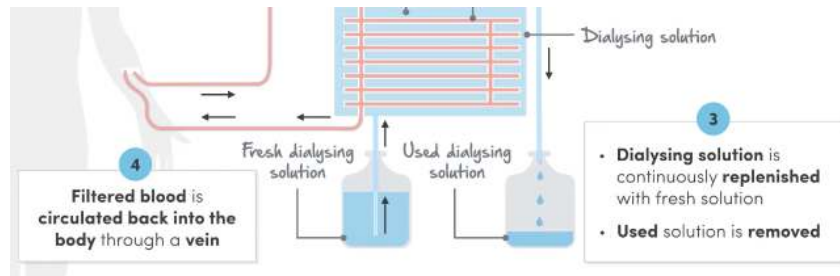
■ **Semi-permeable**

- Pores:
 - Allows for *small molecules* to move out of the blood (e.g. urea to move of the blood)
 - Prevents any *large molecules* form *leaving* the blood



■ **Dialysing solution**

- Promotes the *diffusion* of appropriate substances *into and out* of the blood
 1. Has *higher* concentration of **glucose and ions** to *limit their diffusion* from the blood
 2. Has a *low* concentration of **wastes** to promote their diffusion out of the blood



■ **Advantages:**

- Improved quality of life
- Professionals do the process for you

■ **Disadvantages:**

- Comes with a lot of equipment that is often only available at the hospital
- A person needs to be connected to the dialyser for at least two hours, two to three times a week
- **Blood issues:**
 - Invasion of the individual's blood vessels can cause blood clotting, bleeding and infections

○ **Renal dialysis**

- Used to keep patients with renal failure alive
- The blood is carried from the body via tubes and into a dialyser
- The dialyser consists of a mass of very small diameter semi-permeable fibres through which the blood flows, surrounded by fluid called dialysate
 - Dialysate is similar to plasma but doesn't contain wastes
- The primary forces cleaning the blood in dialysis are diffusion osmosis
 - Since dialysate contains a similar nutrient concentration to the blood plasma, useful substances such as glucose and amino acids remain in the blood
- Nitrogenous wastes move down their concentration gradient, from the blood into the dialysate
- Filtration occurs but not as efficiently as a kidney would do normally
- **Advantages:**
 - Only occurs every 4 days
 - Professionals do the process for you

- Life preserving
- Disadvantages:
 - High in cost
 - Causes the patient to be unwell for a day after haemodialysis occurs
 - The patient must be mindful of the food and drinks they consume
 - Salt and alcohol intake must be minimised
 - High risk of infection with tubes inside the body which may cause blood infections if not treated carefully
 - Can't do active transport (nutrients can be reabsorbed)
- **Judgement:** overall, haemodialysis is life preserving and is good for children until a kidney becomes available for kidney transplant
- *Kidney transplant*
 - Kidneys can be donated from *healthy individuals* on a voluntary basis (usually amongst family members)- places *pressure* on the remaining kidney of the healthy donor
 - There aren't enough kidney donations so the wait time is quite long
 - While a person suffering loss of kidney function waits for a suitable kidney, the wastes that the kidney normally filters will **build up** in the body (toxic effect)
 - Urea build up in the blood impacts the pH and causes the **denaturation of enzymes in cells**
 - Advantages:
 - *Improved quality of life:* when successful, can allow a person to live a longer, more active life
 - *Efficient procedure:* surgery normally takes 2-3 hours and recovery only takes days
 - More effective than dialysis
 - Disadvantages:
 - *Kidney rejection:* the recipient's immune system recognises that the kidney is foreign, and attacks it
 - *Immunosuppressive drugs:* transplant recipient is far more vulnerable to future infections and cancers
 - *Long waiting lists:* transplant requires access to a donated organ
- *Peritoneal dialysis*
 - The *semi-permeable membrane* used to clean the blood is a membrane inside the abdomen of the body- a large sac called the **peritoneum**
 - **How does it work?**
 - Similarly to dialysis, a *sterile glucose solution* is pumped into the *peritoneum* (acting as a glomerulus)
 - Blood vessels flow on the outside of the *peritoneum*, and substances *diffuse in and out* based on their concentration gradients

- The movement of nutrients is *minimised* whereas the movement of nitrogenous wastes is *maximised* by having a large concentration difference
- The fluid is left in the *peritoneal cavity* for several hours and is then drained, removing the nitrogenous waste
- The cavity is then refilled with fresh dialysate solution (cycle is repeated 4-5 times a day)

■ Advantages:

- Life preserving until kidney is available for transplant
- Can be done at home
- Low cost

■ Disadvantages:

- Done 4-5 times a day (time consuming)
- Risk of infection at the abdominal cavity
- Can't do active transport (nutrients can be reabsorbed)
- Done by the individual themselves